“Even in the Middle of the City, I Notice Nature Around Me”: Is Nature Relatedness Associated with Noticing More Nature in Urban Environments?

by

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Abstract

Nature relatedness is an individual difference characteristic defined as an interconnected feeling between the self and nature (Nisbet et al., 2009). Previous meta-analytic research suggests that nature related individuals are happier (Capaldi et al., 2014; Pritchard et al., 2020). The literature suggests nature related individuals may be happier because they are more likely to go out into nature, which promotes well-being. However, it could also be partly because nature-related individuals may be noticing more nature around them at any given moment, which has not been previously tested. In this dissertation, I examined the possibility that highly nature related individuals notice more nature in urban scene images. Items of the Nature Relatedness scale (Nisbet et al., 2009) propose noticing nature behaviours that fit in the self-concept of a person high on the trait. Still, no research has explored behavioural measures of momentary noticing nature in this association. In this dissertation, noticing was explored through two distinct processes, perception and attention, with the research split into two phases. Perception (Phase 1) is characterized as what an individual interprets as nature in an urban scene. Attention (Phase 2) is operationalized in two ways: the first objects seen in an image (first attention) and voluntary attention paid to pleasant objects in the images. I hypothesized that nature related individuals would perceive more nature, notice nature first, and focus more on the nature in the images than individuals lower on nature relatedness. Contrary to the hypothesis, there was no association between perception of nature and nature relatedness in Phase 1. However, there was an association between attention being paid towards nature and nature relatedness in Phase 2, where nature related individuals had their attention drawn to nature first and chose to focus their attention on it. There is also some preliminary support that nature related people's ability to notice nature may be a reason they are happier in general. This study is the first to utilize

behavioural tasks to explore if nature related individuals do, in fact, notice more nature in urban environments.

Keywords: Nature Relatedness; Nature; Noticing; Perception; Attention; Urban Nature; Extinction of Experience; Images; Self-concept

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"Even in the Middle of the City, I Notice Nature Around Me": Is Nature Relatedness Associated with Noticing More Nature in Urban Environments?

One of the largest impacts of the industrial revolution was the transformation of our manufacturing processes and workplaces. This shift led to a surge in people relocating to larger cultural and economic centers. Thus, most people were no longer living in rural environments, but rather in urban areas. For example, in the 1920s, only 50% of Canadians lived in cities; in 2015, this number increased to 80% (Stelter & Artibise, 2015). Within the next 20 years, around the world, it is projected that 70% of people will be living in urban cities (Ritchie & Roser, 2018). As more people move to cities, the more cities must expand and reduce the natural areas to make room for more development (McKinney, 2008). This shift has changed our access to nature, which may be impeding our interactions with it. This reduction of experiences can be detrimental to our well-being because spending time in nature has previously been associated with physical, emotional, social, and spiritual well-being (Chowdhury, 2019). If we spend less time in nature, we may lose out on its benefits.

Even with much research showing the benefits of spending time in nature on our well-being, the definition and metric of nature in research have not been consistent across the literature. The Merriam-Webster Dictionary (n.d.) defines nature as a physical environment free from human production. This physical environment of nature can encompass many things, such as forests, bodies of water, plants, wildlife, and rocks, to name a few. In research, there are many different ways authors define nature in their articles, with nature being characterized as environments such as forests, urban parks, blue spaces (e.g. beaches, marinas, lakes), or even just trees in a neighbourhood. When Taylor and Hochuli (2017) explored the definition of greenspace in 125 articles across multiple disciplines, it was clear there was no single unified definition of

greenspace; across the literature, the authors found that there was a range of operational definitions, interpretations, and even a lack of clear articulation about what the author meant by using the term greenspace. Minimal forms of nature, such as indoor foliage or views of nature outside their windows, also count as nature in the literature. Additionally, some research utilizes nature activities or indirect nature exposure, such as images, videos, art, or biophilic architecture as forms of nature. Altogether this suggests that many different forms of nature may count, but more minimal and indirect forms of nature may still differ from larger, more expansive forms of nature, such as the forest.

Outside of research, individual people can also differ in their definition of nature. For example, when Vining et al. (2008) asked participants how they defined nature, approximately 47% of people reported that nature was defined as untouched by humans or not human-made. However, even if they felt that nature had to be untouched by humans, over 75% of people reported feeling part of nature (when asked if they felt a part or separate from nature; Vining et al., 2008). Therefore, even with less chance to interact with nature, people still seem to connect and feel a part of it. This feeling of being a part of nature is characterized as nature connectedness (nature relatedness). Nature connectedness is a trait defined by an individual's feeling as being one with nature, caring for nature, and wanting to protect it (Schutlz, 2002). Previous research suggests that individuals who identify as connecting more to nature than others tend to be happier in general. Two meta-analyses have shown that nature connection is positively related to well-being indicators, such as pleasant emotional balance, life satisfaction, and purpose in life (Capaldi et al., 2014; Pritchard et al., 2020). There are a few suggestions for why we see

this relationship. It could be that nature connected individuals[1] are interacting with nature more than those who are not as nature connected. One way they could be doing this is by being more likely to be out in nature, which has previously been associated with well-being (see review Capaldi et al., 2015). The other way is that nature connected people may be noticing more nature in their everyday urban lives, with or without differences in objective nature contact. The current dissertation examines the possibility of the latter. Specifically, the main goal of the dissertation is to examine if nature connected individuals are more likely to "*notice*" nature in an urban environment. Three studies, split into two separate phases, were run. The three studies used behavioural measures of noticing nature rather than relying on broad self-report items generally used to measure nature connection. For example, the item highlighted in the title ("Even in the middle of the city, I notice nature around me"; Nisbet et al., 2009) used to score an individual's nature connection (nature relatedness) may be more a belief of the self than actualized behaviours in real-life situations. I used urban images with a controlled amount of nature to explore if nature connected individuals noticed the nature in the urban environments.

Nature and Well-being

One of environmental psychology's many focuses is nature research, and even more specifically, addressing how nature is beneficial for humans. Much research suggests that spending time with nature or connecting with nature is beneficial in many ways. There are physical benefits of nature exposure that can be both mental and bodily. For instance, research has shown that exposing ourselves to nature, such as walking in nature (Basu et al., 2019; Berman et al., 2008), seeing nature images (Gamble et al., 2014), and even nature sounds (Van

[1] When identifying nature related (connected) people/individuals, I am referring to people who score higher on the individual difference characteristic. This implies an implicit comparison with people lower on the dimension, but I am treating nature relatedness as a continuous dimension rather than a binary classification.

Hedger et al., 2019) help with attention control, usually tested through attention tasks in labs (Stenfors et al., 2019). This is the case for both children and adults. Nature exposure is also helpful for child development. For example, children who had access to more greenness had more imaginative play than those in an ordinary school (through green schools; Dowdell et al., 2011), were more self-disciplined (Taylor et al., 2002), developed more motor skills (White, 2004), had better academic performance (Sivarajah et al., 2018), and developed more problem-solving skills (Leiberman et al. 1998). Research also suggests that the amount of time spent in greenspaces in childhood was related to lower levels of psychiatric disorders in adulthood (Engemann et al., 2019). Nature is also used as a therapeutic tool for mental illness (see review: Harper et al., 2021).

There are also physiological and self-report markers of health shown to be better with more nature exposure. For example, Ulrich and colleagues (1991; 1983) found that nature views, images, or videos were good for surgical recovery, pain tolerance, and stress reduction. Additionally, White and colleagues (2019) showed that those who reported spending more time in nature (defined as "open spaces in and around towns and cities, including parks, canals, and nature areas; the coast and beaches; and the countryside, including farmland, woodland, hills, and rivers" p. 8) reported better general physical health and felt more satisfied with their life than those who did not spend any time in nature. The researchers found a significant increase between 0 and 120 minutes spent a week in nature, but there were diminishing returns after 120 minutes (i.e. levelling off). A study conducted in Toronto examined the health benefits of neighbourhood trees. Researchers examined the objective number of trees in a neighbourhood through mapping data and subjective ratings of overall health and diagnosis for numerous medical and mental health conditions. The researchers found that those living in a neighbourhood with more trees

self-reported better overall health and fewer cardio-metabolic disorders (Kardan et al., 2015b). There was no significant relationship between the number of trees in a neighbourhood and other disorders, such as mental disorders and other illnesses (cancer, migraines, asthma, or arthritis). The authors reported that having ten trees in your block would predict an increase of .04 units in a 1 to 5 scale of health perception, which was equivalent to an increase of your annual household income by $10,200 or being seven years younger. Hunter et al. (2019) examined the amount of time spent in a natural environment to see a physiological reduction in stress through cortisol and alpha-amylase hormones. They saw that after participants spent 21 to 30 minutes in nature, there was a reduction in both stress hormones, but there were diminishing returns after 30 minutes of exposure.

Nature and Happiness

Nature is also related to emotional and psychological well-being. One way this can be characterized is a person's feeling of happiness. Diener (1984) proposed that for a person to feel happy, they would emotionally feel good in day-to-day life and have positive cognitive judgements about their life overall. This view encapsulates the term subjective well-being. Positive affect, negative affect, and satisfaction with life generally fall under this term. Affect is defined by experienced feelings, such as moods (Diener et al., 2010). In research, this is usually tested by asking participants to report how often they felt (i.e. recall) or how much they are currently feeling various emotions (i.e. state). A person's trait level affect is an aggregate of state feelings over time. Life satisfaction is an individual's judgment that their life events have gone well and they are happy (satisfied) with their life (Diener et al., 1985). The overall assessment of life satisfaction arrives from each person's idiosyncratic criteria of what it means to be satisfied with life.

However, others suggest that subjective well-being may be too narrow of a characterization of happiness and may be missing out on other factors that describe a good life. For example, Ryff (1989) argues that happiness or positive well-being comes from more extensive psychological markers and proposes the Six-Factor model of Psychological Well-being. Ryff's (1989) model is characterized by six factors: meaning in life, personal growth, autonomy, personal mastery, positive relationship with others, and self-acceptance. To date, there are many other markers or ways to assess how a person is flourishing in their life that scholars describe as eudaimonic indicators (see review by VanderWeele, 2017). The term eudaimonic well-being is widely debated on its definition and the markers of well-being that fit within it (Huta & Waterman, 2014). Still, these markers are commonly explored to examine well-being and its association with nature (see Calpaldi et al., 2015 for a review on the ways nature is associated with a person flourishing in life).

Nature contact relates to a range of psychological and emotional well-being indicators. For one, meta-analytic work on the effect of nature exposure on affect (both positive and negative) across 32 studies in 6 different countries (Japan, USA, Canada, Sweden, Netherlands, and the UK) suggests that being exposed to nature increases positive affect and decreases negative affect; however, the relationship between negative affect and nature is not as strong (but still significant; McMahan & Estes, 2015). They also found no difference between the nature-affect relationship in two types of nature: wild nature (e.g. forest) and managed nature (e.g. urban walking trail). Additionally, both actual nature exposure and lab-simulated nature exposure were found to be beneficial for mood, but actual nature exposure (15 studies) was better than lab-simulated exposure (16 studies; McMahan & Estes, 2015).

Further, White and colleagues (2019) showed that the odds of someone reporting good well-being (measured through a single life satisfaction item) was higher for those spending more time in nature (at least 60 minutes per week) than those not spending any time in nature. This relationship saw a linear increase (of the odds ratio) from 0 minutes to 120 minutes a week and diminishing returns after that. Though, other research suggests that even spending at least 5 minutes outside was shown to increase both positive affect and feelings of awe (Neill et al., 2019).

Even everyday nature experiences have been related to mood. For example, those who travelled more in green and blue spaces experienced a generally higher positive mood (Glasgow et al., 2019). Further, if they travelled through more dense, compact environments, such as more urban-type areas, they were more likely to see a decrease in their moods. Also, those momentarily outdoors or even doing outdoor activities, such as gardening, bird/nature watching, walking, or hiking, reported being happier on a single-item happiness measure (MacKerron & Mourato, 2013). This was examined through GPS data of locations (i.e. more objective indicator of nature). In another study, when people reported being in nature, they reported more state vitality (defined as feeling alive, alert, and energized), over and above social interactions, presence of weekends, and even if the activity they were doing was physical activity (Ryan et al., 2010). This was also seen during the COVID-19 pandemic, where those spending time outdoors felt happier (single item; Stieger et al., 2021), and in particular, those doing activities that may be viewed as an activity engaged with nature, such as walking, gardening and bird watching showed higher levels of positive affect (Lades et al., 2020). Altogether, this suggests that nature contact is related to happiness.

Why do we see benefits from being exposed to nature?

The biophilia hypothesis describes a tendency for humans to have an innate urge to connect with other forms of living things, such as nature and wildlife (Wilson, 1984). Biophilia comes from the Greek term meaning love of life or living things. The hypothesis stems from our understanding of our evolutionary history in nature. Humans have spent the large majority of our existence dependent on nature for our survival. We need to understand our local surroundings to read the weather, the movement of our prey, the local plant life that can be used and consumed, and how to protect ourselves from our predators. Therefore, we have evolved to process information about the local place, its plants, wildlife, and surroundings and be drawn to the healthy parts.

More than just being drawn to healthy nature, there are theories about why we may see the benefits from spending time in nature. One of these theories is the Attention Restoration Theory (Kaplan & Kaplan, 1989). As we have moved away from nature and into a more urban environment, we have to try to process new information about highways, automobiles, shopping centres, workplaces, to name a few, that we have not evolved to process. Therefore, with its bright lights, loud sounds, and stressful everyday life demands, the urban environment over-stimulates and makes us feel fatigued. Kaplan and Kaplan (1989) describe these stimulations of urban living as hard fascinations. They suggest that nature can be a soft fascination that allows one to feel restored, which characterizes the theory. Four characteristics define a soft fascination: the soft fascination must: 1) allow for a sense of being away from the hard fascination; 2) allow you to keep your attention or fascination with minimal effort; 3) be large enough to explore and; 4) it must also elicit feelings of pleasure. Nature fits these four characteristics. The Attention Restoration theory claims that to feel restored, we need to find soft fascination to reduce the

effects of the hard fascinations in our lives. However, the nature effect on cognitive performance has been mixed in the literature (as identified by Stenfors et al., 2019). When Stenfors and colleagues (2019) explored this effect in a pooled dataset of 13 experiments performed in the Berman lab, they found that nature exposure improved attention post-experience compared to urban environments. This effect was even stronger after controlling for the ordering effect of conditions. Therefore, spending time in nature can be beneficial for attention and restoration.

Additionally, our modern world is full of stressors that trouble us. Previously, our ancestors would get physiological responses from dangers around them that would trigger their fight or flight instinct. Once the threat was gone, their bodily response would go back to normal (going through the "stress cycle" as coined by Nagoski & Nagoski, 2020). Stressors in our daily lives can also trigger these physiological responses, but do not necessarily go away in the short term. Therefore, it is hard to indicate to the body that you are safe to move out of the stress response. The Stress Reduction Theory proposes that exposing ourselves to nature helps reduce the psychological (i.e. negative affect) and physiological response (e.g. stress hormones; Ulrich et al., 1991) to stress. The theory proposes that healthy nature (that we are drawn to) tells our bodies that everything is okay and safe to relax. Therefore, we see a physiological reduction in stress hormones by spending time in nature (Yao et al., 2021; Hunter et al., 2019) and thus, go through our stress cycle.

The previously described research on the relationship between nature and well-being is, in general, consistent with all of the theories on why we may see the benefit of nature. Both the Attentional Restoration Theory and the Stress Reduction Theory highlight the evolutionary connection with nature (i.e. the biophilic effect). From these theories, there are three more recent theoretical approaches to the benefits from nature, as highlighted in a book chapter by Joye and

van den Berg (2018). The first is perceptual fluency. For both the Attentional Restoration Theory and the Stress Reduction Theory, natural environments can be easier to process cognitively than the visual information coming from urban environments making them more familiar and less fatiguing to see due to our evolutionary connection to nature. The second approach is nature connection. The authors suggest that "people gain purpose and meaning in life by feeling that they belong in the natural world" (p. 62; also see Passmore & Howell, 2014 review). Again, implying for both the Attentional Restoration Theory and the Stress Reduction Theory, feeling a part of the natural world would make people feel more comfortable and safe in the environment. In addition, research supports the proposition that nature connection (relatedness) is a psychological need (Baxter & Pelletier, 2019). The last approach is micro-experiences with nature. They suggest that over time, people accumulate a large number of micro-experiences with nature, such as viewing natural scenes out of a window, in art, television, or even reading books about nature, that improve a person's well-being and creates a way to mitigate negative feelings from everyday stressors (also see review Passmore & Howell., 2014). Altogether, these theories propose that our connection, dependence, and preference for healthy and safe nature are vital for our well-being (both physical and psychological) because of our evolutionary interconnection with nature. The differences across the theories come from the details of the processes or mechanisms of our interactions of nature (e.g. reduction in stress, feelings of restoration, etc.). However, no matter the theory (or its processes and mechanisms), the benefit of nature are quite robust.

Extinction of Experience

In our urban world, people spend more time indoors than out in nature. For example, Canadians spend approximately 90% of their time inside and report only spending 6% of their

time outdoors (Matz et al., 2015), which does not necessarily indicate they are spending any of that time in natural environments. Additionally, in the Canadians at Work and Home Statistics Canada Social survey, 26% of participants reported they did not participate in any type of outdoor activity (e.g. fishing/hunting, camping, hiking, wildlife viewing) over the past 12 months (Statistics Canada, 2018). When examining the changes in our experiences with nature, research shows that children spend less time in nature over the generations (using both census data and interviews; Gaster, 1991). The generational shift characterizes the Extinction of Experience (Pyles, 1993); it is the generational loss of interaction between humans and nature (Gaston & Soga, 2020; Soga & Gaston, 2016). Even with the term extinction, it does not mean that we have entirely lost our experience with nature. Instead, it is more a discussion about our reduction in our contact and interaction with nature. Research showed that contact with nature in childhood predicted continued contact with nature in adulthood (D'amore, 2016; Pensini et al., 2016; Rosa et al., 2018). If people are being exposed less to nature as children, they are likely to not spend as much time in nature as adults and then not expose their children to nature. This perpetuates a cycle in reducing our contact with nature. Therefore, the more we are missing our experiences with nature, the baseline average of time spent in nature for each generation diminishes. However, some research suggests that there is no difference in the amount of time kids are spending in nature (when comparing two surveys a contemporary one and one from early 1900), but rather it is the differences in the activities they are doing in nature (Novotny et al., 2021).

The reduction of connection with nature can also be seen in indirect ways (compared to only direct contact with nature). We are witnessing the loss of nature connection and importance in cultural products. For instance, research on literature, film, and television has shown a decrease in nature depictions. Between 1937 and 2010, there has been a decrease in nature (even

more so wild, untamed nature) in Disney and Pixar animated feature movies over the years, in addition to a reduction in the biodiversity of the non-anthropomorphized animals present in the films (Prévot-Julliard et al., 2014). Also, after the 1950s, there was a decline in nature words in English fiction novels, song lyrics, and movie storylines internationally (Kesebir & Kesebir, 2017). In 2007, the Oxford Junior Dictionary removed approximately 50 nature terms. Terms such as beaver and dandelion were replaced by words such as Blog and MP3 player (Dhillon, 2008). When Voneeta Gupta, the head of the Oxford Junior Dictionary, was interviewed about the removal of words, she gave other examples of nature words that have previously been shifted. She stated, "When you look back at older versions of dictionaries, there were lots of examples of flowers, for instance. That was because many children lived in semi-rural environments and saw the seasons. Nowadays, the environment has changed" (see the article by Dhillon, 2008 for quote). In general, these examples show the shift of seeing nature in our cultural products that represent our connection (or reduction of it) to nature.

As the generations are losing their experiences in nature, they are also missing out on the many benefits of spending time in nature, such as the health and psychological benefits. Thus, there is clear value to creating awareness about the extinction of experience problem and promoting nature connection. Shanahan and colleagues (2015a) suggest the importance of two interventions to help with the extinction of experience. First, the authors suggest introducing more urban nature to expose to people. Research on five metropolitan cities around the world (Tucson, Arizona, USA; Berlin, Germany; Washington DC, USA; Florence, Italy; and Chiba, Japan) indicates that even if the cities may have a large portion of biodiversity (measured through plants and birds), the majority of the residents live in areas that have a below-average amount of biodiversity (Turner et al., 2004). This is problematic as objective amount of nature,

such as neighbourhood tree canopy, has previously been associated with well-being, such as general health (Kardan et al., 2015b). This suggests that adding more nature to neighbourhoods may benefit residents. To make a city greener, adding greenspaces and trees in neighbourhoods, making nature more accessible, and utilizing biophilic design (characterized as bringing nature exposure through incorporating natural elements in built environments by mimicking or specifically using them; Kellert & Calabrese, 2015) could be implemented. The second intervention suggested by the authors was to find ways to connect people to the nature they already have around them. Richardson and colleagues (2021) propose the importance of showing people how to connect to nature rather than solely giving them the opportunity to be in contact with it. In this study, they had UK participants report their time spent in natural environments (such as visiting local parks, greenspaces, rivers, or forested areas) and also how often they did noticing nature activities (such as looking at natural scenery, taking pictures or videos of nature, listening to bird songs, smelling flowers, or studying natural areas, to name a few). Those who participated in more noticing nature activities were more likely to feel connected to nature than those who just spent a lot of time in nature. Said another way, having people take notice of nature can be more helpful in connecting people with nature (i.e. reducing the effects of the extinction of experience) than solely giving better access for nature contact. Still, it is also important to understand who may already be spending time in the nature available to them and who may be embracing the "magic of mundane" (Dobson et al., 2021) in urban nature (i.e. who is noticing the nature already around them?).

Nature Connectedness

Even with an evolutionary urge to connect to nature, we still see individual differences in the extent one feels connected with nature. Nature connectedness is characterized as the

psychological interaction between an individual and nature, making the individual feel a part or at one with nature (Lengieza & Swim, 2021). Nature connection has been examined as both trait-like and state. State nature connection refers to the experience when an individual feels a sense of connection with nature in the moment, usually stemming from their current exposure to natural surroundings. In contrast, trait-like nature connection is a stable feeling and understanding of connection to nature. For this dissertation, I focus on the trait-like nature connection. Previously, many different measures and scales have been created to assess the trait, such as Emotional Affinity Towards Nature (EATN; Kals et al., 1999), Connectedness to Nature (CNT; Mayer & Frantz, 2004), Commitment to Nature (COM; Davis et al., 2009), Environmental Identity (EID; Clayton, 2003), Connectivity with nature (CWN; Dutcher et al., 2007), Inclusion of Nature in Self (INS; Schultz, 2002), the Extended Inclusion of Nature in Self (EINS; Martin & Czellar, 2016), and Nature Relatedness (NR; Nisbet et al., 2009). Previously, Tam (2013) highlighted the similarities across the different nature connection measures (all but the EINS). First, they are all highly correlated (*r* values all above .40). Second, in factor analyses, they all have an underlying higher-order construct of connection to nature. This indicates they are examining the similar factor of nature connection. Third, the pattern of relationships with discriminant validity measures, such as personality, well-being, and environmentalism, was similar, but of modestly varying degrees. Altogether, this suggests that none of the different measures of nature connection are clearly best to use for this dissertation. I have, however, chosen to explore the relationships with the Nature Relatedness scale.

Nature relatedness is defined as feelings of interconnection between the self and other living beings (Nisbet et al., 2009). Nisbet and colleagues (2009) created the construct to incorporate physical, cognitive, and affective connections with nature. For example, it contains

pro-environmental feelings/views, love for nature, seeing the importance of unappealing nature, and self-identity as an environmental/nature person. The measure includes three distinct facets (derived from exploratory factor analyses): 1) the Self, 2) Experience, and 3) Perspective. The self scale encompasses identification of the interconnection with nature, such as affective feelings towards nature and reflective thought about how actions may affect nature. The experience facet consists of physical contact with nature, such as the degree of comfort and desire to be in nature. The perspective facet embodies worldviews towards nature, such as humans' impacts on the planet.

Some research has explored how trait-like nature relatedness can predict different behaviours and attitudes. For example, nature relatedness has predicted love for animals (Nisbet et al., 2009), environmentalism (Mackay & Schmitt, 2019), physical activity, and less screen-time (Molina-Cando et al., 2021). Previous research also suggests the relationship between nature connection (more general and also specifically with nature relatedness) and subjective well-being. The trait level of nature connection was related to both hedonic well-being (positive affect and life satisfaction) and eudaimonic well-being (Ryff's psychological well-being [1989] and vitality; Capaldi et al., 2014; Pritchard et al., 2020) in meta-analyses. More specifically, nature relatedness had a small mean effect size for happiness (r = .18; aggregating well-being markers; Capaldi et al., 2014), hedonic well-being (r = .19) and eudaimonic well-being (r = .22; Pritchard et al., 2020) across multiple studies examining these relationships. Therefore, not only are those who spend time in nature happy, but those who feel more connected to nature report that they are happier people overall.

It could be that even though the general population is interacting less with nature (extinction of experience), people who identify as nature connected may be finding ways to

interact with nature, which makes them happier people. This would fit with general understandings of the self and the self-concept. The self is the conscious part of you that thinks, decides and acts in a certain way (Myers et al., 2015). The self is made up of the self-esteem (sense of worth), the social self (role with other people), the self-knowledge (how a person describes themselves), and the self-concept (description of whom a person thinks they are). A person who feels or identifies as nature connected would find ways to interact or connect with nature to fit their self-concept. For example, in terms of spending time in nature or connecting with it, it is the self that decides to go out and notice nature. Nature can also help form our identity and self-concept (Passmore & Howell, 2014). For example, nature can do this by allowing us to compare ourselves to other animals (Myers, 2003), find attachment with our local environments (Taylor, 2007), to reflect about ourselves (Kaplan & Kaplan, 1989), and even help heighten our self-awareness (Mayer et al., 2009). Therefore, nature can help us form our self-concept, but if we identify as a nature person, we may be more likely to do activities to connect or interact with nature to fit our self-concept. In terms of research, understanding people's self-concept can also help us distinguish between people (i.e. individual difference characteristics, such as nature relatedness, that create an individual's self-concept).

This differs from understanding nature relatedness as a merely descriptive individual difference characteristic; seeing it as part of the self-concept implies psychological processes of nature activities. By applying this to happiness, nature related people may be happier, in general, because they are doing nature activities that fit with their self-concept, such as seeking out and noticing nature more. These two ways of interacting or experiencing nature may not be exclusive, and may work alongside each other.

Are Nature Related Individuals More Likely to go Out into Nature?

People living in more urbanized environments may be less directly exposed to nature, or the nature they are exposed to may be different. Objectively, even if cities have a large degree of biodiversity in the city limits, people generally live in neighbourhoods with the least biodiversity (Turner et al., 2004). For example, city limits do not generally include forested areas, which may be viewed as quality nature, but may have access to marinas and greenspaces as natural areas. This hints that when people visit more natural areas, it is usually internally motivated by choice or desire (Ajzen & Driver, 1991). This may suggest that individual difference characteristics, such as trait-like nature relatedness, predict individuals going out into nature.

When an individual responds to nature connection questionnaires, some items suggest that nature connected individuals are likely to go out into nature. For example, there are items in the Nature Relatedness scale (Nisbet et al., 2009) such as: “I enjoy being outdoors, even in unpleasant weather”, “My ideal vacation spot would be a remote wilderness area”, “I enjoy digging in the earth and getting dirt on my hands”, “I don’t often go out into nature” (reverse-coded), and “The thought of being deep in the woods, away from civilization, is frightening” (reverse-coded; Nisbet et al., 2009); people who endorse these items from the experience subscale of the Nature Relatedness scale express that they are spending a lot of time in nature. However, these self-report items are not directly assessing, objectively, if individuals high in nature relatedness are indeed more likely to go out into nature. As a broad self-report measure, it is important to corroborate face valid interpretations of the items against other indicators.

In general, research suggests that nature connected individuals are more likely to spend time in nature in different places around the world. Lin et al. (2014) examined the difference in an individual’s nature orientation (i.e. nature relatedness) versus their opportunity (i.e. a person’s

access to the park) on self-report park use in Brisbane, Australia. Nature relatedness was a more important predictor for time spent in the parks and private nature spaces, such as gardens, than was easy access (i.e. opportunity) to these spaces. Further, nature related Australians were more willing to travel farther distances to find parks with a higher percentage of trees and native plant coverage (Shanahan et al., 2015b). In the UK, nature related individuals were more likely to frequent the local greenspaces (Flowers et al., 2016). In Tokyo, Japan, predicting the probability of greenspace use was highlighted by both factors of the landscape, but most importantly, with individuals' level of nature relatedness; furthermore, those with higher levels of nature relatedness were more likely to also report longer duration in urban greenspaces (Soga & Akasaka, 2019). French individuals who felt more connected to nature reported spending more time in natural places (Colléony et al., 2017). When asked which natural areas they were most likely to spend time in, nature connected French people were more likely to report both forests and urban parks than those lower in nature connection. My collaborators and I have also seen that during the COVID-19 pandemic, Canadian nature related students reported spending more time in nature (Desrochers et al., 2022). More specifically, nature relatedness was associated with more time spent in greenspaces, blue spaces, and countryside areas during a usual week of the pandemic. Additionally, when the participants were asked if they felt they were spending more or less time in nature during the pandemic than prior, nature related participants self-reported spending more time in all three natural environments, whereas the overall sample (i.e. no matter their level of nature relatedness) reported spending about the same amount of time in nature during the pandemic than prior.

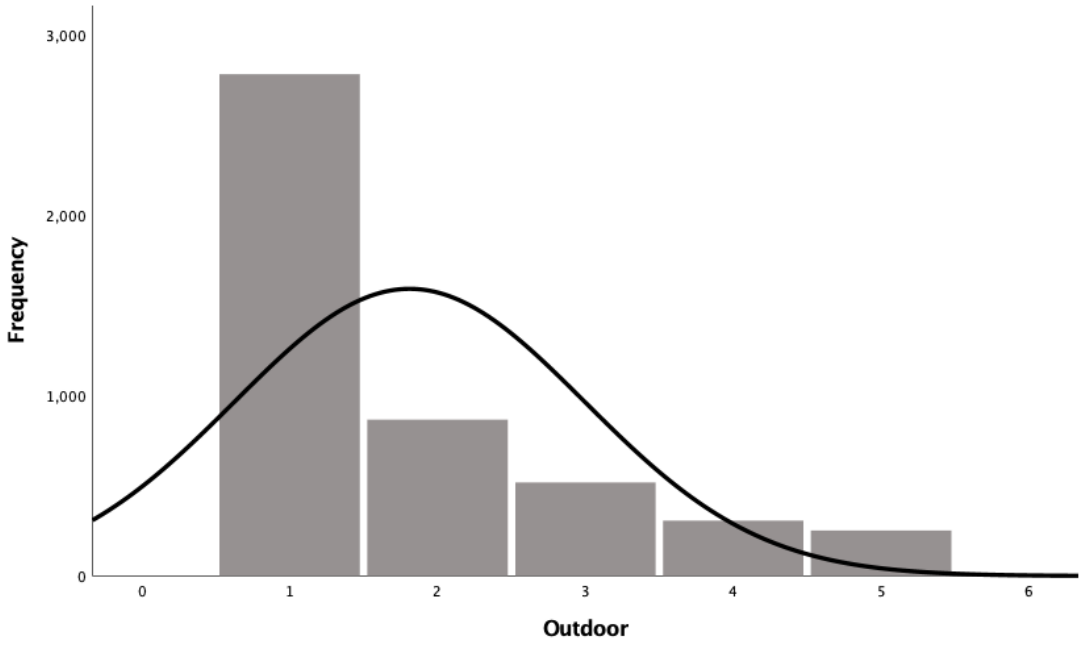
Moreover, nature related people spending more time in nature was also seen in more momentary assessments, such as daily diary or experience sampling data. These measurement

techniques focus on concrete instances of behaviour, avoiding some biases stemming from general beliefs about the self. For example, in an experience sampling study, participants filled out a survey twice a week for eight weeks about the amount of time they spent outdoors and in nature over the previous three days. Here, trait nature relatedness was moderately associated with both time spent outdoors (r = .25) and in nature (r = .30; Nisbet et al., 2009). Similarly, in an unpublished daily diary study, where participants filled out a questionnaire every night about their day for 14 days, nature related people reported spending more time outdoors (Lutz, 2022). In an unpublished experience sampling study, however, my collaborators and I have found that trait nature relatedness did not significantly predict time spent outdoors in experience sampling data measured three times a day for seven days in multilevel modelling analyses (Desrochers et al., 2021). This contradicts previous findings, although this data is highly positively skewed with a low average of how much time the student sample reported spending time outdoors (See Figure 1 for the distribution of responses). This skewed data may be missing out on the variability in the data needed to examine this relationship. The skewness of the data may be due to multiple different reasons. First, this data was collected during the second wave of the COVID-19 pandemic in Ontario (fall 2020), where students were completing their classes virtually, and restrictions were in place, minimizing what they could do. However, the participants here were recruited during the same time as the participants from our lab's pandemic study (Desrochers et al., 2022), which showed a positive association between nature relatedness and time spent in nature. Second, we asked participants to report their time spent outdoors, not specifically in nature. This wording could affect the way the participants responded to this question. Third, it may be harder to gather momentary assessments of time spent outdoors versus daily reports (c.f.

to Lutz, 2022). Even with this dataset's limitations, it suggests we may need better data on objective time spent in nature and nature relatedness.

Figure 1

Distribution of responses of how much the participant spent in nature during the last 2 hours in Desrochers et al. (2021). The scale is anchored from 1 (not at all) to 5 (very much).



The majority of the research proposes that nature-related individuals report spending more time in nature. Still, this line of research relies on self-report items of the time spent in nature. The little bit of research on more objective ratings of time spent in nature through experience sample measurements shows mixed results.

Are Nature Related Individuals more likely to Notice Nature?

Similar to being in nature, items in the Nature Relatedness scale (Nisbet et al., 2009) suggest nature-related individuals may be *noticing* more nature. These items are: “I take notice of wildlife wherever I am” (from the experience subscale) and “Even in the middle of the city, I notice nature around me” (from the Self subscale). People who report higher scores on these two

items would suggest that they notice more nature; however, little research has directly examined this to date. Still, previous research may allude to this in indirect ways.

For one, previous research suggests that noticing nature is a way to connect people to nature, such as in the Five Pathways to connecting to nature (Lumber et al., 2017). The authors explored actions or activities a person can partake in to connect with nature. To figure out these activities, they examined which activities were done by people with high trait nature relatedness. They found in two studies that trait nature relatedness was associated with doing more activities that engage with nature through the senses (contact), perceiving the aesthetic beauty of nature (beauty), using nature to express ideas or metaphors (meaning), feeling an emotional bond with nature (emotion), and feeling compassionate and ethical concern towards nature (compassion). These five categories of activities were titled the Five Pathways to nature connection. The contact pathway includes activities such as seeing flowers, listening to birds, viewing beautiful sights, to name a few. These would be in line with noticing nature. However, it is not clear if these activities are happening while people go out into more remote natural areas or if they are happening in their everyday life. In the third study, the authors examined the Five Pathways as an intervention to see if the activities would indeed increase nature connection, as suggested from the previous studies. Participants were randomly assigned to one of three walking conditions: 1) walking in nature with pathway activities; 2) walking in nature without pathway activities; and 3) walking in an urban environment with pathway activities. The authors supported their hypothesis that nature relatedness was increased post activity from baseline scores, but this effect was only found for those in the nature walk with pathway activities condition (i.e. it was not significant in the built activity or the nature control conditions).

Further, other studies have also explored noticing nature explicitly as an intervention to increase state nature connection (noticing nature to nature relatedness causal direction). For example, having participants participate in a three good things in nature intervention (where they were to report three good things they noticed in their immediate environment) increased feelings of connection towards nature (Richardson & Scheffield., 2017). In this intervention, people generally described the good things as noticing nature through the senses, such as hearing birds chirp and seeing nature sights, similar to the contact through the senses pathway (Lumber et al., 2017). Experimentally, McEwan et al. (2019) showed that creating a noticing good things intervention in urban greenspaces was related to increased feelings of nature connectedness (operationalized through the Inclusion of Nature in Self scale and the Nature Relatedness scale). For seven days, every time a participant was in the space they were randomly assigned to (either urban greenspace or urban built condition), they were prompted on a phone application to report one good thing they noticed in their immediate surrounding. Those in the greenspace condition were only prompted to report the one good thing when they were in a greenspace (and vice versa for urban built environment condition). After seven days of the intervention, the participants in both conditions saw an increase in nature connection scores from baseline values of nature relatedness and nature-self scores. This suggests that noticing good things increases feelings of nature connection no matter the environment. The authors suggest that the unexpected increase in nature connection for those in the built environment condition may come from people noticing nature even in those environments, or that being more mindful or aware of their surroundings may make them feel more connected to it. Additionally, Richardson et al. (2021) showed that noticing nature (measured through the frequency of doing nature noticing activities) was more important for predicting feelings of nature connectedness than solely spending time in nature.

Together, these studies show the association between noticing nature and nature relatedness. However, the direction of the association is mixed. In the experimental studies, the causal direction of the association shows that taking notice of nature through activities (e.g. the Five Pathways and the good things) increases nature relatedness. It is unclear if nature related individuals are already noticing more nature than others (the reverse causal direction). The first two cross-sectional studies by Lumber et al. (2017) do suggest that trait nature related people report doing more noticing nature activities (which they then used to create the Pathways activities in the experimental study). Additionally, research showed that nature related individuals reported that they noticed more wildlife during the pandemic (Desrochers et al., 2022). These studies propose the reverse direction of the association between nature relatedness and noticing nature; however, it may be biased by self-report accounts. No studies have explored behavioural measures of noticing nature in this association.

Noticing Nature Related to Well-Being

One other purpose in exploring if nature related individuals are more likely to notice nature is to begin unpacking the process of noticing nature in everyday life as a way to feel happier. Research does suggest that those who notice nature see benefits of well-being because of it. The strongest claim to this is when Passmore & Holder (2016) asked participants to notice objects or scenes in daily life that evoked strong emotions over the course of two weeks. The participants were randomly assigned to one of three conditions: 1) a natural environment condition; 2) a human-built environment condition; or 3) a control condition. Those in the nature or human-built conditions were both told to notice the objects or scenes that evoked strong emotions, but they differed in telling them to notice in either natural or human-built environments. The control group was told to go about their regular life and were not instructed to

notice anything. The authors found that when participants were asked to notice nature that evoked strong emotions, they reported more positive affect than those in the human-built and the control conditions. After the two-week noticing nature intervention, all participants were asked how much time they spent in nature during the intervention, regardless of condition. Interestingly, there was no difference in the amount of time spent in nature during the two weeks between the conditions. This suggests that the individuals in the nature condition did not merely go out into nature more to be able to report noticing natural objects or scenes that evoked strong emotions. In other words, they noticed nature in their everyday lives, which increased their positive affect from pre-intervention levels. In qualitative data, those in the nature condition described more positive emotions evoked, whereas those in the human-built environment described more negative feelings. The natural object or environment itself generally evoked the descriptions of the strong emotions for those in the natural condition (e.g. "The two rainbows induced feelings of happiness and creativity. Seeing all the colours made me joyous and calm and peaceful." p. 542). In contrast, the strong emotions brought up in the human-built environment were more likely to bring up memories, be in light of activities, or discuss the functions of the object or scenes (e.g. "Jealous, envy, frustration, most emotions due to the fact I cannot afford a vehicle of that sort and see mostly younger individuals driving them." p. 542).

More recently, two replication studies have examined the noticing nature interventions. In China, individuals who participated in the noticing nature intervention had more positive affect (replicating Passmore & Holder, 2016), life satisfaction, meaning in life, and transcendent connectedness, but also showed lower levels of stress, negative affect, and anxiety than those in either control condition (noticing urban condition and unrelated school journaling condition; Passmore et al., 2022a). These associations were not moderated by the participant's level of

nature connectedness nor their time spent in nature. Passmore et al (2022b) also showed that the noticing nature intervention also worked in Canadian winter, where those noticing nature had higher levels of positive affect (replicating Passmore & Holder, 2016), life satisfaction, elevation, hope of agency, and state nature connection than those noticing built environments in urban spaces. Interestingly, in a post-intervention qualitative question asking participants if and what they had learned from the study, over-half of those in the urban condition mentioned nature. They suggested learning a greater affinity for natural settings, that paying attention to built objects made their appreciation for nature stronger, and more concern towards nature. Altogether, these nature intervention studies highlight the benefit of showing people that taking notice of more nature may be more effective than just being in nature to connect with it and seeing emotional benefits from it.

Another study alludes to the relationship between noticing nature and well-being. For instance, when individuals reported one good thing they noticed in either greenspaces or urban environments, they reported more well-being, measured through better quality of life (measured with the Recovering Quality of Life scale; Keetharuth et al. 2018), positive affect, feelings of safety, and feelings of relaxation than baseline levels (McEwan et al., 2019). There was, however, no main effect of condition nor an interaction effect. This suggests that noticing (through the one good thing activity) was beneficial for the well-being markers regardless of the environment (urban or natural). This is similar to their results (McEwan et al., 2019) about noticing in either condition (urban or natural) on nature connection (see section Are Nature Related Individuals more likely to Notice Nature?). However, in this case, previous research on positive psychology interventions, such as the three good things, has been shown to improve feelings of gratitude and well-being. In this positive psychology research, the three good things

activity is found to boost mood, no matter the location of the activity (Seligman et al., 2005). Therefore, the lack of condition main effect or interaction effect for well-being is less surprising than the results on nature connection.

Nisbet et al. (2019) examined the effects of nature exposure on moods and whether mindfulness instructions (i.e. guidance) during the nature exposure would increase the effect. Participants were randomly assigned to an indoor walk, a nature walk, or a nature walk with mindfulness techniques. The nature walk and the nature walk with mindfulness techniques were taken on the same route (therefore, the same nature exposure). All three walks were guided by a research assistant. In the mindfulness condition, participants were instructed first to center their attention on the action of walking. Part of the way through the walk, the participants were then guided to focus their attention on a feature of their immediate surroundings, such as the sounds, the smells, or the sights. In other words, they were instructed to actively notice the surroundings (i.e. nature) through the senses. It was found that even though those in the nature walk reported fewer negative moods than those in the indoor walking condition, those in the mindfulness/noticing condition reported the least negative moods. In other words, the active engagement or noticing of the natural environment through the senses was more beneficial for reducing negative moods than solely the exposure to nature. However, the effect on positive mood was less clear. Those who walked outdoors (both with and without mindfulness) had better moods than those walking indoors, but the mindfulness technique did not significantly boost the effect more than the location. The mindfulness techniques did increase feelings of nature connection (measured through INS), similar to the other studies above.

Overall, there is a suggestion for the potential benefits of noticing nature. Even with the benefits shown for noticing nature on nature connection and well-being, it is not clear if trait nature relatedness would, in fact, predict noticing nature.

What is Noticing Nature?

Noticing is a colloquial term that could mean many different things. For example, the three definitions that come up from Oxford Languages Dictionary (n.d.) are: 1) "become aware of"; 2) "treat (someone) with some degree of attention or recognition"; and 3) "remark upon". These definitions conclude that the term is about paying attention, perceiving, or explicitly acknowledging something or someone. Previously when discussing noticing nature, researchers have described this as paying attention to the nature around you and turning your senses to conscious thought. For example, Richardson et al. (2021) describe that when walking in an environment with nature, a person can be exposed to sensory information from natural stimuli, such as birds singing, smells of flowers, and sights of trees (to name a few). This is in accordance with Lumber et al.'s (2017) contact through the senses Pathways of nature connection.

Previously, Franco et al. (2017) reviewed the literature on the benefits of nature exposure through the senses with either explicit awareness of it or not. Many studies have shown that the benefits of nature can come without any awareness of the nature. As an illustration, an Ottawa study showed that individuals' self-reported neighbourhood satisfaction in general and satisfaction with the neighbourhood's nature were associated with the objective diversity of bird species of the neighbourhood (Hepburn et al., 2021). Interestingly, however, the residents who participated in the study did not seem aware of the bird diversity as a reason they felt satisfied with their area of residence (this was reported through open-ended questions). This suggests that

merely having the biodiversity of nature (such as the birds) may be beneficial even if people cannot articulate the reasons. Other studies (some previously identified early) show that the objective measure of greenspace, greenness, or even access to nature is associated with well-being. In these studies, participants may still be aware of the nature exposure, but the assessment of nature is independent of their self-report. For example, in Glasgow et al.'s (2019) study, participants who travelled through more green or blue spaces had more positive moods. Green and blue spaces were assessed by GPS locations and motion-sensing data through the Daynamica phone application. Therefore, it is not clear if participants were aware or not of the environments that made them have more positive moods.

However, other studies rely on a person's self-report nature exposure (e.g. time spent in nature) and their well-being. These studies (many previously mentioned) depend on a person's awareness of nature. For example, White and colleagues (2019) had participants report how much time they spent outdoors, which they defined as natural areas outside of the city. To recall the time spent in these environments, the participants would have needed to notice the environments when they were being exposed to them. Due to this recalled awareness, there was an association between time spent in nature and life satisfaction. There is also research that examines more in-depth awareness (i.e. noticing of nature) and well-being. For instance, being mindful in nature spaces (Nisbet et al., 2019) and being aware of natural objects or scenes that cause strong emotions (Passmore & Holder, 2016; Passmore et al., 2022ab) are associated with well-being. Some researchers, such as Lumber et al. (2017) and Richardson et al. (2021), suggest that paying attention and using the senses is a way nature connected people may be immersing themselves in nature.

I believe it is important to examine more objective measures that align with the Nature Relatedness scale than solely relying on the self-report nature of the scale. As highlighted, the Nature Relatedness scale has items linked to spending time in nature and noticing nature. Research has mostly supported the idea that nature related people spend more time in nature, though there are some method limitations. Even less research has examined if nature related individuals are indeed noticing more nature. This dissertation extends previous research by exploring subjective experience of noticing nature (i.e. consciously noticing nature). More specifically, I examined if nature related individuals do, in fact, consciously notice more nature, even when the objective circumstances are held constant across people. To do this, I did not specialize in one specific process that could define noticing nature, but instead, took a wider approach by examining two processes that fit the colloquial term. Behavioural measures were created to tap into the two processes of perception and attention. The goal was to create behavioural measures to explore noticing nature that is feasible for other researchers to use. The research was split into two phases: Phase 1 examined perception, and Phase 2 examined attention.

Perception

Phase 1 focused on examining the process of perception. In cognitive psychology, perception is defined as the conscious experience that comes from sensory information (Goldstein, 2011). For the purpose of the dissertation, I took a step further from this definition to also include some knowledge and judgement. Thus, when discussing perception, I mean a person's interpretation of nature. If a person does not define an object or scene as nature, they will not recognize that they *notice* nature in an environment. Not only is the definition of nature different across studies in the literature, but the perception of what counts as nature can also

differ by individual people. Hartig and Kahn (2016) highlighted that "the term 'nature' means different things in different contexts. Recognizing personal and cultural aspects of the experience of nature, psychological research considers how different people encounter nature in diverse contexts, from viewing indoor plants in urban offices to walking in wilderness areas" (p. 936). Therefore, it is important to consider an individual's interpretation of what nature is when exploring if they are noticing everyday nature. This was taken into consideration by exploring the individual difference in what is perceived as nature in urban environments.

Previously, researchers have also discussed the potential of bottom-up versus top-down processing a person may use to characterize nature. Bottom-up processing of nature is defined by building the perception of nature based on sensory information from the external environment. For example, Berman et al. (2014) explored low-level visual features of images of urban parks and natural scenery that were inferred to describe them as natural. Participants rated on a seven-point scale how unnatural/human-made (value of 1) to natural (value of 7) the scene in the image was, and the authors estimated low-level features of the images. The results suggest that images with fewer straight edges, more non-straight edges, and less colour saturation were reported as more natural (Berman et al., 2014). In this line of research, the low-level features are discovered by examining which features are associated with the naturalness ratings in the images. However, it may also suggest that these features are what a person uses in the image to create their rating of naturalness. Additionally, not only did the reported naturalness of the image predict a person's preference for the image, so did the similar low-level features that Berman et al. (2014) found (Kardan et al., 2015a). However, other research suggests that the semantics of nature is important for preference ratings. Kotabe et al. (2017) explored the nature-disorder paradox regarding ratings of aesthetic preferences for nature images (the same images used in Berman et al., 2014

and Kardan et al., 2015a). More disorder in environments seems to reduce a person's preference for it, but natural environments, that are generally preferred, are also identified as disorderly. The researchers wanted to examine this paradox. They found that being able to fully picture the images in their original form (i.e. having semantics) instead of solely relying on low-level visual features was essential for the relationship between the level of naturalness with aesthetic preferences. In addition, the combination of low-level features (previously highlighted as inferring naturalness; Berman et al., 2014) and the semantics of the scene were essential to naturalness's influence on aesthetic preference over and above a natural image's self-report rating of disorder. This study exemplifies that bottom-up and top-down processing may both be used when characterizing nature and be important to nature's benefits.

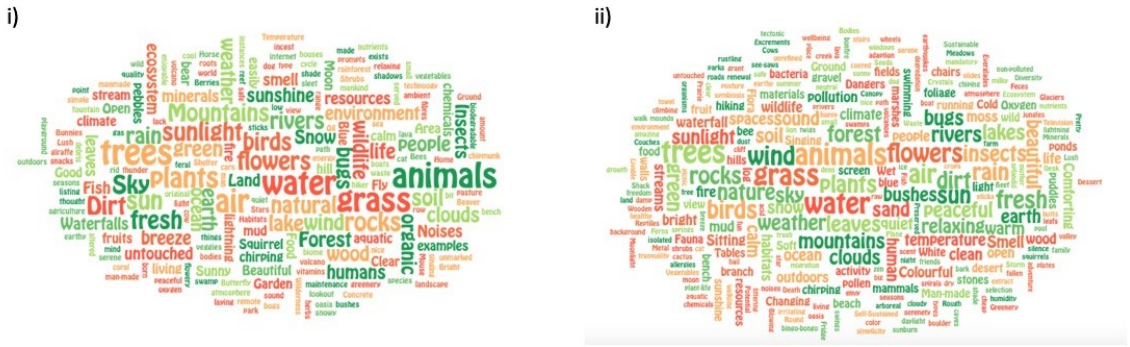
Top-down processing is defined as the perception created by previous experience. Some research suggests that the benefits of nature come from using top-down processing to characterize nature. For example, when participants listened to pink noise (sound that is consistent in noise frequency for each octave interval), they were more likely to report the sound as pleasant, relaxing, and restorative if they were told the sound came from nature, compared to an industrial environment (Haga et al., 2016). Said another way, the benefit of nature was dependent on people's understanding of nature. Moreover, people had an aesthetic preference for nature sounds over urban sounds when easily distinguishable (Van Hedger et al., 2019). Hartig and Kahn's (2016) suggestion of the ever-changing characterization of nature depending on context suggests top-down processing; further, Hartig et al. (2014) put forward that previous nature exposure or experience help define the perception and evaluation of what is nature. All this suggests that an individual's previous experience or subjective perceptions can help define

nature (i.e. top-down processing). In other words, there is potential for individual difference characteristics, such as nature relatedness, that could be used to help define nature.

Vining et al. (2008) also showed that individuals define nature differently. Still, no published research has examined how the definition of what nature is depends on a person's level of nature connection. In an unpublished study by Dan Shaw (2018) from the Happiness Lab, there was a difference in magnitude of what people described as nature depending on their levels of nature relatedness. He asked undergraduate students to report as many features as possible of a natural environment in three minutes. I split the data by the median score on nature relatedness (into high and low nature relatedness). See Figure 2 for Word clouds of these responses. From this research example, the number of words in the high nature relatedness group was more extensive (859) than for those lower in nature relatedness (503). This may suggest that nature related individuals highlight a larger range of features that fit the definition of nature.

Figure 2

Word cloud of words used to describe nature separated by levels of Nature Relatedness (i: low NR; ii: high NR), using a median split in Shaw (2018).



Even with a variation in what exactly is defined as nature or natural, there are key things that have a large consensus about what is considered nature. For example, in Figure 2, animals, flowers, and trees have a high frequency of being highlighted as nature in both groups. Other research shows that people define nature as undisturbed by humans and separate from civilization (Fehnker et al., 2021; Haluza-DeLay, 2001; Pointon, 2014; Vining et al., 2008). In

this phase, I explored if nature related individuals perceive a broader range of features as nature in an urban setting.

Attention

Phase 2 explored the process of attention. As a social psychologist with limited expertise in cognitive psychology, I wanted to explore a more colloquial definition of the term attention than what cognitive psychologists have defined as the cognitive process of attention. As defined by the Oxford Dictionary (n.d.), the colloquial definition of attention is taking notice of, being aware, or consciously observing something or someone. These definitional terms of attention relate to what cognitive researchers consider explicit processing of attention, where there is a conscious processing of the information.

For the purpose of the dissertation, I focused on how natural objects in a scene of primarily urban/non-natural scenes draw and sustain participants' preferred attention. Specifically, I explored where the eyes are being drawn to and what a person wants to focus on in an urban setting. I labelled these two "processes" of attention as first attention (where the eyes were being drawn) and voluntary attention (where the participant wanted to focus their attention). My definition of voluntary attention is quite different from the one used by cognitive psychologists. Voluntary (selective or endogenous) attention is defined in cognitive psychology as conscious attention driven by the individual towards the stimuli (Goldstein, 2011). In other words, the attention is a top-down processing (the mind intentionally focuses on the stimuli).

Much of the research on attention and natural environments fits within the Attention Restoration Theory. In this theory, the features of nature make it softly fascinating, enabling involuntary attention (defined as attention that requires little to no effort to focus on; Kaplan & Berman, 2010). This soft fascination allows people to not focus solely on nature but also on other

thoughts (i.e., allowing the mind to wander), making it restorative from the hard fascinations in one's life. In this research, nature reduces feelings of fatigue and helps regain attentional capacity. For example, many research studies examine the benefits of nature on attention control in laboratory attention tasks (Basu et al., 2019; Gamble et al., 2014; Berman et al., 2008; Stenfors et al., 2019). This line of research is in light of cognitive psychology's definition of attention. However, little to no research has examined the attention as the focus being paid towards nature.

The limited research on paying attention to nature has utilized eye-tracking techniques. Some eye-tracking research suggests that individual difference characteristics can differentiate between fixation points for natural environments. For example, students majoring in forestry were more likely to fixate on natural images, like forests and wetlands; in contrast, art students were more likely to fixate on images of humanistic attributes, such as urban and recreational areas (Wang et al., 2021). This suggests that an individual who may be more familiar or comfortable with the surroundings may pay more attention to it. Said another way, this could indicate that a person may be looking at objects or scenes that fit within their self-concept.

A single study has examined the difference in voluntary attention being paid to nature for people with varying degrees of nature relatedness through eye-tracking techniques. Bingjing et al. (2022) recorded the eye movement and fixation of participants wearing virtual reality headsets, with half of the scene set in an urban setting and half in a natural environment. The 360° virtual reality consisted of 180° of buildings (built/urban environment) and 180° of trees (natural environment). Individuals with high levels of nature relatedness (top 31% of the sample) spent more time looking at the natural environment, whereas those low in nature relatedness (bottom 33% of the sample) spent more time looking at the urban setting. There was no

difference in the time spent looking at natural or urban settings for those with a medium level of nature relatedness (middle 36% of the sample). There was also a significant association between high nature relatedness and time spent looking at trees. This is the first study to directly examine how nature related individuals were objectively more likely to focus their attention on nature. However, this study does not consider the subjective awareness of their noticing. By this, I mean that the participants may subconsciously spend more time looking at nature, but they may not be aware they are doing this. Additionally, this study had an equal amount of natural and urban stimuli in the study. This may suggest that nature related individuals are purposely exposing themselves to the natural setting more than the urban one, but it does not consider noticing nature elements within the urban part of the environment.

The Current Studies

The current dissertation explores whether nature-related individuals notice more nature. To do this, I examined how an individuals' level of trait nature relatedness relates to two processes I deemed essential for exploring behavourial noticing of nature in one's life. To accomplish this, I ran three studies in two separate phases. Phase 1 focused on the process of perception, defined as the interpretation of what is nature. Phase 2 focused on the process of attention, described as focus being drawn to or being held by nature. In all three studies, to control for the level of naturalness, images of urban scenes with a degree of nature were used. This way, all participants were exposed to the same amount of nature, but I predicted that some would perceive or pay attention to the nature in the scenes more than others. Across both Phases, I hypothesized that highly nature related people would be likely to *notice* more nature.

In Phase 1, participants were exposed to many images of urban scenes with a degree of nature. To explore the perception of nature, the participants reported if they notice the nature in

the image by rating how much nature they see in the urban space and by rating the percentage of nature in the images. This phase explored how nature relatedness may be associated with nature perception (if they notice the nature or not). I hypothesized (*H1*) a positive association with nature relatedness and each nature perception item, where those higher in nature relatedness would be more likely to report that they perceived more nature in the urban environment images.

In Phase 2, to explore attention to nature, the participants were also exposed to many images with an urban scene containing some nature. The participants were asked to report on two attention questions for each image. First, participants were asked to indicate where their attention was drawn to the image. Second, participants were requested to indicate where they would like to focus their attention on in the image. I hypothesized that those higher in nature relatedness would be more likely to pay attention to nature. More specifically, when examining an urban scene, I hypothesized (H2) a positive association between nature relatedness and first attention, where those with higher levels of nature relatedness would be more likely to have their attention drawn toward the little bit of nature in the image; I also hypothesized (H3) a positive association between nature relatedness and voluntary attention, where those with higher levels of nature relatedness would be more likely to want to focus their attention on the nature in the image. This fits with research on eye-tracking techniques. However, eye-tracking techniques consider unconscious attention on the natural environment, such as in Bingjing et al.'s (2022) study. In this dissertation, I focused on conscious attention (i.e., the participants being aware that they are paying attention to the nature by asking the participants to report on where they focused their attention first and where they want to focus it).

Altogether, this dissertation investigated if nature relatedness was associated with the subjective noticing of nature through two phases with three hypotheses. To accomplish this, I

developed new tasks to measure behavioural indicators of noticing nature. Previous research relied on self-report measures of time spent in nature or activities related to noticing nature, which lacked objectivity. By assessing noticing through these novel tasks, I extended this line of research by examining if the self-report responses align with the behaviour of noticing nature. Thus, with the creation of new behavioural tasks of noticing nature, another goal of the dissertation was the tool development. It was crucial to properly evaluate the new tasks, as any negative results could be attributed to poor task design rather than a lack of effect. Further, with the proper tool development, other researchers may use the task to explore similar relationships. An Open Science Framework (OSF) database was created to provide readers with pre-registrations, materials, syntaxes, and other relevant information for the current studies: https://osf.io/hcdp2/?view_only=b2ef552766df41f9aa7b221827cac9bf .

Phase 1

Study 1

Study 1 explores if individuals high in nature relatedness perceive nature in images portraying urban scenes with some nature. To do this, I designed two tasks of nature perception. Before examining the hypothesized relationship between nature relatedness and nature perception, it is important to assess the reliability of the experimental tools (i.e. the two nature perception item-types). If there is not a proper distribution for each image, it would be hard to aggregate all images reliably, and it would also be hard to explore further relationships (see review by Clark, 1995). That is, if there is not enough between-people variability in the image ratings, it will be hard to examine if nature relatedness predicts the nature perceptions. Said another way, if everyone either *notices* or does not *notice* the nature in the images (regardless of their nature relatedness levels), then it will not allow me to investigate the correlation between

nature relatedness and the nature perceptions of the images. Therefore, I employed optional stopping techniques to stop my recruitment at 100 participants to allow me to test the variability of the nature perception ratings of the images. After properly assessing the variability of my tools, I continued recruitment until I reach 300 participants before running more psychometric analyses to optimize my experimental tools. Lastly, after the best images and item(s) were chosen, the correlations between nature relatedness and nature perceptions were assessed.

Methods

Participants. Due to the study's exploratory features, it is difficult to properly choose a likely effect size or the smallest effect size of interest to run an *a priori* power analysis. However, it is still essential to justify the required sample size. In addition, it is important to have good power to allow for exploration (Lakens, 2022). A previous simulation study has indicated that correlations generally stabilize with a sample of at least 250 individuals for a small effect size of .1 with 80% confidence (Schönbrodt & Perugini, 2013). Therefore, a sample size of 300 will be run to allow for the deletion of poor data. A sensitivity analysis was run in *G*Power* to explore the effect size that can be identified with this sample size. With alpha set at .05, 90% power, and a sample size of 250, an effect size of as small as .12, regardless of direction, can be identified.

Three hundred participants were recruited. Ten participants did not complete the study (36% or less completion rate) and thus, were removed from the sample. After identifying the problematic participants through outliers and string responding, the final sample was 282 ($M_{age} = 20.44, SD = 4.41, \min = 17, \max = 49$). Out of the eight participants who were removed: one was removed for being identified as an outlier for nature relatedness (± 3 standard deviations away from the mean), four for having no variability in the image ratings for item 1

(i.e. string responding), and six for having no variance in their nature relatedness responses (i.e. string responding).

For the final sample of 282, 197 identified their gender as women (70%), 67 identified it as man (24%), 13 identified as gender fluid/non-binary/two-spirited (5%), one identified as a trans-man (<1%), one as a non-binary trans-femme (<1%), one preferred not to say (<1%), and two did not respond to the question (<1%). The majority of the sample identified as White (61%), with a smaller percentage of South Asian (11%), Chinese (9%), Black (8%), Arab (5%), Indigenous (3%), West Asian (2%), Southeast Asian (1%), Filipino (1%), Latin American (1%), Korean (<1%), Japanese (<1%), other (3%), and 3% preferred not to identify their ethnicity. The majority of participants grew up and currently lived in the suburbs (see the frequency of each area in Table 1). The majority of participants filled out the questionnaire on laptops/desktops (90%), with a smaller percentage on phones (5%) and tablets (4%).

Table 1

Frequency of where Study 1 participants identified growing up and currently living.

Area	Grew Up	Currently Living
Campus Residence	N/A	49
City (Downtown)	41	50
City (Suburbs)	159	153
Exurban area	2	2
Small Town	48	17
Rural or Farm	26	7
Other	3	2
Missing	3	2

Procedure. The study was posted on Carleton's psychology department research recruitment system SONA with the name *Life in the city: Perceptions of urban scenes*. Participants were informed that the study aimed to examine how people perceive nature within images of urban environments. After participants filled out informed consent, they were shown images of urban scenes and then asked to report how much nature they see in the space. There were two ways the

participants were asked about their perception of the nature in the urban spaces ([1] Rate how much nature you perceive in this urban scene? [2] What percentage of the image is nature?). The two nature perception questions were set up into two distinct blocks (Block A and B). In each block, the participants were shown each urban image, one at a time in random order, and asked to respond to the item. The order of the blocks (i.e. the two tasks) was counterbalanced, and participants completed both, seeing each image twice (once for each block). By using this method, participants were able to encounter both questions and prevent the potential impact of question ordering. After the two task blocks, participants were asked to fill out the Nature Relatedness 21-item measure (NRS; Nisbet et al., 2009), Noticing Nature Activities survey (NNA; Richardson et al., 2021), and the Scale of Positive and Negative Experience (SPANE; Diener et al., 2010). Lastly, participants filled out demographic questions, such as age, gender, ethnicity, where they grew up, where they lived now, and the device they completed the survey. See Appendix A for all the items collected.

Measures.

Perceived Nature. To assess individuals' perceptions of nature in urban images, two different question types were asked:

1) "Rate how much nature you perceive in this urban scene?". The participants rated this item on a 7-point scale anchored from 1 = *there is no nature for an urban scene* to 7 = *there is a lot of nature for an urban scene*. This item is identified as the 7-point scale rating item.
2) "What percentage of the image is nature?". The participant responded to this question from 0 (*no nature at all*) to 100 (*all nature*) to allow for variability in the rating of nature. This item is identified as the percentage rating item.

Nature Relatedness. The Nature Relatedness scale (NRS; Nisbet et al., 2009) was used to measure a person's feeling of nature connectedness. This scale consists of 21-items examining three subscales: Self (nine items), Experience (six items), and Perspective (six items). The Self subscale examines an individual's intrapersonal feelings towards connecting and feeling a part of nature; the Experience subscale explores experiential moments, noticing, and attention in nature; and the Perspective subscale measures attitudes toward nature worldviews. Each item was rated on a Likert-type scale from 1 = *disagree strongly* to 5 = *agree strongly*. I computed the average score for each subscale (Experience $\propto$ = .76; Self $\propto$ = .83; Perspectives $\propto$ = .59) and an overall NRS score (across all 21-items; $\propto$ = .83). Additionally, the items "Even in the middle of the city, I notice nature around me" and "I take notice of wildlife wherever I am" of the NRS were used as individual face valid items of noticing nature.

SPANE. As an extra analysis, not pre-registered in my prospectus, I wanted to also examine positive and negative affect. This measure was added because it aligns with my reasoning behind the goal of my dissertation. One of my suggested reasons for examining if nature related individuals were more likely to perceive nature in my dissertation was to examine if this was a potential reason why nature related individuals were happier in general. I, therefore, measured affect through the Scale of Positive and Negative Experience (SPANE; Diener et al., 2010). The scale contains 12 items, with six items about positive emotions and six negative emotions. The participants were asked to rate how often they experienced each of the 12 emotions in the last four weeks. Average scores were created for positive affect (PA; $\propto$ = .87) and negative affect (NA; $\propto$ = .83). Three items were added ad-hoc from SPANE to measure feelings of awe. These three items were *awe*, *wonder*, and *amazement*. An average across all three items was computed ($\propto$ = .82).

Noticing Nature Activities. An additional extra measure I added outside of my pre-registered prospectus was to test a face valid noticing nature measure. To do this, the Noticing Nature Activities (NNA; Richardson et al., 2021) scale was measured. Richardson and colleagues (2021) defined noticing nature as actively paying attention and engaging with the sensory information from nature and bringing "elements of attention, awareness, and intention" (p. 2) to it (i.e. bringing it to mind). The NNA is a 13-item scale asked participants to indicate how often they did each of the noticing nature activities, on a 4-point scale (1 = *Never* to 4 = *Often*). An average of how frequent participants did all 13 activities was computed ($\propto$ = .87). An example noticing nature activity is "taken time to notice butterflies and/or bees". This face valid measure allows me to see if an individual who is self-reporting doing noticing nature activities in their real-life (i.e. higher score on the NNA scale) also reports perceiving more nature with my nature perception items I created.

Materials.

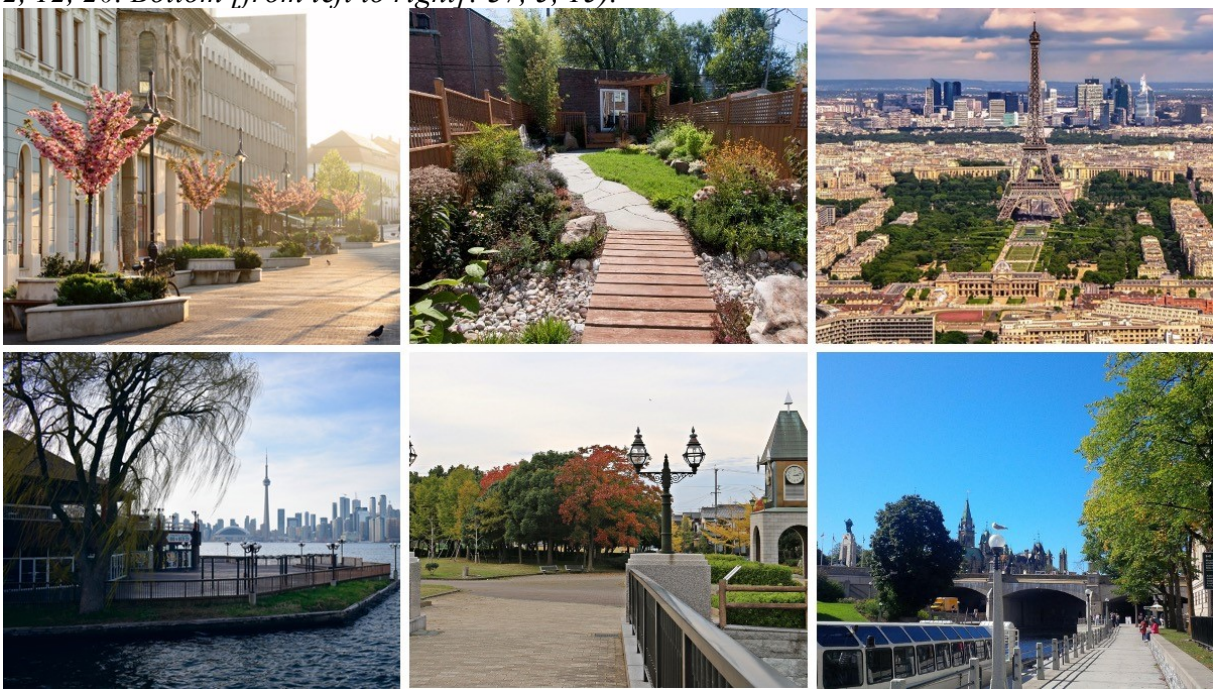
Images. Participants were exposed to 50 images for each nature perception item-type. The decision to use 50 images came from the consideration of wanting to have a large number of images to have higher reliability when aggregating the image ratings into a single score, but not wanting to make the task too tedious for the participants by exposing them to too many images. I came to my conclusion by examining how many images and item ratings previous research used with nature/urban images (full details on this decision can be seen in my pre-registration on OSF).

The 50 images portrayed urban scenes, but include some degree of nature. The nature in the urban scenes were plants, flowers, animals, birds, water, and/or mountains (to name a few). The images vary in the amount of nature in the images. See Figure 3 for example images (all

images can be seen on OSF). The images were labelled with number IDs to help identify the images for the psychometric analyses.

Figure 3

Example images that were shown to participants in Study 1. (Image IDs Top [from left to right]: 2, 12, 20. Bottom [from left to right]: 37, 5, 15).



Psychometric Analysis Results: Assessing the Experimental Tools

At 100 participants, across both item-types, there were eight images identified as potentially problematic (see Appendix B for a full report on the psychometric analysis results for the sample of 100 participants). Seven images were left-skewed with low means for both items, whereas one was right-skewed with a large mean. These eight images, however, had appropriate standard deviation (7-point scale rating item: between 0.99 and 1.43; percentage rating item: between 11.16 to 20.32) and a good range of values (7-point scale rating item: smallest range 1 to 5; percentage rating item: smallest range 30 to 100). Overall, the initial set of 50 images were acceptable to continue using, thus, a full sample ($N \sim 300$) was recruited to explore their variability and potentially use them to examine the association with nature relatedness.

After recruiting 300 participants and data cleaning (see details in the Participants section of Study 1 Methods), a second round of psychometric analyses was performed. Similar to the first round of psychometric analyses, the variability of how much nature the participants perceived in the urban scene images was assessed through the descriptive statistics of the distribution, a visual assessment of the distribution, reliability analyses, and inter-image correlations. Additionally, before running the main analysis, multilevel models were examined to explore if there was substantial between-participants variability in the ratings across the images and not solely within-participant variability. Lastly, principal component analyses and individual correlations between each image and nature relatedness were explored to choose the best images to aggregate.

7-Point Scale Rating Item – "Rate how much nature you perceive in this urban scene?". For the total sample, the average rating of nature perception across all 50 images was 3.68 (*SD* = 1.00), which is close to the middle value of the scale (4) and similar to the mean across all images for the psychometric analyses at 100 participants (*M* = 3.67). Descriptive statistics for each individual image can be seen in Table 2. The minimum mean rating of nature perception was 1.86 (Image 30), and the maximum mean was 6.45 (Image 23). The pre-registered criterion for the mean was to be around the middle value of the scale, which is 4. There were 13 images with means below the value of 3 and five above the value of 5. The visual representation of the distributions for each image can be seen in Figure 4. Aggregating all the images together showed good reliability, Cronbach's $\propto =$ .96, Omega = .95, and split-half reliability $r(280) = .91$ $p < .001$, 95% CI [.88, .93]. The reliability, through Cronbach's ∝, was robust from removing each individual image (all images showed either a decrease or no change in ∝ if removed), indicating that this was not an appropriate criterion to use when examining the

variability of the images. The range of inter-image correlations was from r = -.22 to r = .64. There were 11 images that had some negative correlations with other images (Images 2, 3, 6, 10, 11, 21, 23, 30, 41, 43, & 46).

Table 2

Descriptive statistics of the 7-point scale rating item of nature perception ratings for the full sample of Study 1 (N = 284).

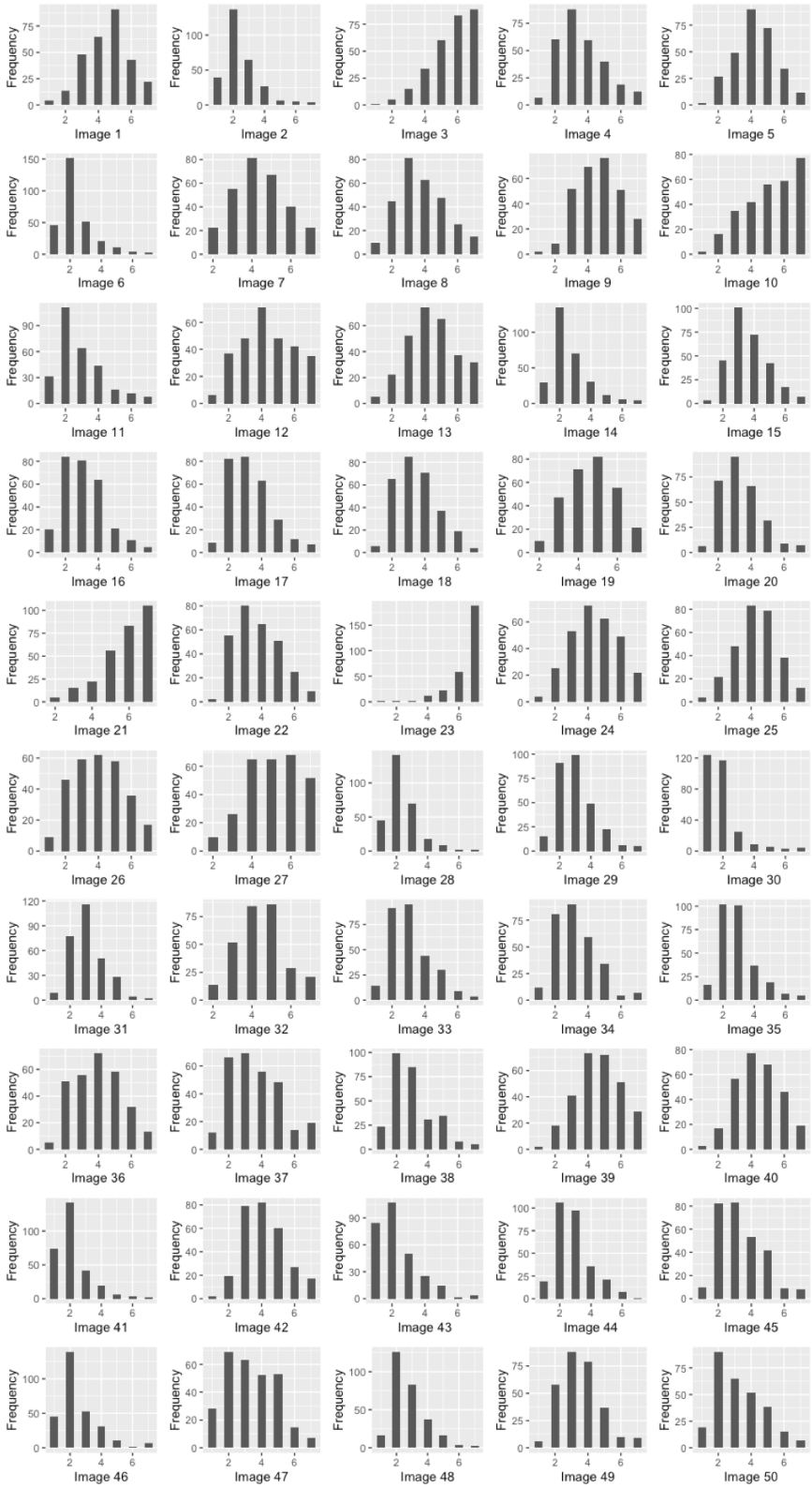
Image	*N*	Mean	Median	Mode	SD	Min	Max	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlation with NR
1	280	4.53	5	5	1.35	1	7	.954	.05	.53	-.01
2	278	2.47	2	2	1.13	1	7	.954	-.09	.54	-.06
3	280	5.64	6	7	1.30	1	7	.955	-.004	.41	.03
4	279	3.58	3	3	1.41	1	7	.954	.10	.56	.01
5	280	4.22	4	4	1.29	1	7	.954	.16	.51	-.03
6	280	2.34	2	2	1.10	1	7	.954	-.05	.50	.01
7	280	4.4	4	4	1.35	2	7	.954	.15	.56	-.01
8	280	3.76	4	3	1.46	1	7	.955	.09	.41	.05
9	280	4.65	5	5	1.34	1	7	.954	.06	.51	.04
10	280	5.18	5	7	1.58	1	7	.955	-.05	.43	.09
11	279	2.88	3	2	1.41	1	7	.954	-.01	.55	.03
12	280	4.33	4	4	1.62	1	7	.954	.10	.49	.05
13	280	4.41	4	4	1.47	1	7	.954	.08	.52	-.04
14	280	2.61	2	2	1.17	1	7	.954	.03	.63	.02
15	280	3.63	3	3	1.24	1	7	.954	.18	.44	-.002
16	280	3.11	3	2	1.30	1	7	.954	.12	.52	-.02
17	280	3.28	3	3	1.30	1	7	.954	.04	.52	.05
18	280	3.47	3	3	1.28	1	7	.953	.15	.59	.02
19	279	4.66	5	5	1.27	2	7	.954	.16	.53	-.02
20	280	3.34	3	3	1.24	1	7	.954	.06	.55	.01
21	280	5.79	6	7	1.26	2	7	.954	-.02	.51	.00
22	280	3.75	4	3	1.36	1	7	.954	.13	.54	.08
23	280	6.45	7	7	0.96	1	7	.955	-.22	.51	.06
24	280	4.38	4	4	1.45	1	7	.954	.11	.50	.13*
25	278	4.31	4	4[a]	1.30	1	7	.954	.10	.48	-.10
26	280	3.99	4	4	1.54	1	7	.955	.08	.40	-.02
27	280	5.08	5	6	1.37	2	7	.954	.11	.47	.03
28	280	2.34	2	2	1.01	1	7	.954	.004	.56	-.01
29	280	3.00	3	3	1.20	1	7	.954	.02	.62	.02
30	280	1.86	2	1	1.10	1	7	.955	-.22	.48	.01
31	280	3.10	3	3	1.07	1	7	.954	.06	.64	.03

32	279	4.43	4	5	1.25	2	7	.954	.14	.55	.04
33	280	3.09	3	3	1.24	1	7	.954	.14	.48	.06
34	280	3.19	3	3	1.24	1	7	.954	.04	.53	-.01
35	279	2.90	3	2	1.17	1	7	.954	.11	.57	.05
36	280	3.94	4	4	1.44	1	7	.954	.13	.56	.02
37	277	3.63	3	3	1.55	1	7	.955	.11	.56	.02
38	280	2.98	3	2	1.34	1	7	.954	.11	.50	.002
39	280	4.62	5	5	1.40	1	7	.954	.15	.50	.002
40	279	4.39	4	4	1.36	1	7	.954	.20	.44	.004
41	280	2.12	2	2	1.04	1	7	.954	-.13	.56	-.08
42	280	4.14	4	3	1.31	1	7	.954	.13	.48	.01
43	279	2.29	2	2	1.25	1	7	.955	-.06	.48	-.001
44	280	2.84	3	2	1.11	1	6	.954	.16	.57	.05
45	280	3.33	3	3	1.34	1	7	.955	.07	.40	.03
46	280	2.46	2	2	1.22	1	7	.954	-.05	.53	.04
47	280	3.36	3	2	1.51	1	7	.954	.12	.52	-.001
48	280	2.76	2.5	2	1.11	1	7	.954	.03	.64	-.04
49	280	3.51	3	3	1.27	1	7	.954	.11	.49	.05
50	279	3.23	3	2	1.44	1	7	.954	.12	.49	.11

Note: NR – Nature relatedness. [a] – identifies that there was more than one mode and the lowest mode was reported. * $p < .05$, ** $p < .01$, *** $p < .001$.

Figure 4

Histograms of the distribution of the 7-point scale rating item of nature perception for each image in Study 1(N =282).



Next, an unconditional multilevel model was run to explore if there was enough between-participants variability in the ratings of the 7-point scale rating item. To do this, the data was nested within individual participants. The intraclass correlation (ICC) of the unconditional model indicated that 18.99% of the data variance is due to the within-person effect. This percentage indicates that there is a portion of the variance due to the variation in the image (i.e. an individual person varied in their ratings for the different images), but that there is still enough between-participants variability to explore the individual difference characteristic nature relatedness. This within-person variability may suggest that the variation in the rating is due to the differences in urban images because each image has a different degree of nature.

Overall, the psychometric analyses suggest the images have decent variability to aggregate for the association. However, prior to aggregating the images, I also looked at ways to identify the best images to aggregate for the hypothesized correlation. First, a principle components analysis (PCA) was explored. Images were vetted as appropriate for PCA by exploring inter-image correlations before running the PCA. A detailed description of the PCA results can be seen in Appendix C. Forty-four images were deemed suitable for PCA (Images 3, 10, 21, 23, 30, & 46 were deemed unsuitable for PCA). A single-factor solution was identified as the best suited for the observed responses. Three additional items (Images 8, 9, & 43) showed higher factor loadings on other factors (with small Eigenvalues) and were therefore suggested to be removed. The reliability of the 41 images was identified as good, but reduced slightly from the reliability of the full set of images: Cronbach's $\propto = .95$, Omega = .95 95% CI [.94, .96], split-half reliability $r(280) = .90$ $p < .001$ 95% CI [.88, .93].

Additionally, I also explored the correlation between nature relatedness and each of the individual images ratings. The majority of images were not significantly correlated with nature

relatedness. A single correlation was significant; Image 24 had a significant correlation, $r(278) = .13$ $p = .04$. (Note that 2.5 significant correlations would be expected by chance under the null.) There were 31 images with non-significant positive correlations, 17 with non-significant negative correlations, and 1 with a correlation of 0. The average correlation between each image and nature relatedness was $z = .02$ (as computed with Fisher's r-to-z transformed correlations). The range of correlations was between $r = -.10$ (Image 25) and $r = .13$ (Image 24). When splitting the data into the images with a negative correlation and positive correlation, the average of the 31 images with a positive correlation (not including the correlations of 0) was $z = .04$, and the average of the other 15 images with a negative correlation was $z = -.03$.

Percentage Rating Item – What percentage of nature do you perceive? The quantitative values of these analyses can be viewed in Table 3, and the histograms for each image can be seen in Figure 5. The variability of the images was compared to the value across all the images. The mean value of nature perception was 35.94% ($SD = 15.57\%$). This mean value is similar to the mean identified at 100 participants ($M = 35.74$). The minimum mean percentage was 12.31% (Image 30) and the maximum was 81.32% (Image 23). The reliability across all 50 images was good, Cronbach's $\propto = .96$, Omega = .96, and spit half-reliability $r(279) = .91$ $p < .001$, 95% CI [.88, .94]. The reliability estimates were also robust in removing each image (all images showed either a decrease or no change in $\propto$ if removed). The distribution of inter-image correlations was from $r = -.05$ to $r = .65$. Only four images had negative inter-image correlations with other images (Images 9, 10, 34, & 48).

Table 3

Descriptive statistics of the percentage rating item of nature perception for the full sample of Study 1 (N = 284).

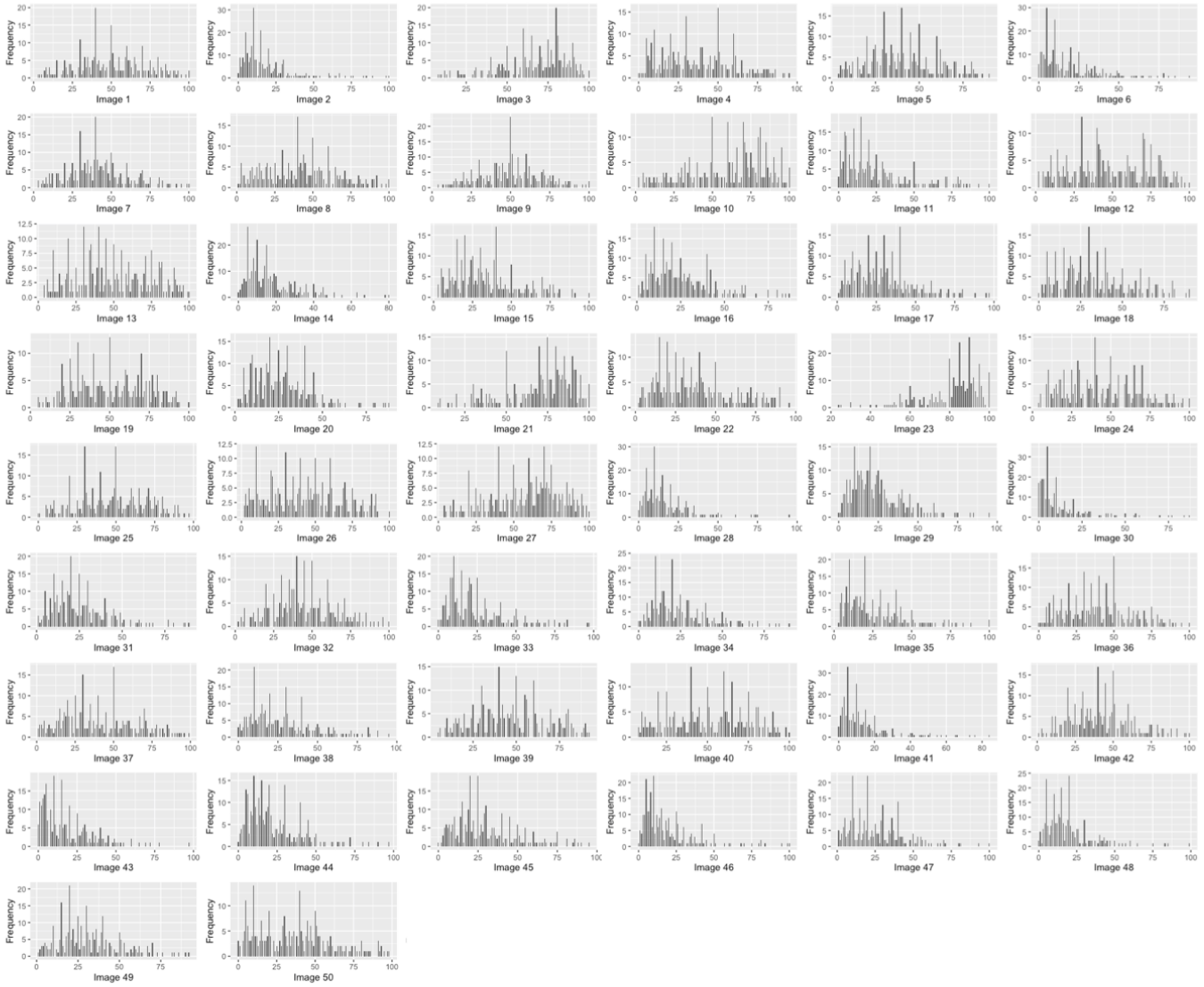
Image	*N*	M	Median	Mode	SD	Min	Max	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlations with NR
1	274	49.62	50	40	21.68	3	100	.962	.17	.54	-.06
2	272	18.00	12	10	17.41	0	100	.962	.13	.60	-.01
3	276	67.53	71	80	19.45	9	100	.962	.14	.50	-.03
4	272	36.96	35	50	22.47	0	95	.962	.16	.55	.02
5	275	40.78	40	40	19.32	4	90	.962	.21	.52	-.02
6	270	18.41	11	5	17.88	0	94	.962	.00	.54	-.10
7	275	41.97	40	40	18.84	3	100	.962	.14	.61	-.01
8	274	42.16	41	40	23.26	1	100	.962	.21	.51	-.04
9	276	50.20	50	50	19.45	4	100	.962	-.01	.48	-.17**
10	275	59.92	66	50	24.63	2	100	.963	-.05	.47	.08
11	266	24.00	17.5	15	21.73	0	100	.962	.11	.56	.05
12	275	48.46	46	30	25.10	2	100	.962	.16	.51	-.02
13	273	49.78	48	30[a]	24.27	4	100	.962	.11	.52	-.04
14	272	17.24	14	5	13.50	0	80	.962	.11	.64	-.07
15	275	34.05	30	40	19.95	3	100	.962	.15	.55	-.13*
16	268	25.01	20	10	17.18	1	87	.962	.20	.52	-.01
17	274	29.66	28	40	18.48	1	97	.962	.15	.61	-.07
18	272	32.44	30	30	20.23	0	90	.961	.22	.65	-.04
19	272	50.53	50	50	21.72	5	94	.962	.14	.55	-.06
20	273	26.89	25	20	15.73	2	88	.962	.15	.64	-.04
21	277	70.10	75	75	20.28	9	100	.962	.08	.54	-.14*
22	275	36.74	33	15	22.63	2	96	.962	.19	.55	.01
23	275	81.32	85	90	14.59	24	100	.963	.02	.50	.01
24	276	44.58	41	40	23.23	4	100	.962	.18	.43	.04
25	275	47.10	48	30[a]	21.80	0	97	.962	.18	.51	-.06
26	274	42.14	40	10	24.19	2	100	.963	.09	.40	-.04
27	275	57.98	61	40[a]	22.36	4	98	.962	.22	.54	-.08
28	268	16.27	13	10	13.88	0	95	.962	.17	.64	-.05
29	272	23.63	20	20	16.25	0	95	.962	.13	.63	-.01
30	249	12.31	6	5	15.24	0	87	.962	.08	.49	-.06
31	274	24.86	21	20	16.21	1	89	.962	.06	.58	-.04
32	275	44.48	43	40	19.86	2	100	.962	.23	.57	-.07
33	273	23.93	19	10	18.71	0	97	.962	.15	.55	-.01
34	270	24.12	20	10	16.46	0	90	.962	-.05	.49	-.06
35	275	25.46	20	20	18.49	3	100	.962	.18	.62	-.04
36	275	38.71	38	50	21.28	0	100	.962	.17	.60	-.03

37	275	40.49	37	50	23.25	1	99	.962	.12	.55	-.07
38	274	27.06	22	10	19.59	0	95	.962	.15	.51	-.12
39	275	46.20	45	40	20.75	5	92	.962	.24	.51	.00
40	275	51.04	51	40	23.19	8	100	.962	.16	.54	-.06
41	261	13.49	10	5	14.13	0	84	.962	.09	.62	-.14*
42	276	41.61	40	40	19.60	1	100	.962	.14	.48	-.02
43	256	17.94	14.5	10	16.05	0	97	.962	.10	.48	-.05
44	275	22.58	18	10[a]	16.97	0	97	.962	.20	.65	.02
45	275	29.97	25	20	19.66	0	95	.962	.20	.53	-.02
46	268	18.86	13	10	17.90	0	100	.962	.07	.60	-.07
47	268	25.40	21	10	17.38	1	100	.962	.18	.56	.01
48	271	18.62	15	20	14.76	0	99	.962	-.01	.63	-.01
49	273	30.88	28	20	17.66	1	92	.962	.13	.60	-.03
50	272	35.46	34	10	23.79	0	98	.962	.18	.56	-.02

Note: NR – Nature relatedness. [a] – identifies that there was more than one mode and the lowest mode was reported. * $p < .05$, ** $p < .01$, *** $p < .001$.

Figure 5

Histograms of the distribution of the percentage rating item of nature perception for each image in Study 1 (N = 284).



To explore if there is enough between-participants variability to explore individual differences in the ratings, an unconditional multilevel model was run, where the data was nested within individual participants. The ICC of the model indicated that 20.67% of the variance in the ratings was due to within-person variation. This may be more about the variation in the amount of nature in the individual images. However, it suggests that there is approximately 80% of between-participants variability to examine in the ratings, which is an appropriate amount to explore individual differences in the ratings.

To explore the best images to aggregate, a PCA was run. Forty-eight images were identified as appropriate for the PCA (see full details in Appendix C; Images 10 & 23 removed). A single-factor solution was deemed the most appropriate. All 48 images loaded on the single factor. The reliability for these 48 images was good: Cronbach's $\propto = .96$, Omega = .96 95% CI [.95, .97], split-half reliability $r(279) = .92$ $p < .001$ 95% CI [.89, .95]. The values of reliability through Cronbach's $\propto$ and Omega are the same as when using the sample of 50 images; however, the split-half reliability was slightly better for this set.

When exploring how each image correlated with nature relatedness individually, the majority of images (42 images) had a negative correlation. Four of the 42 negative correlations were significant (Images 9, 15, 21, & 41). The average across all images with nature relatedness was $z = -.04$. The correlations ranged from $r = -.17$ (Image 9) to $r = .08$ (Image 10). When splitting the images into positively and negatively correlated sets, the average positive correlation was $z = .03$, and the average negative correlation was $z = -.05$. The average across the five images with significant negative correlations was $z = -.15$.

Both Nature Perception Items. The two item-types were developed with the intention of exploring a similar concept of nature perception. With that in mind, the scores across the two item-types should be highly correlated. This was the case when looking at the correlation between the aggregates for each item-type, $r(278) = .72$ $p < .001$. Additionally, when looking at the correlation between both item-types for each individual image, there were 50 significant correlations (all $p < .001$). These correlations ranged from $r = .34$ (Image 19) to $r = .67$ (Image 11).

Statistical Analysis Results: Correlations with Nature Perception Items

The pre-registered data analysis plan was to examine the correlation between the aggregated items (separately) with average scored nature relatedness; however, it is not clear which set of images is the best to use for this aggregation. With this said, the statistical analysis was explored with different sets of aggregated images to see if the results change. There were four different combinations of image sets computed and examined. For each set, there was a separate set created for each item. The first set for both items was the full set of images (i.e., all 50). The second sets were created by removing the images that were identified as poor in the individual PCAs (Table 4ii). The third sets were created with only the images that positively (but not necessarily significantly) correlated with nature relatedness (Table 4iii). The last sets were created with only images negatively (but not necessarily significantly) correlated with nature relatedness (Table 4iv).

Table 4

Description of the different sets of images for the 7-point scale and percentage rating item-types of nature perception for Study 1.

		7-point Scale Rating Item		Percentage Rating Item	
Image set	Description	Number of Images	Images removed	Number of Images	Images removed
i	Overall aggregate of all 50 images	50	N/A	50	N/A
ii	Aggregate of images identified as good for PCA	41	3, 8, 9, 10, 21, 23, 30, 43, 46	48	10, 23
iii	Aggregate of images positively correlated with NR	33	1, 2, 5, 6, 7, 13, 15, 16, 19, 21, 25, 26, 28, 34, 41, 47, 48	8	1, 2, 3, 5, 6, 7, 8, 9, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 45, 46, 48, 49, 50
iv	Aggregate of images negatively correlated with NR	17	3, 4, 8, 9, 10, 11, 12, 14, 17, 18, 20, 22, 23, 24, 27, 29, 30, 21, 32, 33, 35, 36, 37, 38, 39, 40, 42, 43, 44, 45, 46, 49, 50	42	4, 10, 11, 22, 23, 24, 44, 47

Note: PCA = Principle components analysis. NR = Nature relatedness.

The different aggregate sets were then correlated with nature relatedness. Ninety-five percent confidence intervals were run for these correlations with 1000 sample Bootstrapping. All statistical analysis results can be seen in Table 5. The overall correlations of nature relatedness with both item-types were non-significant (see Table 5i). This pattern was also seen across the other sets of aggregated images (see Table 5 ii, iii, & iv). Thus, hypothesis 1 (*H1*) was not supported.

Next, as exploratory analyses, I also examined how the two nature perception item-types correlated with affect (positive, negative, and awe), noticing nature activity, the nature relatedness subscales and the two individual items that were face valid to my research hypothesis ("Even in the middle of the city, I notice nature around me" and "I take notice of wildlife wherever I am"). Only positive affect was significantly associated with the 7-point scale rating item across all different sets of images. Negative affect was negatively correlated to the 7-point scale rating item for the forced negatively correlated images to nature relatedness. For the percentage rating item, the Self subscale of nature relatedness was positively associated with the forced positively correlated to nature relatedness images. The Experience subscale of nature relatedness was negatively correlated with the forced negatively correlated to nature relatedness images. All other correlations with the two nature perception items were non-significant.

Overall, the correlations between nature relatedness with the nature perception item-types across the different set of images were non-significant. With the current tools, the effect of nature related individuals perceiving more nature in images appears to be null. Correlations between all other variables than the perception items can be seen in Appendix F.

Table 5

Correlations with the nature perceptions items for Phase 1 Study 1 (with different image sets).

Variables	7-point Scale Rating Item				Percentage Rating Item			
(i) Overall Aggregate	*n*	*r*	*p*	CI	*n*	*r*	*p*	95% CI
NR	278	.03	.67	-.09, .16	278	-.07	.25	-.19, .06
PA	278	.17**	.01	.05, .28	278	.05	.39	-.08, .17
NA	278	-.11	.07	-.23, .003	278	-.05	.40	-.17, .06
Awe	278	.02	.70	-.11, .15	278	.08	.19	-.05, .21
NNA	278	-.04	.50	-.18, .08	278	-.07	.27	-.18, .06
NR Item 8	278	.01	.83	-.11, .15	278	.02	.74	-.12, .16
NR Item 21	278	.06	.35	-.07, .19	278	-.01	.84	-.14, .11
NR Experience	278	-.03	.64	-.14, .09	278	-.10	.10	-.21, .02
NRS Self	278	.08	.18	-.05, .21	278	.02	.78	-.11, .13
NR Perspectives	278	-.01	.85	-.15, .12	278	-.10	.10	-.24, .04
(ii) Only Images Identified as Good in PCA	*n*	*r*	*p*	CI	*n*	*r*	*P*	95% CI
NR	278	.02	.70	-.10, .16	278	-.08	.21	-.20, .05
PA	278	.17**	.01	.05, .28	278	.05	.45	-.09, .16
NA	278	-.12*	.05	-.24, -.003	278	-.05	.39	-.17, .06
Awe	278	.04	.55	-.10, .15	279	.08	.19	-.05, .21
NNA	278	-.04	.51	-.18, .09	278	-.07	.23	-.19, .06
NR Item 8	278	.01	.86	-.11, .15	278	.02	.77	-.13, .16
NR Item 21	278	.06	.34	-.07, .18	278	-.02	.73	-.15, .10
NR Experience	278	-.03	.62	-.14, .08	278	-.01	.09	-.22, .01
NR Self	278	.08	.19	-.05, .21	278	.01	.85	-.12, .13
NR Perspectives	278	-.01	.83	-.15, .12	278	-.10	.09	-.24, .03
(iii) Only Images Positively Correlated with NR	*n*	*r*	*p*	CI	*n*	*r*	*p*	95% CI
NR	278	.07	.28	-.05, .19	278	.05	.38	-.06, .18
PA	278	.16*	.01	.04, .28	278	.08	.19	-.05, .21
NA	278	-.09	.14	-.21, .02	278	.001	.99	-.12, .11
Awe	278	.01	.83	-.12, .13	278	.07	.27	-.06, .19
NNA	278	-.01	.87	-.14, .10	278	.00	1.00	-.11, .12

NR Item 8	278	.05	.41	-.07, .17	278	.05	.44	-.07, .17
NR Item 21	278	.09	.14	-.03, .21	278	.12	.05	-.01, .23
NR Experience	278	.00	1.00	-.11, .12	278	.02	.78	-.09, .14
NR Self	278	.12	.05	.001, .24	278	.13*	.04	.01, .24
NR Perspectives	278	.01	.84	-.12, .15	278	-.05	.39	-.18, .07
(iv) Only Images Negatively Correlated with NR	*n*	*r*	*p*	CI	*n*	*r*	*p*	95% CI
NR	278	-.05	.41	-.17, .08	278	-.09	.12	-.22, .03
PA	278	.18**	.003	.04, .30	278	.04	.46	-.10, .17
NA	278	-.14*	.02	-.26, -.02	278	-.06	.32	-.17, .05
Awe	278	.04	.53	-.09, .17	278	.08	.18	-.05, .20
NNA	278	-.10	.11	-.23, .05	278	-.08	.20	-.20, .05
NR Item 8	278	-.06	.36	-.19, .08	278	.01	.83	-.14, .16
NR Item 21	278	-.01	.89	-.14, .12	278	-.04	.52	-.17, .09
NR Experience	278	-.08	.19	-.19, .03	278	-.12*	.05	-.24, -.01
NR Self	278	.01	.91	-.11, .13	278	-.01	.90	-.13, .11
NR Perspectives	278	-.05	.37	-.19, .09	278	-.11	.08	-.24, .03

Note: NR – Nature relatedness. PA – Positive Affect. NA – Negative Affect. NNA – Noticing Nature Activity. NR Item 8: “Even in the middle of the city, I notice nature around me”. NR Item 21: “I take notice of wildlife wherever I am”. * $p < .05$, ** $p < .01$, *** $p < .001$.

Discussion

In Phase 1 of the dissertation, I wanted to explore if nature related individuals would notice more nature through the process of perception. I asked participants to rate 50 images on how much nature they perceived with a 7-point scale and a percentage rating. Due to the tool-building nature of the study, I ran psychometric analyses to ascertain the extent of the variability among images to allow the exploration of the association with nature relatedness. Across the initial sample of 100 participants and the final sample of 284 participants, the images showed reasonable variability through descriptive statistics, reliability analyses (Cronbach's alpha, Omega, and split-half reliability), and inter-image correlations. Further, MLM results indicated that the images had enough between-person variability to explore an association between an individual difference characteristic and the image ratings. Lastly, principal component analyses and individual correlations with nature relatedness were run to identify the best images to use when aggregating the two nature perception item-types prior to data analysis. Despite reasonable variability, links with trait nature relatedness did not emerge. More specifically, when looking at both individual images and aggregated means of nature perception item-types across all images, a null association with nature relatedness was evinced. Additionally, this null association was robust across the exploration of different sets of images, identified through the psychometric analyses. This suggests that, with my current tools, nature related individuals do not, in fact, notice more nature, through perception, in urban environment photos than those low on nature relatedness. Therefore, the results did not support *H1*.

It is unclear with this single study whether the results suggest a lack of nature perception differences across the nature relatedness dimension, or if the research tool I built did not accurately measure the concept of nature perception (or noticing nature, in general). Though

psychometric analyses suggested the items were, in fact, measuring something reliably and that there was enough between-person variability in the ratings between the images, the tool I created may have been measuring something outside of noticing nature. For one, there was no association between more face-valid noticing nature items and the items of nature perception. The face valid items from the NRS ("Even in the middle of the city, I notice nature around me" and "I take notice of wildlife wherever I am") and the Noticing Nature Activities scale (Richardson et al., 2021) were added to explore the validity of my newly created tool for measuring noticing nature. This lack of correlation between these face-valid measures and my nature perception items indicates that the tool may explore something different than noticing nature. In other words, nature *perception* (or whatever this task measures) may differ from the concept of *noticing* nature.

There remains some uncertainty about what the perception task is assessing, yet it may still be measuring participants' perception of how much nature is in the images. A key driver of differences in nature perceptions may be differences in participants' *definitions* of nature; for example, (how much) is a potted plant actually nature? The individual differences (as identified through reliability and the MLM results) in these responses could highlight that the conceptualization of nature differs across people and contexts (Hartig & Kahn, 2016). In other words, the conceptualization of nature may come more from top-down processing (Haga et al., 2016; Van Hedger et al., 2019; Hartig et al., 2014) than bottom-up processing, as suggested by research on low-level features of nature (Berman et al., 2014; Kardan et al., 2015a; Kotabe et al., 2017).

However, little research has examined, generally, what people conceptualize as nature. From the minimal work, the main themes identify nature is separate from or untouched by

humans (Fehnker et al., 2021; Vining et al., 2008) and “out there” away from civilization (Haluza-Delay, 2001; Wilson, 2019). This suggests that the context of an urban environment may reduce a person’s perception of nature within it.

Some studies suggest that differences in conceptualizations of nature may be due to a person’s levels of nature relatedness. An unpublished study by a previous Happiness Lab graduate student Dan Shaw (2018) asked participants to identify key features of nature in three minutes. Results showed that not only did those high in nature relatedness identify substantially more features than those low in nature relatedness (see Figure 1), but they also differed on their ratings of prototypicality (i.e. a rating of how characteristic the term was to natural environment) of 100 of the most frequent features of natural environments. Those higher in nature relatedness reported the features, in general, as higher in prototypicality. This was also the case across the level of prototypicality (splitting the terms into high, medium, and low prototypicality based on average scores across all participants). Together, this may suggest that nature related individuals may have a richer language for nature and that they have more strong feelings towards what is nature.

Although the conceptualization of nature and its differences by people (level of nature relatedness) may be one part at play for perception, it is most likely not the whole picture. There may be other biases that influence the understanding of perception. For example, where a participant focuses their attention may also influence their response to nature perception. If Participant A focused their attention on the tree in the background of the image, whereas Participant B focused their attention on the car that drove by the tree, participant A might have rated perceiving more nature than Participant B. This other potential aspect of noticing nature (attention) will be explored in Phase 2 of the dissertation.

Other biases to perception may be how much weight participants put on the urban context of the images reducing the perception of nature. This abstract weight to the responses may be influenced by a participant's level of nature relatedness. In a study by a previous undergrad thesis student from the Happiness lab, Tristan Marshall (2022) asked participants to rate how representative 137 images, that fit into seven categories (green wilderness, arid wilderness, urban, people in nature, nature in urban areas, expansive nature, and nature as utility), were to their concept of either nature or natural environment. Marshall (2022) found some differences in the "perception" of natural environments by levels of nature relatedness. For the image category most similar to my images (e.g. urban greenspaces, tree-lined neighbourhoods), there was a difference in the conceptualization fit where highly nature related individuals reported that images were *less* representative of their concept of nature/natural environments. There was also a similar difference for urban images with little to no greenery and images of nature that were viewed as more utilitarian (e.g. agriculture and natural resources). However, there was no association between nature relatedness and image categories with more nature (humanity included nature, expansive nature, arid nature, and wilderness). Even though nature related individuals felt some aspects of nature were more characteristic of a natural environment (i.e. the prototype study), including the context of urban environments may influence natural features to be perceived as less natural due to the context (i.e. the photo study). In other words, two competing processes (i.e., highly nature related people often defining more things as nature, but less so in urban contexts) may combine to create a null association in the perception task here.

One significant association found in the study was between the 7-point scale nature item and affect. More specifically, when participants reported noticing more nature in the urban environment images, they also reported feeling more positive affect in the last four months. This

is interesting as the proposed hypothesis of nature related individuals *noticing* more nature stemmed from research showing that nature related individuals were happier people in general (Capaldi et al., 2014; Pritchard et al., 2020) and that happiness may be found through being in contact with nature (McMahan & Estes, 2015) and also noticing it (Passmore & Holder, 2016; Passmore et al., 2022ab). In the current study, nature related individuals did report more positive affect, and even though nature relatedness was not associated with nature perception, affect was associated with nature perception. This may still suggest that nature related individuals, when asked to perceive nature in urban environments, report not noticing more because of their high threshold for what constitutes nature there, but maybe do, still, *notice it* in their everyday life. Once more, this suggests that my new tool may not be an accurate way of assessing noticing nature through the process of perception.

Altogether, the findings of this phase suggest a null association between nature relatedness and nature perception. This may stem from the urban context of the images reducing the perception of nature, or it may be that the perception items were not appropriate measures for *noticing nature*. Nevertheless, due to this null association, a replication study was not run with Amazon Mechanical Turk employees.

Phase 2

Phase 2 explores the process of attention. Attention, here, is defined as what an individual notices first in a conscious way (first attention) and what in the image they want to focus their attention on more (voluntary attention). I hypothesized that an individual high on trait nature relatedness would notice nature first in an urban scene image more often and that they would want to focus on the nature more. This phase was run simultaneously to Phase 1.

Study 2

Methods

Participants. Much like Phase 1, running a proper *a priori* power analysis to justify the sample size is challenging due to the lack of previous work done in this area. However, a sample of 300 participants was identified as appropriate to recruit. This aligns with a simulation suggesting that correlations stabilize at a sample of 250 participants with 80% confidence (Schönbrodt & Perugini, 2013). With a sample of 250 participants, alpha set at .05, and power set at 90%, an effect size of .12, in either direction, can be identified (as highlighted by a sensitivity power analysis run in G*Power).

Optional stopping techniques were used to stop the recruitment of participants at 100 to check the variability of the two attention items for each image. This strengthened the tools used to examine the association between nature attention and nature relatedness. This phase had more difficulty in finding images with an appropriate amount of variability in the items. This may be because the items were measured on a dichotomy of clicking on the nature area or not. With this, it took three rounds of psychometric analyses to properly find the appropriate set of 50 images to run a full sample. Therefore, the data was collected in four rounds. During Rounds 1 and 2 of optional stopping recruitment, 100 participants were recruited for each round, respectively. During Round 3, again at a sample of 100 participants, appropriate psychometric properties of the images were observed, and the recruitment of the other 200 participants was begun in Round 4. Thus, a total of 497 participants were recruited to participate. The data reported here, however, only includes the sample of 300 who saw the final, refined set of images.

Three-hundred undergraduate students were recruited for the final sample. Participants were excluded if they were identified as outliers or as demonstrating string responses on the

NRS. String responding is defined as participants responding in the same manner across items of the NRS. Outliers were identified if the participants had scores of the NRS that were ±3 standard deviations away from the mean. Out of the 300 participants, five were identified as poor and were removed. Another participant was identified as problematic in a manual assessment of the data. They responded that their age was 98 and their responding was mostly string responding with a single response as different. They were also removed. The final sample was 294.

The age ranged from 16 to 56 with an average of 19.76 (*SD* = 3.47). The majority of participants identified their gender as woman (206; 70.1%), with a smaller percentage identifying their gender as man (76; 25.9%), gender fluid/non-binary/two-spirited (8; 2.7%), self-described (1; 0.3%), and three participants preferred not to say (3; 1%). Fifty-six percent of participants self-described as White, 12% as Black, 12% South Asian, 10% as Arab, and a smaller percent self-described as Indigenous (3%), Chinese (4%), Filipino (2%), Latin American (2%), Southeast Asian (2%), West Asian (1%), Korean (> 1%), and other (3%). The bulk of participants both grew up and currently lived in a city (See Table 6 for frequency of this demographic. information). Eighty eight percent of participants filled out the survey on a computer (i.e. laptop or desktop).

Table 6

Frequency of where Study 2 participants identified growing up and currently living.

Area	Grew Up	Currently Living
Campus Residence	N/A	86
City (Downtown)	48	40
City (Suburbs)	151	139
Exurban area	9	5
Small Town	57	11
Rural or Farm	23	9
Other	6	4
Missing	0	0

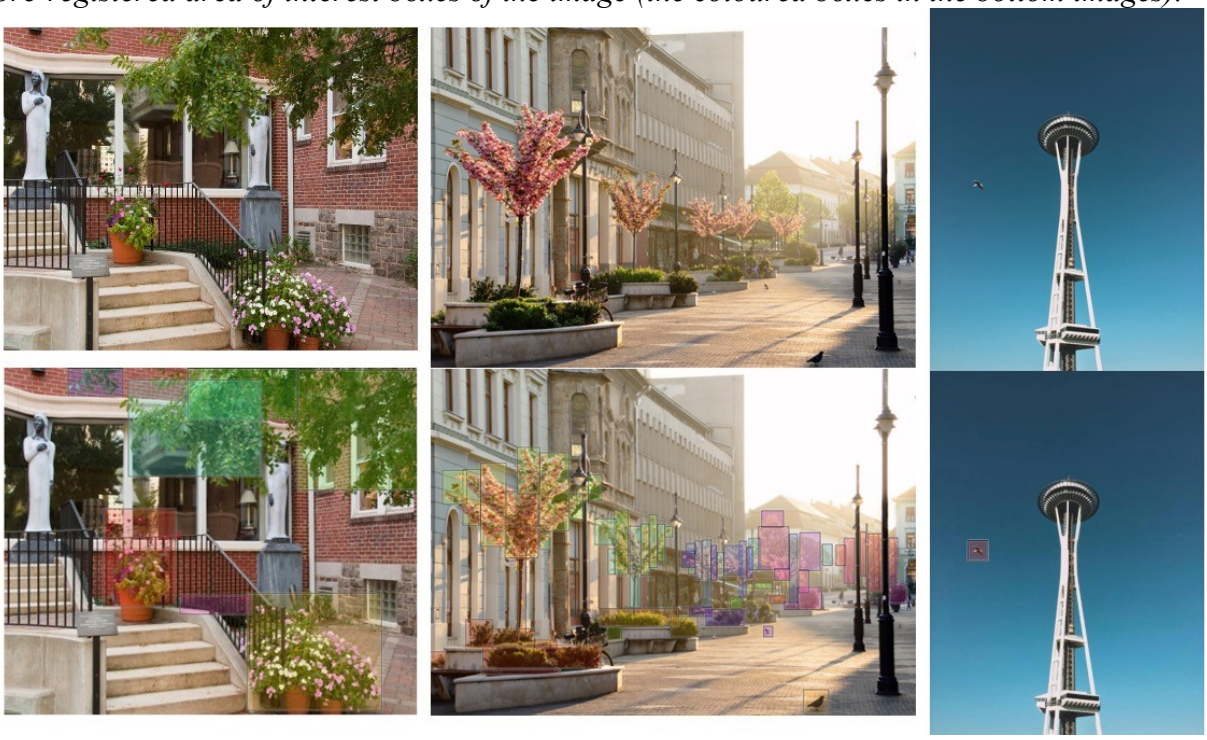
Procedure. The students were recruited through the SONA research management system to participate in an online study titled "*What do you see? A study of visual stimulus in urban photographs*". Once recruited, participants first filled out informed consent before completing the study. They were informed that the study aimed to gather information about what captures their attention in images. The study consisted of three sections. The first two sections were the blocked nature attention tasks. There were two different types of nature attention tasks to explore the two distinct versions of attention being paid to nature in urban images. The different attention questions were presented to participants in blocks, similar to Phase 1 Study 1. Participants were shown the set of images and asked a single attention question. Then, they were shown the images again and asked the other question. Participants were exposed to the same set of images for both blocks, randomly ordered. Because the two different attention questions can affect each other (i.e. ordering effect), the different questions were presented to participants in a set order. In Block A, participants were asked about the first thing they noticed in the image. This is known as first attention. In Block B, the participants were asked to choose an object/thing in the image that they wanted to focus their attention on. This is known as voluntary attention. After completing the two blocks, participants filled out the demographic questionnaire, the 21-item Nature Relatedness scale (NRS; Nisbet et al., 2009; $\propto = .79$), Noticing Nature Activities Survey (NNA; Richardson et al., 2021; $\propto = .88$), and the Scale of Positive and Negative Experiences (SPANE; Diener et al., 2010; PA $\propto = .88$ NA $\propto = .80$ Awe $\propto = .83$). Finally, participants answered demographic questions, such as age, gender, ethnicity, where they grew up, where they were currently living, and what type of device they used to fill out the questionnaire.

Materials.

Images. Similar to Phase 1, images of urban scenes with some degree of nature were used in Phase 2. The images were of scenes usually seen in urban North American environments, with some degree of nature, such as trees, grass, water, flowers, animals, etc. The participants were exposed to 50 images. For data analysis, areas of interest were specified in the images as those containing nature. This technique is similar to eye-tracking methods. The regions were created by making boxes around the areas of interest (i.e. the nature) in Qualtrics prior to data collection. For example, if a flower is in the image, a region box was placed around the flower. Example images and their area of interest can be seen in Figure 6. All pre-registered areas can be seen on OSF.

Figure 6

Example images that were shown to participants in Study 2 (Images: 11, 23, and 41) and the pre-registered area of interest boxes of the image (the coloured boxes in the bottom images).



Measures.

First attention Item. The first nature attention item focused on the initial attention being drawn. Participants were requested to report the first thing they noticed by clicking on the image. To ensure participants were prepared for the task, they were given instructions and a practice image before continuing. Participants had 7 seconds to click on the thing that drew their attention first, and then it moved on. If a participant clicked on one of the areas of interest (i.e. nature in the image), the participant received a score of 1 for that image. If they clicked on any other area of the image, they received a score of 0. After all psychometric analyses, a sum score of all the images was created.

Voluntary Attention Item. The voluntary attention question explored what the participants wanted to focus their attention on in the image. Before the images were shown for Block B, participants were told, "For the next task, you will be shown 50 images of urban environments. Please choose something in the image that you would want to focus your attention on. The something you choose can be anything in the image. Imagine you were in this environment, what would you want to focus your attention on? You can take as long as you need to look at each image. Please click on the thing you want to focus on in the image and then click next". If the participant clicked the nature area of the image, a score of 1 was given, while if another area was clicked, a score of 0 was given. After all psychometric analyses, a sum score across all images was created for the count data.

Psychometric Analyses Results

Three rounds of optional stopping and psychometric analyses were performed before a set of 50 images with appropriate variability was found. For every round, 100 participants were recruited and psychometric analyses were run to test the variability of each item. For both items

separately, the variability for each image was tested by examining descriptive statistics (such as minimum, maximum, mean, standard deviation, and frequency of the area clicked), a standardized distribution of the images created with Z-transformed mean scores for each image[2], reliability analysis, and inter-image correlations. These analyses identified a set of unsuitable images. These unsuitable images were replaced, and a new sample of 100 participants was recruited. During Round 1, 32 images were deemed unsuitable and needed to be replaced. During Round 2, 11 images needed replacing. During Round 3, the images demonstrated appropriate variability to continue recruitment to the full sample size of 300. More details for each round can be found in Appendix E. The same psychometric analyses, and a few extra, were examined for each nature attention item separately during the fourth round (i.e. full sample).

First Attention. All quantitative results of the psychometric analyses for the first attention item can be seen in Table 7. Approximately 32% of all the clicks were on the nature areas. The percentage of nature clicks did vary across images with a range of 2% (Image 70) to 93% (Image 85). The mean of 0.32 is the largest mean across all psychometric rounds, but relatively similar in size. Additionally, all images showed some variance in the responses (i.e. no image had 0 participants clicking the nature area). The range of inter-image correlations was from $r = -.26$ to $r = .38$. All images had a minimum of six negative correlations with other images (i.e. negative inter-image correlations).

[2] The Z-scores were created for each image in comparison to the overall mean across all the images, thereby, allowing the comparison of means across the different images.

Table 7

Descriptive statistics of the first attention item for the full sample of Study 2 (N = 294).

Image	N	*M*	*SD*	Frequency of Other Clicks	Frequency of Nature Clicks	% of Nature Clicks	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlation with NR
2	279	0.51	0.50	136	143	51.25	0.95	.700	-.22	.25	.17**
4	281	0.21	0.41	222	59	21.00	-0.52	.693	-.11	.21	.05
7	284	0.35	0.48	184	100	35.21	0.17	.697	-.16	.27	.09
8	288	0.60	0.49	115	173	60.07	1.37	.696	-.26	.24	.05
11	283	0.50	0.50	141	142	50.18	0.89	.687	-.14	.20	.08
12	277	0.46	0.50	149	128	46.21	0.70	.694	-.17	.19	.04
22	278	0.14	0.35	238	40	14.39	-0.83	.692	-.19	.19	-.02
23	278	0.87	0.33	35	243	87.41	2.69	.692	-.16	.27	.17**
25	279	0.24	0.43	213	66	23.66	-0.39	.691	-.13	.22	-.08
26	280	0.20	0.40	224	56	20.00	-0.56	.693	-.16	.26	.06
28	282	0.58	0.49	119	163	57.80	1.26	.689	-.20	.18	.05
30	279	0.14	0.35	240	39	13.98	-0.85	.694	-.18	.30	.02
35	288	0.47	0.50	153	135	46.88	0.73	.690	-.14	.34	.04
41	286	0.22	0.42	223	63	22.03	-0.47	.693	-.22	.34	.02
42	285	0.39	0.49	173	112	39.30	0.37	.680	-.18	.31	-.08
43	277	0.26	0.44	206	71	25.63	-0.29	.696	-.12	.18	.02
46	281	0.31	0.46	195	86	30.60	-0.05	.698	-.11	.23	.00
47	285	0.29	0.45	203	82	28.77	-0.14	.688	-.10	.25	.03
52	276	0.38	0.49	171	105	38.04	0.31	.704	-.19	.13	.02
53	284	0.04	0.19	273	11	3.87	-1.34	.693	-.14	.16	.06
54	289	0.17	0.38	240	49	16.96	-0.71	.692	-.21	.30	.02
55	281	0.12	0.32	248	33	11.74	-0.96	.694	-.15	.27	-.05
57	283	0.31	0.47	194	89	31.45	-0.01	.689	-.13	.23	.00
58	288	0.17	0.38	239	49	17.01	-0.71	.694	-.17	.37	.04
59	286	0.05	0.22	271	15	5.24	-1.28	.698	-.19	.23	.02
60	284	0.17	0.37	237	47	16.55	-0.73	.692	-.16	.32	.03

62	282	0.56	0.50	124	158	56.03	1.18	.687	-.16	.36	.07
63	282	0.26	0.44	209	73	25.89	-0.28	.689	-.19	.22	.05
64	284	0.48	0.50	148	136	47.89	0.78	.707	-.19	.19	.07
67	281	0.36	0.48	180	101	35.94	0.21	.693	-.21	.27	.08
70	281	0.02	0.16	274	7	2.49	-1.41	.693	-.16	.37	.04
71	284	0.40	0.49	169	115	40.49	0.43	.688	-.09	.34	.03
73	284	0.20	0.40	228	56	19.72	-0.58	.685	-.12	.36	-.04
74	286	0.07	0.26	266	20	6.99	-1.19	.699	-.22	.29	.06
75	276	0.27	0.44	202	74	26.81	-0.23	.694	-.20	.21	.11
77	287	0.17	0.38	237	50	17.42	-0.69	.686	-.15	.34	-.04
79	284	0.12	0.32	251	33	11.62	-0.97	.692	-.12	.34	-.06
81	282	0.22	0.41	220	62	21.99	-0.47	.685	-.09	.30	.06
82	278	0.17	0.37	232	46	16.55	-0.73	.694	-.11	.18	.06
83	284	0.89	0.31	30	254	89.44	2.79	.695	-.20	.18	.09
84	286	0.23	0.42	219	67	23.43	-0.40	.690	-.18	.38	.01
85	284	0.94	0.24	18	266	93.66	2.99	.693	-.18	.19	.20***
86	273	0.43	0.50	156	117	42.86	0.54	.696	-.20	.24	-.04
87	280	0.28	0.45	201	79	28.21	-0.17	.679	-.05	.33	.08
88	289	0.19	0.39	234	55	19.03	-0.61	.701	-.26	.37	-.05
89	283	0.40	0.49	170	113	39.93	0.40	.694	-.16	.23	.03
90	283	0.36	0.48	180	103	36.40	0.23	.686	-.15	.30	.00
91	279	0.19	0.39	226	53	19.00	-0.61	.688	-.14	.38	.11
92	280	0.18	0.39	229	51	18.21	-0.65	.691	-.12	.27	.22***
93	282	0.28	0.45	202	80	28.37	-0.16	.696	-.09	.28	0.12

Note: All images had a minimum of 0 and maximum 1; therefore, this information was redacted from the table. NR = Nature relatedness. * $p < .05$, ** $p < .01$, *** $p < .001$.

The 50 images showed good reliability, Cronbach's ∝ = .70, which was an improvement from Round 3 (∝ = .67). The reliability was quite robust to the removal of the individual images. The majority of images (42 images) were identified as good (i.e. Cronbach's ∝ was reduced when the image was removed) with a change range of .001 to .02. Seven images (2, 46, 52, 59, 64, 74 & 88) were considered poor to aggregate because the reliability score was increased when removing each individually, with a maximum increase of Cronbach's ∝ of .01. A single image showed no change in Cronbach's ∝ when removed (Image 7). When running other measures of reliability, such as split-half reliability and Omega, the reliability was identified to be low, split-half reliability, $r(291) = .52$ $p < .001$ 95% CI [.42, .61], Omega = .65.

The standardized distribution of the Z-transformed means for each image was examined. The distribution of all 50 images was right-skewed (see Figure 7) and identified as non-normal (Table 8). The robustness of the distribution was examined by removing the images with the lowest percentage of nature clicks. Four robustness tests were run: 1) removing images with 5% or less nature clicks; 2) removing the images with 10% or less nature clicks; 3) removing images with 15% or less nature clicks; and 4) removing images with 20% or less nature clicks. The distribution did not become normal after removing the images for each robustness test (see Table 8). Additionally, outliers were identified (approximate +3SD above mean, more specifically above the value of 2.7), and thus, the normality robustness tests were re-run without these outliers. The distribution became statistically normal after removing the outliers from the full sample of 50 images. A visual of this distribution with 47 images can be seen in Figure 8.

Figure 7

Visual representation of the standardized distribution of all 50 images for the first attention item in Study 2.

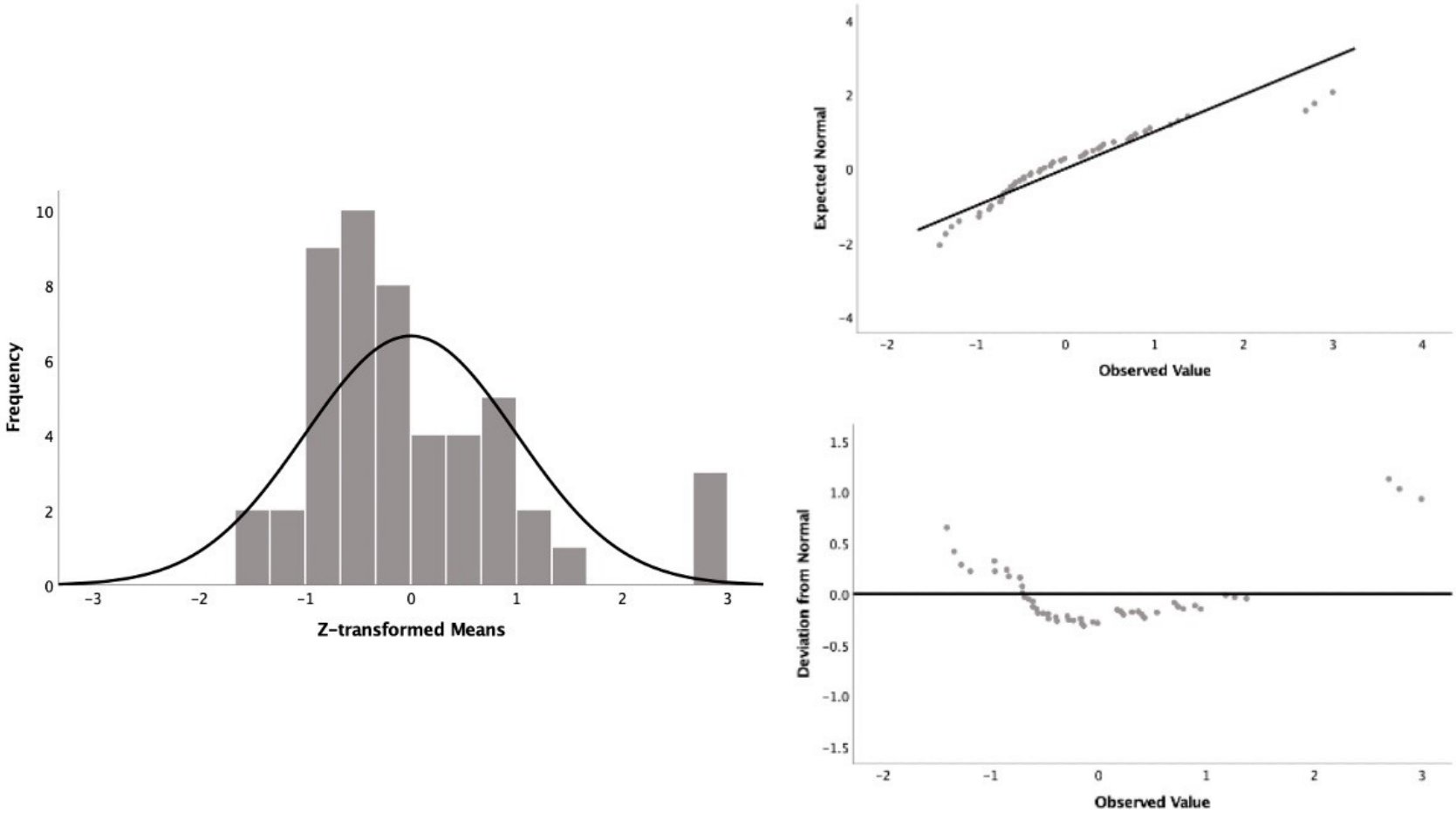


Table 8

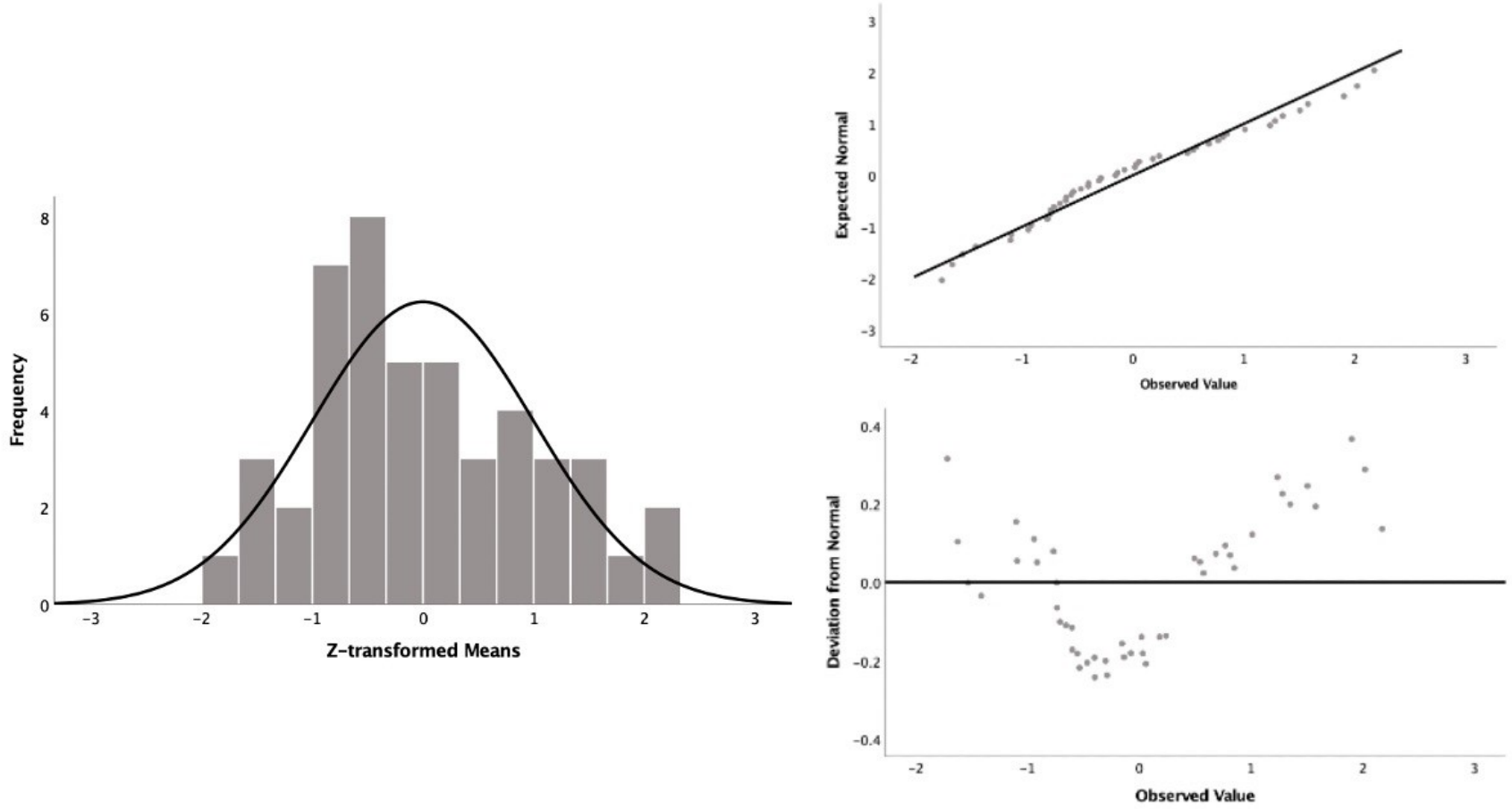
Robustness tests for normality of the first attention task standardized distribution for Study 2.

	Low Percentage Image Removed				Images Outliers Removed			
Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	Number of images	Image Outlier removed	Shapiro-Wilks Estimate	*p*
None	50	N/A	0.89	< .001	47	23[a], 83[a], 85	0.99	.16
Images with 5% or less clicks on the nature area	48	53, 70	0.87	< .001				
Images with 10% or less clicks on the nature area	46	53, 59, 70, 74	0.85	< .001				
Images with 15% or less clicks on the nature area	42	20, 30, 53, 55, 59, 70, 74, 79	0.84	< .001				
Images with 20% or less clicks on the nature area	32	12, 26, 30, 53, 54, 55, 58, 59, 60, 70, 73, 74, 77, 79, 82, 88, 91, 92	0.84	< .001				

Note: Images were first removed for low percentages and then removed as outliers. [a] These images were not initially identified as outliers, but after the removal of the other images became outliers.

Figure 8

Visual representation of standardized distribution of 47 images identified as best for distribution from robustness tests for the first attention item in Study 2.



Before running the hypothesized correlation, I examined how much between-participants variability was in the responses through a multilevel unconditional model, where the data is nested within each participant on the ratings of the first attention item. Unlike Phase 1, the unconditional multilevel model was run as a logistic multilevel model on SAS. This type of model assumes the dichotomous outcome variable has a Bernoulli distribution (Guo & Zhao, 2003). The ICC suggested that ~4.47% of the variation was due to within-person effect. In other words, the correlation between responses across the images is relatively small for a participant. Therefore, there is a large amount of between-participants variability to explore the association with nature relatedness.

Lastly, as an extra psychometric analysis at this final step, I examined the correlation between each image on the first attention item and nature relatedness. This will allow me to choose the best images to aggregate before correlating the aggregate with NR. The correlations ranges from r = -.08 (Image 42) to r = .22 (Image 92), with an average z = .04 (z-transformed correlations). The majority of the images showed a positive correlation, with four of the 38 images had significant correlations with nature relatedness (Images 2, 23, 85, & 92). Only 9 images had a negative correlation (Images 22, 25, 42, 55, 73, 77, 79, 86, & 88), and three images with a correlation of 0 (Image 46, 57 & 90). This would suggest examining only the correlation between the sum of 38 images and nature relatedness.

Voluntary Attention. All quantitative results of the psychometric analyses for the voluntary attention item can be seen in Table 9. Across all clicks, approximately 40% of the clicks were on the nature areas of the urban images. This is higher than the first attention clicks and all the other rounds of psychometrics for the voluntary attention item. No images had 0 clicks on nature areas, indicating all images had some variance in the responses. The percentage

of clicks on nature areas did, however, vary across the images, with a range of 10% (Image 70) to 93% (Image 83). The inter-image correlations ranged from r = -.15 to r = .53. All images had a minimum of one negative (inter-image) correlation with another image.

The 50 images showed good reliability to aggregate, Cronbach's $\propto = .83$. The reliability was also robust to the removal of individual images, with mainly very small changes in Cronbach's $\propto$ (a maximum change of .006 in either direction). The majority of images were identified as good items, where the reliability was reduced with their removal (44 images) or stayed the same (5 images). Only a single image slightly reduced the Cronbach's $\propto$ (.002 change; Image 8). Omega also suggested good reliability of the images, Omega = .83. However, the split-half reliability was below the suggested threshold of 0.80 (recommended by DeVellis, 1991), $r(291) = .69$ $p < .001$, 95% CI [.62, .75].

Table 9

Descriptive statistics of the voluntary attention task for the full sample of Study 2 (N = 294).

Image	N	Mean	SD	Frequency of Other Clicks	Frequency of Nature Clicks	% of Nature Clicks	Z	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlation with NR
2	291	0.52	0.50	139	152	52.23	0.64	.828	-.12	.21	.07
4	292	0.23	0.42	225	67	22.95	-0.84	.827	-.11	.19	.07
7	293	0.16	0.37	246	47	16.04	-1.18	.827	-.05	.21	.10
8	293	0.74	0.44	77	216	73.72	1.73	.831	-.13	.15	.10
11	292	0.4	0.49	174	118	40.41	0.05	.826	-.03	.26	.02
12	292	0.49	0.50	149	143	48.97	0.48	.828	-.04	.23	.20***
22	293	0.32	0.47	199	94	32.08	-0.37	.827	-.08	.23	.06
23	292	0.76	0.43	71	221	75.68	1.82	.829	-.12	.18	.02
25	293	0.28	0.45	212	81	27.65	-0.60	.828	-.07	.21	.06
26	293	0.37	0.48	185	108	36.86	-0.13	.823	-.07	.26	.14*
28	292	0.57	0.50	127	165	56.51	0.86	.829	-.07	.18	.03
30	293	0.16	0.36	247	46	15.70	-1.20	.828	-.11	.19	.12*
35	293	0.20	0.40	233	60	20.48	-0.96	.829	-.15	.25	.09
41	293	0.36	0.48	187	106	36.18	-0.17	.828	-.09	.28	.05
42	293	0.52	0.50	142	151	51.54	0.61	.825	-.07	.28	.12*
43	292	0.29	0.46	206	86	29.45	-0.51	.826	-.05	.24	.19**
46	293	0.21	0.41	231	62	21.16	-0.93	.825	.01	.26	.02
47	293	0.50	0.50	147	146	49.83	0.52	.823	-.02	.29	.17**
52	293	0.34	0.47	194	99	33.79	-0.29	.826	-.12	.20	.06
53	293	0.19	0.40	236	57	19.45	-1.01	.825	-.05	.27	.13*
54	293	0.29	0.45	209	84	28.67	-0.55	.822	-.08	.38	.06
55	293	0.33	0.47	195	98	33.45	-0.31	.827	-.11	.20	.03
57	293	0.31	0.46	201	92	31.40	-0.41	.825	-.03	.31	.14*
58	293	0.57	0.50	126	167	57.00	0.88	.822	-.12	.53	.17**
59	293	0.58	0.49	123	170	58.02	0.93	.822	-.15	.53	.17**
60	293	0.42	0.49	169	124	42.32	0.14	.827	-.11	.27	.07

62	292	0.68	0.47	92	200	68.49	1.46	.827	-.04	.24	.07
63	291	0.33	0.47	196	95	32.65	-0.35	.825	-.07	.26	.03
64	293	0.14	0.34	253	40	13.65	-1.30	.828	-.14	.25	.06
67	292	0.28	0.45	211	81	27.74	-0.59	.825	-.01	.26	.06
70	293	0.10	0.30	263	30	10.24	-1.48	.825	-.03	.30	.10
71	293	0.59	0.49	120	173	59.04	0.99	.826	-.04	.22	.12*
73	292	0.18	0.39	238	54	18.49	-1.06	.826	-.13	.30	-.04
74	293	0.22	0.42	228	65	22.18	-0.87	.825	-.06	.29	.16**
75	291	0.24	0.43	221	70	24.05	-0.78	.825	-.09	.31	.08
77	292	0.33	0.47	196	96	32.88	-0.33	.823	-.11	.30	.06
79	291	0.41	0.49	173	118	40.55	0.05	.824	-.12	.38	.05
81	293	0.14	0.34	253	40	13.65	-1.30	.826	-.07	.27	.14*
82	293	0.40	0.49	176	117	39.93	0.02	.825	-.08	.28	.13*
83	293	0.93	0.26	21	272	92.83	2.69	.829	-.15	.20	.10
84	293	0.59	0.49	121	172	58.70	0.97	.824	-.06	1.00	.14*
85	293	0.88	0.32	34	259	88.40	2.47	.829	-.11	.20	.05
86	293	0.77	0.42	67	226	77.13	1.90	.826	-.14	.22	.06
87	293	0.52	0.50	141	152	51.88	0.62	.823	-.07	.28	.02
88	292	0.23	0.42	225	67	22.95	-0.84	.825	-.09	.34	.09
89	292	0.36	0.48	188	104	35.62	-0.20	.827	-.06	.24	.07
90	293	0.30	0.46	206	87	29.69	-0.50	.828	-.06	.22	-.07
91	292	0.34	0.47	194	98	33.56	-0.30	.826	-.08	.24	.09
92	293	0.34	0.47	193	100	34.13	-0.27	.824	-.04	.28	.05
93	293	0.35	0.48	189	104	35.49	-0.20	.822	-.10	.28	.11

Note: All images had a minimum of 0 and maximum 1; therefore, this information was redacted from the table. NR = Nature relatedness. * $p < .05$, ** $p < .01$, *** $p < .001$.

The standardized distribution of all 50 images demonstrates a non-normal distribution, both visually (see Figure 9) and quantitatively (Table 10). When running two robustness tests (1: removing images with 15% or less nature clicks[3]; 2: removing images with 20% or less nature clicks) the distribution becomes less normal. This can be seen visually in Figure 10 and statically in Table 10. Additionally, no outliers were identified. This recommends that removing any of the images does not improve the normality of the distribution; thus, suggests keeping the images.

Table 10
Robustness tests for normality of the voluntary attention task standardized distribution for Study 2.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*
None	50	N/A	0.93	.01
Images with 15% or less clicks on the nature area	47	64, 70, 81	0.92	.002
Images with 20% or less clicks on the nature area	43	7, 30, 53, 64, 70, 73, 81	0.90	.001

[3] The removal of images started at 15% or less nature clicks because all images showed variability and no images had less than 10% clicks on the nature area of the image.

Figure 9

Visual representation of the standardized distribution of all 50 images for the voluntary attention item in Study 2.

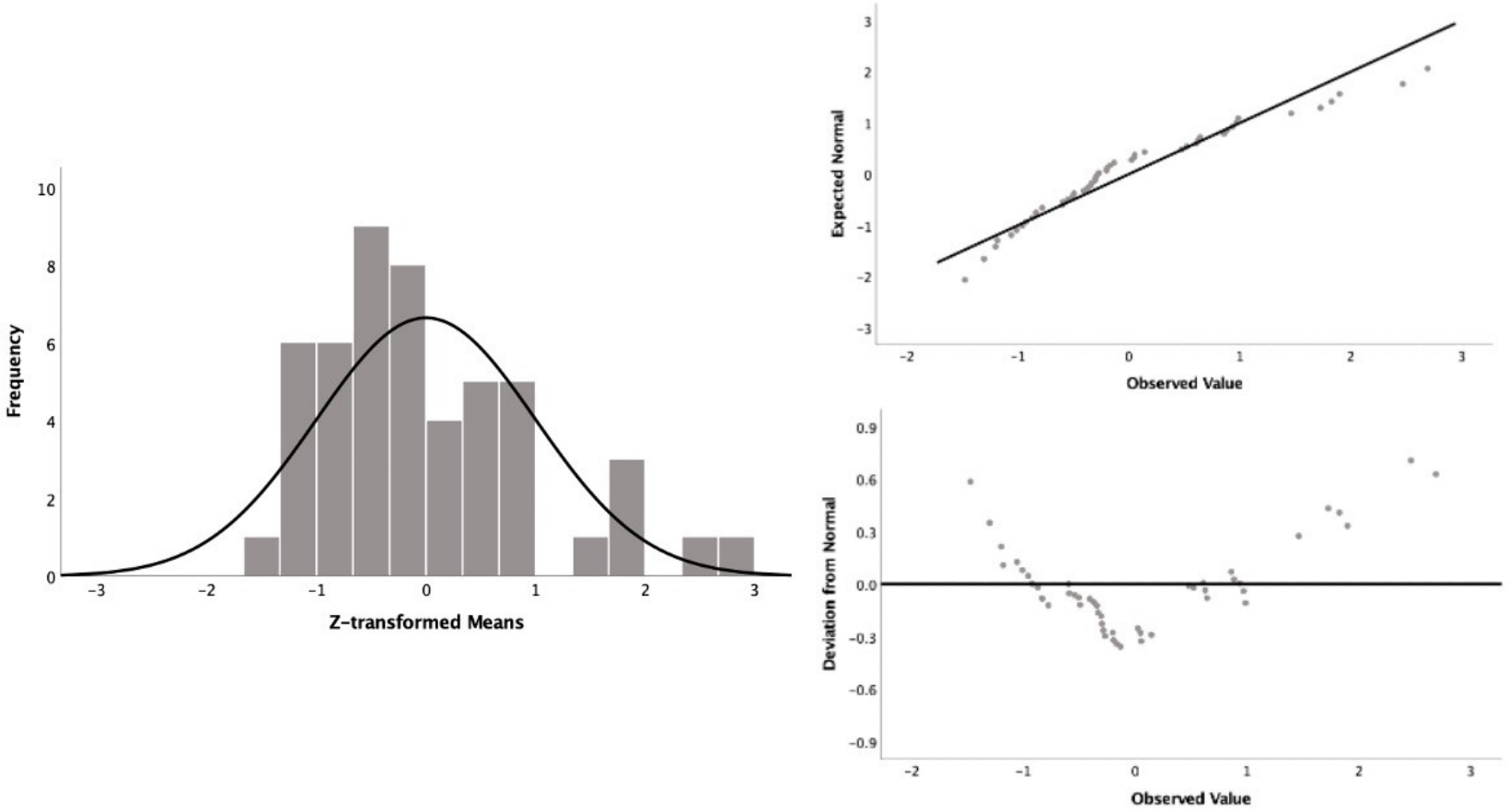
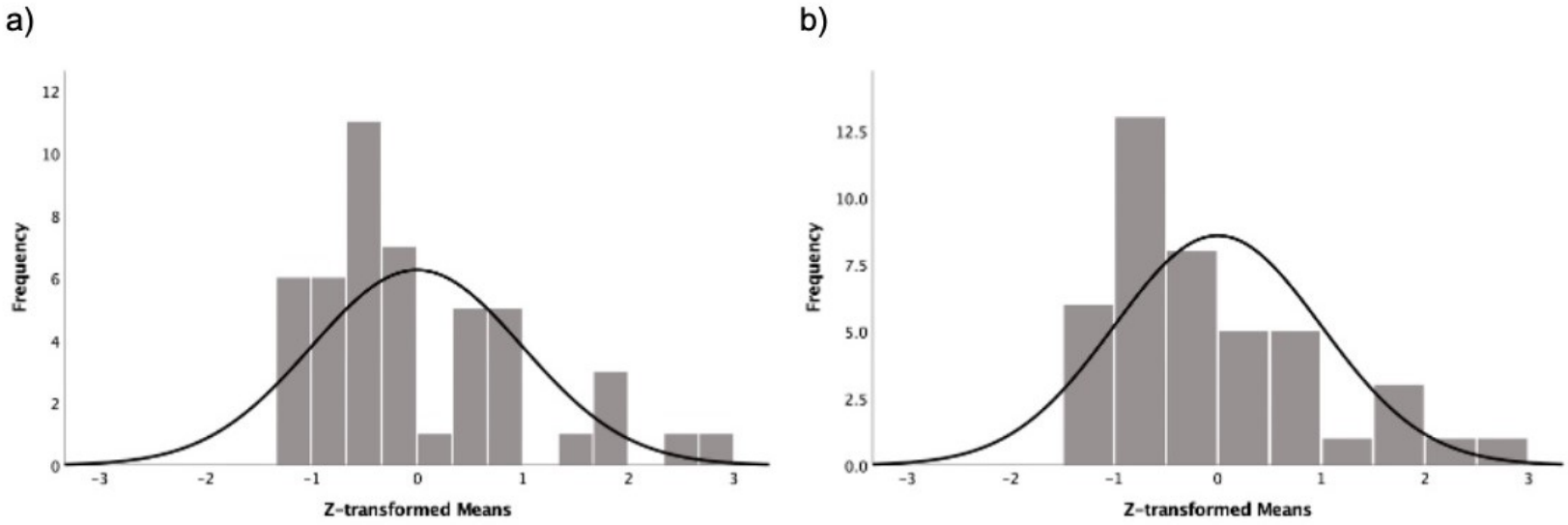


Figure 10

Visual representation of the standardize distribution of the voluntary attention for both robustness tests in Study 2: a) removing images with 15% or less nature click (47 images); b) removing images with 20% or less nature clicks (43 images).



A logistic multilevel unconditional logistic model was run to assess the amount of between-participants variability in the responses. The data was nested by participants. The ICC suggested that ~8.94% of the variance in voluntary attention response was due to the within-person effect. In other words, there was little variation in the responses across the images for a participant. This indicates a lot of between-subjects variability to explore the individual difference characteristic of nature relatedness. With that, the hypothesis tests can proceed.

Lastly, the correlation between the individual images and nature relatedness were examined. The correlations ranged from r = -.07 (Image 90) to r = .20 (Image 12), with an average correlation of z = .08 (z-transformed correlations). The majority of images showed a positive correlation (all but two images: 73 & 90). Of the 48 images with positive correlations, 15 were significant. This recommends the removal of the two images with negative correlations.

Statistical Analysis Results: Correlations with Nature Attention Items

The pre-registered data analysis plan to test the hypotheses was to examine bivariate correlational analyses between the aggregates of the 50 images for each attention item and nature relatedness; however, similar to Study 1, the psychometric analysis results may suggest better images to aggregate. Thus, the robustness of the correlations between the two attention items and nature relatedness was explored. There were four different combinations of image sets computed and examined. For each set, there was a separate set created for each item. The first set was the complete set of 50 images (Table 11i). The second sets were created with only the images identified as suitable for distribution (Table 11ii). The third sets were created with only the images identified as suitable for reliability (Table 11iii). Lastly, the fourth sets were created with only images positively correlated with nature relatedness (Table 11iv). A description of the different sets can be seen in Table 11.

Table 11
Description of the different sets of images for the first attention and voluntary attention tasks for Study 2.

		First Attention		Voluntary Attention	
Image set	Description	Number of Images	Images removed	Number of Images	Images removed
i	Overall aggregate of all 50 images	50	N/A	50	N/A
ii	Aggregate of images identified as good for distribution	47	23, 83, 85	-	-
iii	Aggregate of images identified as good for reliability	43	2, 46, 52, 59, 64, 74, 88	49	8
iv	Aggregate of images positively correlated with NR	40	22, 25, 42, 46, 55, 73, 77, 79, 86, 88	48	73 & 90

Note: For voluntary attention, there were no images identified through psychometric analyses to remove for the distribution criteria. Thus, this image set were disregarded for voluntary attention task.

The different aggregate sets were then correlated with nature relatedness. Ninety-five percent confidence intervals were run for these correlations with 1000 sample Bootstrapping. All statistical analysis results can be seen in Table 12. The correlations between nature relatedness and both nature attention tasks were significant. This pattern was also seen across the other aggregated images (i.e. the correlation was robust across different sets of images). Thus, hypotheses 2 and 3 were supported, suggesting that nature related individuals are more likely to notice nature first and want to focus their attention on nature in an urban scene image.

More exploratory correlations were also explored, in addition to the main correlations between nature relatedness and each attention item. The exploratory correlations were between the two nature attention items with affect (positive, negative, and awe), noticing nature activity, the nature relatedness subscales and the two individual items that were face valid to my research

hypothesis (item 8: “Even in the middle of the city, I notice nature around me” and item 21: “I take notice of wildlife wherever I am”).

The association of nature relatedness with the two attention items was the strongest and most robust for the Self subscale of the NRS. The significant correlations of voluntary attention with both Perspective and Experience subscales of NRS were robust across the different sets. The Experience subscale was consistently not significantly associated with first attention across the different sets, whereas there were more mixed results for the correlation of the Perspective subscale and first attention. It (the Perspective subscale) was significant at the overall level (Table 12i) and with the images positively correlated to nature relatedness (Table 12iv); however, it was not significant for the other two image sets: images that fit a normal distribution (Table 12ii) and images deemed good for reliability (Table 12iii).

For the face-valid noticing nature items and measures (NRS Items 8 & 21; NNA), there was a consistent positive correlation between first attention and voluntary attention with NNA across all sets of images. Less consistency was seen for the two NRS items. Both NRS items were positively correlated with voluntary attention. In contrast, item 8 was correlated with the first attention task at the overall level (Table 12i) and for image sets with only positively correlated to nature relatedness images (Table 12iv), whereas item 21 was only correlated with the last set (Table 12iv).

Positive affect was not significantly associated with either attention item across all sets. There was a consistent significant positive association with negative affect for the first attention item; the affect associations with voluntary attention were all consistently non-significant.

Overall, there were few differences across the different sets of images, indicating that the other scoring sets do not improve criterion correlations. Correlations between all other variables than the attention items can be seen in Appendix F.

Table 12

Correlations with the attention tasks for Phase 2 Study 2 (with different image sets).

Variables	First Attention				Voluntary Attention			
(i) Overall Aggregate	*n*	*r*	*p*	CI	*n*	*r*	*p*	95% CI
NR	292	.18**	.002	.05, .30	292	.25***	< .001	.13, .37
PA	292	-.09	.12	-.22, .04	292	-.07	.25	-.20, .06
NA	292	.13*	.02	.02, .24	292	.01	.88	-.10, .12
Awe	292	.01	.84	-.12, .13	292	.05	.45	-.09, .18
NNA	292	.14*	.02	.02, .23	292	.16**	.01	.02, .28
NR Item 8	292	.12*	.05	.003, .23	292	.14*	.02	.02, .26
NR item 21	292	.10	.10	-.02, .20	292	.13*	.03	.01, .24
NR Experience	292	.07	.27	-.06, .19	292	.16*	.01	.04, .28
NR Self	292	.21***	< .001	.07, .32	292	.25***	< .001	.13, .36
NR Perspectives	292	.12*	.04	.004, .23	292	.12*	.04	.01, .24
(ii) Only Images Identified as Good for Distribution	*n*	*R*	*p*	CI				
NR	292	.16*	.01	.02, .29				
PA	292	-.09	.12	-.21, .04				
NA	292	.14*	.02	.02, .25				
Awe	292	.02	.77	-.11, .14				
NNA	292	.12*	.04	.004, .22				
NR Item 8	292	.11	.07	-.01, .22				
NR item 21	292	.09	.14	-.03, .19				
NR Experience	292	.06	.30	-.06, .18				
NR Self	292	.19**	.001	.06, .31				
NR Perspectives	292	.10	.10	-.03, .20				
(iii) Only Images Identified as Good for Reliability	*N*	*R*	*p*	CI	*N*	*r*	*p*	95% CI
NR	292	.17**	.004	.03, .29	292	.24***	<.001	.12, .36
PA	292	-.10	.08	-.22, .02	292	-.07	.24	.-.20, .06
NA	292	.12*	.03	.01, .24	292	.01	.81	-.10, .13
Awe	292	.01	.93	-.12, .13	292	.05	.43	-.09, .17
NNA	292	.13*	.02	.02, .23	292	.16**	.01	.02, .28
NR Item 8	292	.09	.12	-.03, .21	292	.14*	.02	.01, .26
NR item 21	292	.10	.10	-.02, .21	292	.13*	.03	.00, .23

NR Experience	292	.05	.37	-.08, .18	292	.16**	.01	.03, .27
NR Self	292	.20***	< .001	.07, .32	292	.25***	<.001	.13, .36
NR Perspectives	292	.10	.08	-.02, .21	292	.12*	.05	.005, .23
(iv) Only Images Positively Correlated with NR	*N*	*R*	*p*	CI	*N*	*r*	*p*	95% CI
NR	292	.24***	< .001	.11, .36	292	.26***	< .001	.14, .38
PA	292	-.08	.17	-.21, .05	292	-.06	.28	-.20, .06
NA	292	.14*	.02	.03, .26	292	.00	.99	-.11, .11
Awe	292	.03	.57	-.10, .16	292	.04	.49	-.09, .17
NNA	292	.17**	.004	.05, .27	292	.17**	.004	.03, .29
NR Item 8	292	.15*	.01	.04, .26	292	.15**	.01	.03, .27
NR item 21	292	.14*	.02	.03, .25	292	.14*	.02	.02, .25
NR Experience	292	.10	.08	-.02, .23	292	.17**	.003	.05, .29
NR Self	292	.24***	< .001	.11, .35	292	.26***	< .001	.14, .37
NR Perspectives	292	.18**	.003	.06, .28	292	.13*	.03	.02, .24

Note: NR – Nature relatedness. PA – Positive Affect. NA – Negative Affect. NNA – Noticing Nature Activity. NR item 8: "Even in the middle of the city, I notice nature around me". NR Item 21: "I take notice of wildlife wherever I am". * $p < .05$, ** $p < .01$, *** $p < .001$.

Discussion

In this phase of the dissertation, I explored if nature related individuals would notice more nature through the process of attention. Participants were asked to click on the first thing they noticed in the image (first attention) and what in the image they would want to focus their attention on (voluntary attention) for 50 images of urban environments with varying degrees of nature. Given the tool-building nature of the investigation, psychometric analyses were conducted to examine the extent of variability among the images. This step was deemed more difficult to accomplish than for Study 1. Three optional stopping rounds were conducted with 100 participants each before finding 50 images that demonstrated enough variability through descriptive statistics, a standardized distribution of the images, reliability analyses (Cronbach's alpha, Omega, and split-half reliability), and inter-image correlations. This variability was also considered acceptable with the full sample of 294 participants. Additionally, unconditional logistic multilevel models identified that there was enough between-people variability in both attention items to explore the association with nature relatedness. The overall results showed a positive association between both attention items and nature relatedness; these relationships were robust across the different sets of images. This suggests that, with my current tools, participants higher in nature relatedness were more likely to notice nature first and also choose to focus their attention on it in an urban environment. Therefore, the results did support *H2* and *H3*.

Be that as it may, a single study showing small associations between noticing nature through attention and nature relatedness may stem from a sampling error. The correlations of both first attention and voluntary attention with nature relatedness are less than modest in size. These correlations may be due to the sample of students and may not be generalizable to a larger population. Even though an appropriate sample size allowed for the detection of a small

correlation (r = .12), the sampling size of approximately 300 participants may also increase sampling error and reduce the correlation's accuracy. To strive for more reliability in the robustness of the correlations and validity of the measurement tool, running a replication study with a larger and different sample is important. Thus, to further explore the significant associations, a replication study was run to try to confirm the results from this study. This allows consideration of the consistency of the results in a different sample of community members in Canada and the United States.

Study 3

The pre-registration for all hypotheses and research questions of this replication study can be found on Aspredicted https://aspredicted.org/1TV_XRZ .

Methods

Participants. To justify the sample of Study 3, using the information from Schönbrodt and Perugini's (2013) simulation, a sample of at least 470 is needed for the most stable correlation (smallest width of stability) for a small effect size of .10 with 95% confidence (this is higher confidence than for Study 2). A sensitivity analysis was run with G*Power to explore the smallest effect size that can be detected with this sample size. With an alpha set to .05, power set to 90%, and a sample size of 470, an effect size of .09, regardless of direction, can be detected.

Differently from Study 2, participants were only exposed to one of the two attention tasks. This was chosen to shorten the participation time for each participant, and it helps ensure that the results for voluntary attention do not depend on the initial task. Because of this, I needed to double my sample size to ensure an appropriate sample size was used to explore the association of each attention item with nature relatedness. Therefore, the target sample size was

1100 (i.e. a sample of 550 participants for each attention task). This sample size will also allow for the deletion of some poor data.

A sample of 1128 participants was recruited. As pre-registered, participants were excluded if they: 1) did not complete the study survey (n = 63), 2) were identified as an outlier on the NRS (n = 5), 3) demonstrated string responding on the NRS items (n = 3), and/or 4) responded that they could not commit quality to their responses (n = 6). A total of 76 participants were removed[4]. Of the final sample of 1052, 544 identified as a man (52%), 489 as a woman (47%), eight as gender-fluid/non-binary/two-spirited (<1%), two as a trans man (<1%), two as a trans woman (<1%), two self-described (<1%), and five did not want to identify (<1%). The ages of the participants ranged from 19 to 85 (M = 42.33, SD = 12.16). The majority of participants identified as White (80%). A smaller percentage identified their ethnicity as Black (11%), Latin American (5%), Chinese (2%), Filipino (1%), Korean (1%), Indigenous (1%), South Asian (1%), and other (1%). All other ethnicity categories were reported less than 1% (Arab, Southeast Asian, West Asian, Japanese, and prefer not to say). The bulk of the participants reported growing up and currently living in a city (see Table 13). Approximately 95% of the participants completed the survey on a computer (i.e. desktop and laptop).

Table 13

Frequency of where Study 3 participants identified growing up and currently living.

Area	Grew Up	Currently Living
City (Downtown)	164	199
City (Suburbs)	481	489
Exurban area	53	66
Small Town	230	191
Rural or Farm	113	98
Other	8	6
Missing	3	3

[4] The results of Study 3 are unchanged with or without outliers and string responders on the NRS. This can be seen in the Supplement on OSF.

Procedure. The study was run on May 11th from 10:03 AM to 12:44 PM. After consenting to participate in the study, participants were randomly assigned to either the first attention (n = 528) or the voluntary attention task (n = 524). A practice image was given for each attention task before beginning. The 50 images identified as having appropriate variability in Study 2 were used. After completing the given attention task, participants filled out the NRS (Nisbet et al., 2009; $\propto = .89$), NNA (Richardson et al., 2021; $\propto = .91$), SPANE (Dierner et al., 2010; PA: $\propto = .94$; NA: $\propto = .93$; Awe: $\propto = .93$) measures, and answered demographic questionnaire, similar to Study 2. No additional items were added to the study.

Psychometric Analyses Results

First Attention. All quantitative results for the psychometric analyses for the first attention task can be seen in Table 14. The average clicks on the nature area for the first attention item across all 50 images was ~27% (SD = 19.12%). This percentage ranged from 2% (Image 70) to 85% (Image 83). All images had variability (i.e. no single image had all nature or other area clicks). The inter-image correlations ranged from r = -.25 to r = .46. All images had a minimum of one negative inter-image correlation.

Table 14

Descriptive statistics of the first attention task for the full sample of Study 3 (n = 528).

Image	N	*M*	*SD*	Frequency of Other Clicks	Frequency of Nature Clicks	% of Nature Clicks	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlation with NR
2	515	0.34	0.47	341	174	33.79	0.39	.781	-.06	.17	.02
4	519	0.18	0.38	426	93	17.92	-0.45	.779	-.06	.19	.08
7	520	0.24	0.43	396	124	23.85	-0.13	.778	-.07	.29	.09*
8	520	0.64	0.48	188	332	63.85	1.96	.785	-.23	.30	.05
11	522	0.43	0.50	298	224	42.91	0.86	.781	-.13	.17	.07
12	520	0.36	0.48	332	188	36.15	0.50	.780	-.13	.30	.04
22	519	0.15	0.35	443	76	14.64	-0.60	.779	-.11	.22	.12**
23	519	0.82	0.38	91	428	82.47	2.90	.784	-.25	.18	.08
25	517	0.13	0.34	449	68	13.15	-0.71	.777	-.07	.26	-.03
26	520	0.16	0.37	436	84	16.15	-0.55	.772	-.02	.30	.08
28	519	0.56	0.50	230	289	55.68	1.54	.781	-.16	.19	-.01
30	516	0.14	0.34	445	71	13.76	-0.65	.776	-.05	.27	.19*
35	514	0.38	0.49	317	197	38.33	0.60	.779	-.11	.23	.03
41	520	0.14	0.35	448	72	13.85	-0.65	.780	-.13	.19	.02
42	516	0.25	0.44	385	131	25.39	-0.08	.774	-.03	.24	.05
43	520	0.25	0.43	391	129	24.81	-0.08	.781	-.13	.21	.04
46	518	0.23	0.42	397	121	23.36	-0.18	.779	-.07	.19	-.04
47	518	0.25	0.44	387	131	25.29	-0.08	.774	-.04	.31	.05
52	513	0.27	0.45	373	140	27.29	0.03	.778	-.07	.23	.08
53	517	0.03	0.16	503	14	2.71	-1.23	.778	-.04	.46	.10*
54	520	0.13	0.34	451	69	13.27	-0.71	.776	-.19	.34	-.02
55	516	0.13	0.34	449	67	12.98	-0.71	.777	-.08	.24	.03
57	520	0.19	0.39	420	100	19.23	-0.39	.776	-.03	.25	.06
58	519	0.19	0.39	423	96	18.50	-0.39	.778	-.23	.36	.02
59	522	0.07	0.25	488	34	6.51	-1.02	.778	-.25	.36	-.002
60	519	0.12	0.33	456	63	12.14	-0.76	.779	-.16	.31	-.03
62	520	0.40	0.49	313	207	39.81	0.71	.778	-.08	.24	.08
63	520	0.19	0.39	420	100	19.23	-0.39	.775	-.09	.29	-.004

64	521	0.39	0.49	316	205	39.35	0.65	.782	-.17	.24	.05
67	522	0.28	0.45	376	146	27.97	0.08	.776	-.07	.24	.08
70	521	0.02	0.14	510	11	2.11	-1.28	.777	-.01	.46	.05
71	520	0.32	0.47	356	164	31.54	0.29	.774	-.03	.28	.14**
73	519	0.14	0.35	445	74	14.26	-0.65	.776	-.05	.30	.13**
74	523	0.10	0.29	473	50	9.56	-0.86	.780	-.13	.30	-.01
75	516	0.27	0.44	379	137	26.55	0.03	.780	-.10	.24	.11*
77	522	0.15	0.35	445	77	14.75	-0.60	.774	-.03	.27	.12**
79	521	0.14	0.34	450	71	13.63	-0.65	.779	-.17	.26	-.05
81	516	0.14	0.35	442	74	14.34	-0.65	.776	-.07	.26	.03
82	515	0.15	0.35	440	75	14.56	-0.60	.779	-.05	.23	.10*
83	514	0.85	0.36	76	438	85.21	3.06	.780	-.13	.24	.02
84	519	0.20	0.40	416	103	19.85	-0.34	.776	-.13	.32	.01
85	522	0.84	0.36	81	441	84.48	3.01	.779	-.10	.24	.06
86	519	0.40	0.49	309	210	40.46	0.71	.775	-.05	.25	.03
87	520	0.24	0.43	396	124	23.85	-0.13	.774	-.09	.32	.02
88	518	0.15	0.36	441	77	14.86	-0.60	.777	-.10	.27	.03
89	518	0.30	0.46	363	155	29.92	0.18	.774	-.10	.28	-.02
90	517	0.31	0.46	355	162	31.33	0.23	.782	-.10	.14	.03
91	517	0.16	0.37	433	84	16.25	-0.55	.773	-.05	.32	.09
92	517	0.10	0.30	467	50	9.67	-0.86	.775	-.04	.27	.11*
93	516	0.24	0.43	392	124	24.03	-0.13	.777	-.06	.25	.06

Note: All images had a minimum of 0 and maximum 1; therefore, this information was redacted from the table. NR = Nature relatedness. * $p < .05$, ** $p < .01$, *** $p < .001$.

Overall, the first attention task showed good reliability across all 50 images, Cronbach's $\propto$ = .78. Looking at the robustness test for reliability, the majority of images were identified as good (i.e. Cronbach's $\propto$ was reduced if removed; 42 images), whereas four showed no difference (i.e. Cronbach's $\propto$ as unchanged; Images 2, 11, 28, & 43), and four were identified as bad (i.e. Cronbach's $\propto$ were increased if removed; Images 8, 23, 64, & 90). However, the changes in Cronbach's $\propto$ were minimal, with a maximum increase of .004 or a decrease of .009. Additionally, the reliability was identified as good through Omega, Omega = .77, but still below the threshold of 0.80 for split-half reliability, $r(528) = .64$ $p < .001$ 95% CI [.56, .71].

When looking at the standardized distribution of means for all 50 images, the distribution was positively skewed (see Figure 11) and significantly not normal (Table 15). Four robustness tests were run to see if removing images with lower percentage clicks on the nature areas would make the distribution normal: 1) removing images with 5% or less nature clicks; 2) removing images with 10% or less nature clicks; 3) removing images with 15% or less nature clicks; 4) removing images with 20% or less nature clicks. As seen in Table 15, the distribution does not become normal after removing images with up to less than 20% nature clicks. Outliers (z-score of +3) were identified and thus, robustness tests were re-run without outliers. The distribution becomes normal (i.e. Shapiro-Wilkes estimate is reduced to non-significant) after removing the five images with high outlier means (Z-transformed means[5]; see Table 15). This normal distribution can be seen in Figure 12.

[5] In the initial standardization of the distribution (i.e. transforming the means into Z-scores), only Images 83 and 85 were identified as outliers (as seen in Table 12); however, after removing those two images, other images were identified as outliers (Images 8, 23, and 28). The distribution became quantitively normal after the removal of these five images and no other outliers were identified.

Figure 11

Visual representation of the standardized distribution of all 50 images for the first attention item for Study 3.

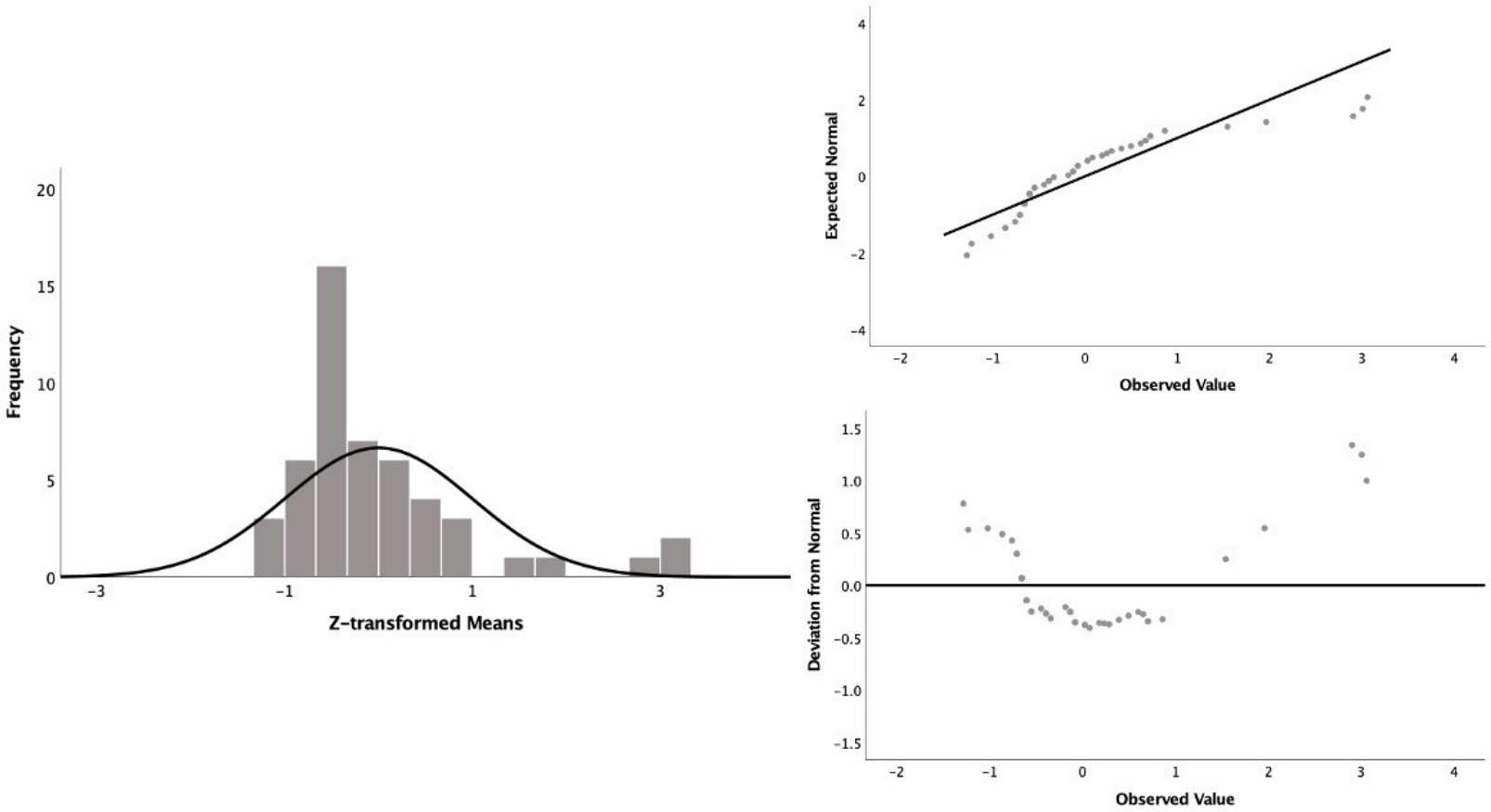


Table 15

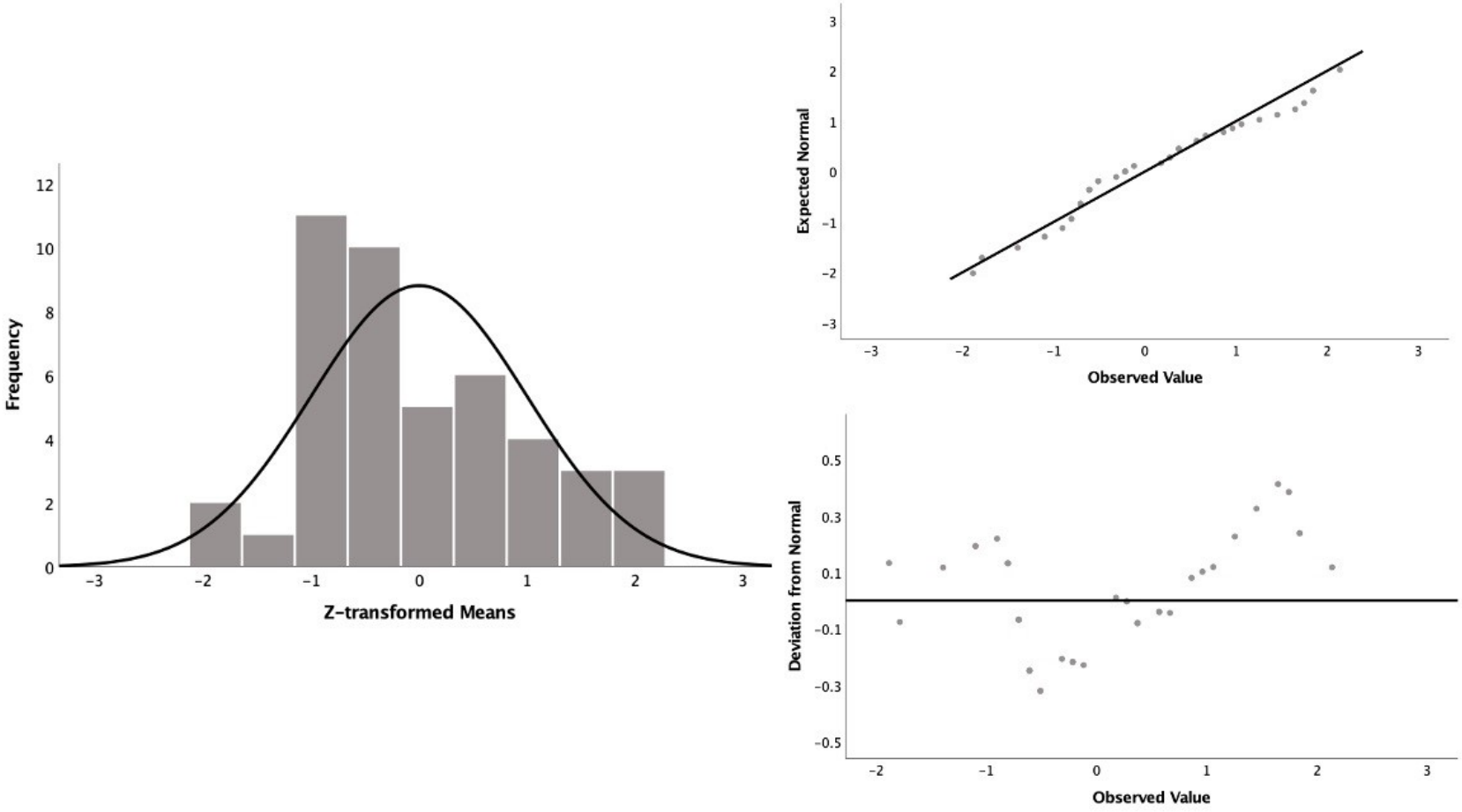
Robustness test for normality of the first attention task standardized distribution for Study 3.

	Low Percentage Image Removed				Outlier (3 SD) Images Removed			
Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	Number of images	Images removed	Shapiro-Wilks Estimate	*p*
None	50	N/A	0.81	< .001	45	8[a], 23[a], 28[a], 83, 85	0.96	.09
Images with 5% or less clicks on the nature area	48	53, 70	0.78	< .001				
Images with 10% or less clicks on the nature area	45	53, 59, 70, 74, 92	0.77	< .001				
Images with 15% or less clicks on the nature area	33	25, 30, 41, 53, 54, 55, 59, 60, 70, 72, 73, 74, 77, 79, 81, 88, 92	0.78	< .001				
Images with 20% or less clicks on the nature area	24	4, 22, 25, 26, 30, 41, 53, 54, 55, 57, 58, 59, 60, 63, 70, 72, 73, 74, 77, 79, 81, 82, 84, 88, 91, 92	0.76	< .001				

Note: Images were first removed for low percentages and then removed as outliers. [a] These images were not initially identified as outliers, but after the removal of the other images became outliers.

Figure 12

Visual representation of the standardized distribution of the z-transformed means for the first attention item in Study 3 after the removal of the five images identified as outliers (z-score > 3).



A logistic multilevel unconditional model was run with the data nested within participants to assess the amount of between-participants variability in the first attention responses. The ICC suggested that 6.92% of the variance in the first attention responses was due to within-person effect. This suggested that there is enough between-participants variability to explore an individual difference characteristic.

Lastly, the correlation of each image of the first attention item and the participant's level of nature relatedness was examined to identify the best set of images to aggregate for the correlation. The correlations with nature relatedness ranged from r = -.05 (Image 79) to r = .14 (Image 71), with an average of z = .05 (z-transformed correlations). Eighty percent of the images had positive correlations with nature relatedness, with 10 of the 40 images having a significant correlation (Images 7, 22, 30, 53, 71, 73, 75, 77, 82, & 92). The other 10 images had a negative correlation (Images 25, 28, 46, 54, 59, 60, 63, 74, 79, & 89). None of the negative correlations were significant.

Voluntary Attention. All quantitative results for the psychometric analyses on voluntary attention item can be seen in Table 16. On average, participants clicked the nature area ~29% of the time (*SD* = 18%). This percentage of nature clicks ranged from 6% (Image 70) to 82% (Image 83). The inter-image correlations ranged from r = -.15 to r = .45. Thirty-nine images had negative correlations with other images (i.e. negative inter-image correlations).

Table 16

Descriptive statistics of the voluntary attention task for the full sample of Study 3 (n = 524).

Image	N	*M*	*SD*	Frequency of Other Clicks	Frequency of Nature Clicks	% of Nature Clicks	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlation with NR
2	505	0.37	0.48	319	186	36.83	0.42	.879	-.10	.28	.11*
4	511	0.16	0.37	428	83	16.24	-0.72	.878	-.01	.28	.002
7	509	0.16	0.37	426	83	16.31	-0.72	.877	-.03	.33	.09*
8	507	0.76	0.43	120	387	76.33	2.54	.882	-.15	.17	.07
11	510	0.25	0.43	383	127	24.90	-0.23	.879	-.04	.29	.01
12	512	0.50	0.50	257	255	49.80	1.13	.879	-.03	.24	.01
22	509	0.32	0.47	346	163	32.02	0.15	.879	-.03	.25	.15***
23	505	0.68	0.47	161	344	68.12	2.10	.881	-.10	.24	.04
25	505	0.20	0.40	404	101	20.00	-0.50	.878	-.03	.25	.12**
26	507	0.22	0.42	394	113	22.29	-0.39	.877	-.01	.31	.12**
28	506	0.52	0.50	241	265	52.37	1.24	.880	-.01	.21	.02
30	505	0.15	0.35	431	74	14.65	-0.77	.878	-.01	.30	-.02
35	507	0.18	0.38	417	90	17.75	-0.61	.879	-.08	.31	.07
41	505	0.22	0.41	394	111	21.98	-0.39	.881	-.07	.18	.03
42	507	0.40	0.49	306	201	39.64	0.58	.875	.07	.33	.15***
43	508	0.27	0.44	371	137	26.97	-0.12	.878	-.02	.23	.15***
46	510	0.15	0.36	434	76	14.90	-0.77	.879	.00	.21	.04
47	510	0.28	0.45	366	144	28.24	-0.07	.876	.00	.32	.12**
52	511	0.20	0.40	409	102	19.96	-0.50	.877	-.05	.33	.12**
53	511	0.08	0.27	470	41	8.02	-1.15	.877	.03	.29	.08
54	505	0.14	0.35	432	73	14.46	-0.83	.879	-.15	.31	.05
55	509	0.22	0.41	399	110	21.61	-0.39	.879	-.07	.24	.04
57	509	0.18	0.38	418	91	17.88	-0.61	.878	-.02	.28	.08
58	510	0.39	0.49	310	200	39.22	0.53	.878	-.11	.48	.14**
59	508	0.26	0.44	378	130	25.59	-0.18	.877	-.08	.48	.12**
60	506	0.25	0.43	382	124	24.51	-0.23	.881	-.10	.23	.09*
62	507	0.44	0.50	283	224	44.18	0.80	.879	.01	.22	.11*
63	508	0.22	0.42	395	113	22.24	-0.39	.876	-.02	.33	.06

64	510	0.15	0.36	431	79	15.49	-0.77	.879	-.02	.23	.06
67	509	0.25	0.43	384	125	24.56	-0.23	.875	.03	.33	.14**
70	509	0.06	0.24	478	31	6.09	-1.26	.878	-.04	.31	.04
71	507	0.46	0.50	274	233	45.96	0.91	.877	-.04	.29	.12**
73	508	0.15	0.36	432	76	14.96	-0.77	.877	-.03	.30	-.01
74	514	0.12	0.32	454	60	11.67	-0.94	.878	-.14	.40	.05
75	507	0.23	0.42	392	115	22.68	-0.34	.877	.03	.28	.08
77	512	0.23	0.42	396	116	22.66	-0.34	.878	-.03	.32	.12**
79	509	0.23	0.42	392	117	22.99	-0.34	.878	-.10	.35	.05
81	511	0.10	0.30	459	52	10.18	-1.04	.877	-.01	.29	-.02
82	511	0.31	0.46	351	160	31.31	0.10	.879	.01	.22	.07
83	507	0.82	0.38	89	418	82.45	2.86	.880	-.10	.30	-.003
84	512	0.39	0.49	312	200	39.06	0.53	.876	.02	.32	.12**
85	505	0.79	0.41	107	398	78.81	2.70	.879	-.03	.30	.07
86	512	0.66	0.47	175	337	65.82	2.00	.877	.00	.32	.11*
87	508	0.33	0.47	339	169	33.27	0.20	.877	-.05	.30	.10*
88	508	0.13	0.34	441	67	13.19	-0.88	.877	-.12	.40	.04
89	509	0.21	0.41	400	109	21.41	-0.45	.877	.00	.33	.06
90	510	0.22	0.42	396	114	22.35	-0.39	.878	.00	.22	.002
91	511	0.21	0.41	404	107	20.94	-0.45	.876	-.09	.33	.08
92	507	0.20	0.40	407	100	19.72	-0.50	.876	-.01	.30	.16***
93	509	0.20	0.40	406	103	20.24	-0.50	.876	-.01	.33	.08

Note: All images had a minimum of 0 and maximum 1; therefore, this information was redacted from the table. NR = Nature relatedness. * $p < .05$, ** $p < .01$, *** $p < .001$.

The 50 images showed appropriate internal consistency, Cronbach's $\propto$ = .88. When running robustness tests, there were 46 images identified as good (i.e. 44 that decreased $\propto$ when removed and two showed no change in $\propto$) and four images identified as poor (maximum increase in $\propto$ of .002 when removed; Images 8, 23, 41, and 60). The reliability of the images identified through Omega and split-half reliability was also good, Omega = .88, split-half $r(515) = .78$ $p <$ .001 95% CI [.73, .83] (close to the threshold of 0.80 as recommended by DeVellis, 1991).

The standardized distribution of the Z-transformed means for all 50 images was not normal (see Figure 13; Table 17). Robustness tests were run to see if removing images with low percentage clicks on the nature area would improve the standardized distribution. This was done in steps: 1) removing images with 10% or less clicks on the nature areas of the image[6]; 2) removing images with 15% or less nature clicks; and 3) removing images with less than 20% nature clicks. The distribution did not improve during the robustness tests (see Table 17 and Figure 14). Additionally, no images were identified as outliers (Z-transformed means of ± 3)[7]. Removing images did not improve the distribution; thus, the psychometric analysis results for this criteria would suggest keeping all 50 images.

[6] The removal of images started at 10% or less nature clicks because all images showed variability and no images had less than 5% clicks on the nature area of the image.

[7] As an extra sensitivity analyses, I also explored how removing approximate outliers (Z-transformed means above 2.70) would affect the distribution for each robustness test. Removing approximate outliers did not improve the distribution.

Figure 13

Visual representation of the standardized distribution of all 50 images for the voluntary attention item for Study 3.

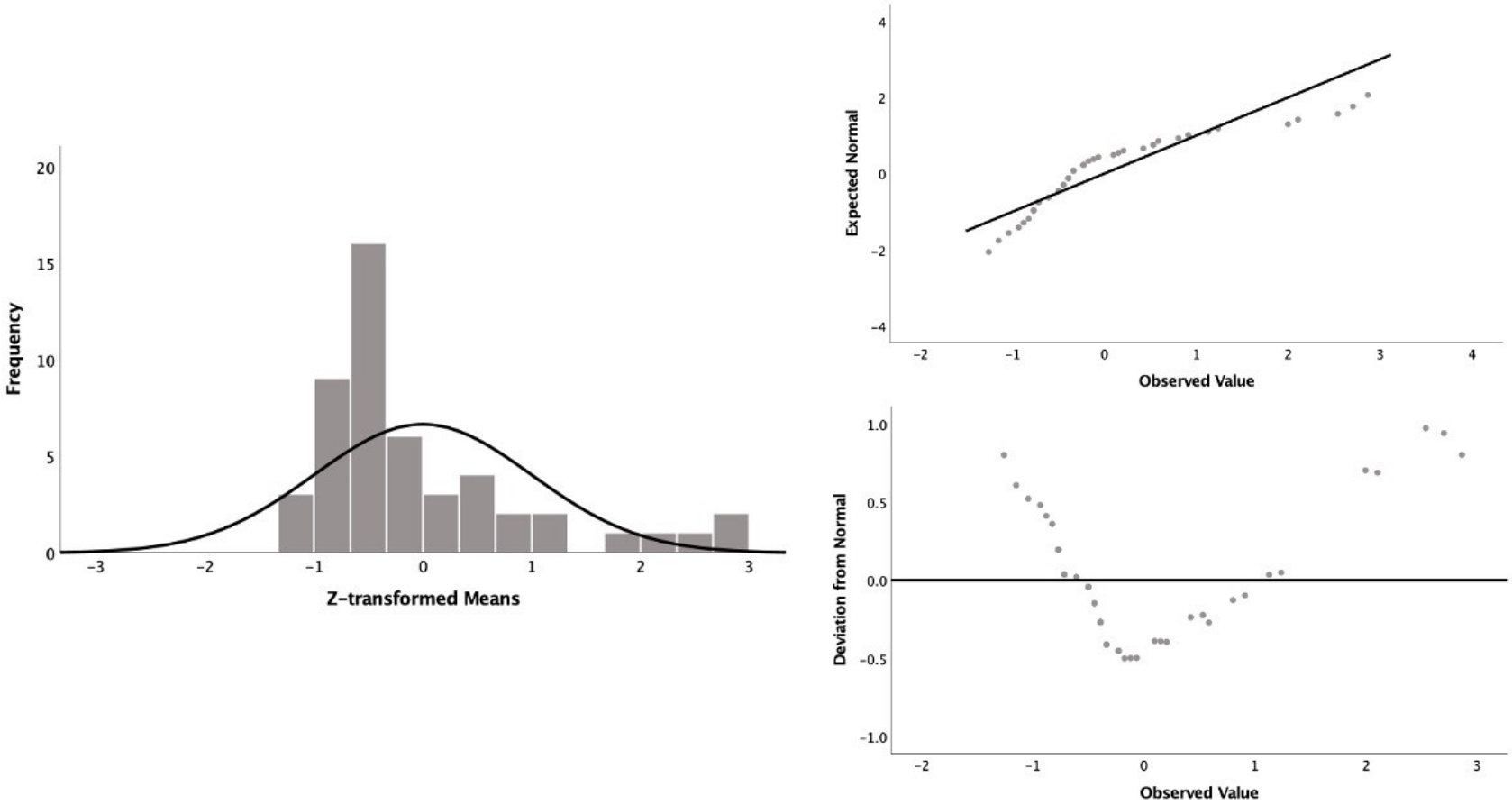


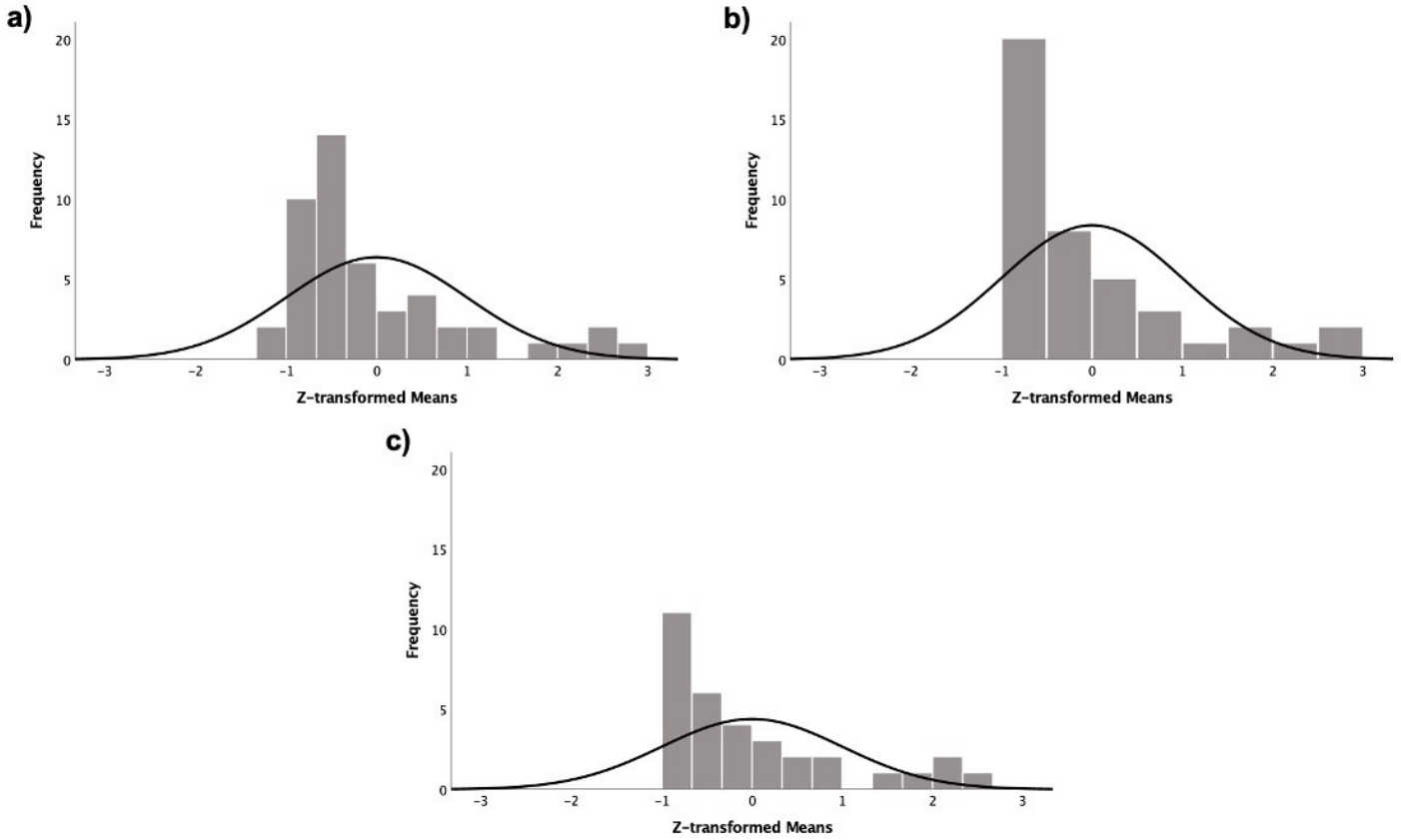
Table 17

Robustness test for normality of the voluntary attention task standardized distribution for Study 3.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*
None	50	N/A	0.83	< .001
Images with 10% or less clicks on the nature area	48	53, 70	0.81	< .001
Images with 15% or less clicks on the nature area	41	30, 46, 53, 54, 70, 73, 74, 81, 88	0.80	< .001
Images with 20% or less clicks on the nature area	33	4, 7, 25, 30, 35, 46, 52, 53, 54, 57, 64, 70, 73, 74, 81, 88, 92	0.80	< .001

Figure 14

Visual representation of the voluntary attention standardized distribution for the three robustness tests of Study 3: a) removing images with 10% or less nature click (48 images); b) removing images with 15% or less nature clicks (42 images); c) removing images with 20% or less nature clicks (33 images).



A logistic unconditional multilevel model was run to assess the amount of between-participants variability in the voluntary attention scores. The multilevel model had the dichotomous responses on the voluntary attention (0 – other, 1 – Nature) for each image nested within participants. The ICC indicated that ~14.08% of the variability in the voluntary attention score was due to within-person variability. This provides enough between-participants variability to assess links with the individual difference of nature relatedness.

Lastly, when exploring the correlation between the individual images and nature relatedness, 92% of the images had positive correlations. Of the 46 images with positive correlations, 20 were significant. The other four correlations were negative, but non-significant (Images 30, 73, 81, & 83). The correlations ranged from r = -.02 (Images 30 & 81) to r = .16 (Image 92), with an average of z = .07 (z-transformed correlations).

Statistical Analysis Results: Correlations with Nature Attention Items

Similar to Study 2, the proposed bivariate correlations between both attention items and nature relatedness were explored with the aggregated attention items across all 50 images, in addition to different sets of images identified as better suited for aggregation from the psychometric analyses. Due to a different number of images being identified as good or poor for aggregation through psychometric analyses, different sets of images were created for each attention item. The first sets of images were the overall aggregation of all 50 items (Table 18i). The second set (only created for the first attention item) was created with images identified as good for a normal standardized distribution (Table 18ii). The third sets were with the images identified as suitable through Cronbach's ∝ reliability robustness test (Table 18iii). The fourth sets had only images positively correlated with nature relatedness (Table 18iv). Descriptions for each image set can be seen in Table 18.

Table 18
Description of the different sets of images for the first attention and voluntary attention tasks for Study 3.

		First Attention		Voluntary Attention	
Image set	Description	Number of Images	Images removed	Number of Images	Images removed
i	The overall aggregate of all 50 images	50	N/A	50	N/A
ii	The aggregate of images identified as suitable for distribution	45	8, 23, 28, 83, 85	-	-
iii	The aggregate of images identified as suitable for reliability	46	8, 23, 64, 90	46	8, 23, 41, 60
iv	The aggregate of images positively correlated with NR	40	25, 28, 46, 54, 59, 60, 63, 74, 79, 89	46	30, 73, 81, 83

Note: For voluntary attention, there were no images identified through psychometric analyses to remove for the distribution criteria. Thus, this image set were disregarded for voluntary attention task.

I hypothesized that the results from Study 2 would be replicated, where nature relatedness was significantly positively correlated with both first attention and voluntary attention scores. To test this, the different aggregate sets of both attention items were correlated with nature relatedness. Additionally, correlations were explored between the various aggregates of the attention items and exploratory variables (affect, face-valid items/measures, and the NRS subscales). Ninety-five percent confidence intervals were run for those correlations with 100 bootstrapping. All the correlations with the attention tasks can be seen in Table 19.

Nature relatedness significantly correlated with the first attention and voluntary attention overall aggregates. This correlation was also robust across the different sets of images. These results suggest that high nature relatedness individuals are more likely to notice nature first and want to focus their attention on nature in an urban scene. This also replicates the results from

Study 2. The association of nature relatedness and the two attention items was the strongest for the Self subscale of the NRS across all sets. The Experience subscale was consistently significant across all sets for both attention items. The Perspective subscale was significant across all sets for voluntary attention, whereas only significant for overall and for the set with only images significantly correlated with NR (Table 19i & iv), but not with the other two sets for first attention.

Positive affect was significantly associated with the first attention item across all sets of images, where those who identified noticing the nature in the image first were more likely to report higher positive affect in the last four weeks. There was also a significant negative association with negative affect. This association was seen across two of three of the sets of images Both positive and negative affect were not significantly associated with the voluntary attention item across all sets of images.

Both attention items were associated with all face valid items (NRS items 8 & 21) and the noticing nature activity measure (NNA). This was also consistent across all the different sets of images. Correlations between all other variables than the attention tasks can be seen in Appendix F.

Table 19

Correlations with attention tasks for Phase 2 Study 3 (with different image sets).

Variables	First Attention				Voluntary Attention			
(i) Overall Aggregate	*n*	*r*	*p*	95% CI	*n*	*r*	*p*	95% CI
NR	528	.17***	< .001	.08, .25	515	.21***	< .001	.11, .30
PA	528	.11*	.02	.01, .19	515	.02	.69	-.06, .11
NA	528	-.10*	.02	-.18, -.01	515	-.01	.78	-.10, .07
Awe	528	.18***	< .001	.08, .27	515	.03	.44	-.04, .12
NNA	528	.17***	< .001	.08, .24	515	.16***	< .001	.08, .25
NR Item 8	528	.17***	< .001	.09, .24	515	.17***	< .001	.10, .25
NR item 21	528	.14**	.002	.05, .22	515	.15***	< .001	.06, .23
NR Experience	528	.13**	.002	.04, .21	515	.17***	< .001	.08, .25
NR Self	528	.17***	< .001	.08, .25	515	.17***	< .001	.08, .26
NR Perspectives	528	.10*	.02	.02, .19	515	.16***	< .001	.07, .26
ii) Only Images Identified as Suitable for Distribution	*n*	*r*	*p*	95% CI	*n*	*r*	*p*	95% CI
NR	528	.16***	< .001	.08, .24				
PA	528	.13**	.002	.04, .22				
NA	528	-.08	.07	-.16, .01				
Awe	528	.21***	< .001	.12, .30				
NNA	528	.17***	< .001	.08, .25				
NR Item 8	528	.17***	< .001	.10, .24				
NR item 21	528	.14**	.002	.05, .22				
NR Experience	528	.13**	.003	.04, .21				
NR Self	528	.18***	< .001	.10, .26				
NR Perspectives	528	.08	.07	-.01, .17				
(iii) Only Images Identified as Suitable for Reliability	*n*	*r*	*p*	95% CI	*n*	*r*	*p*	95% CI
NR	528	.16***	< .001	.07, .24	515	.20***	< .001	.11, .30
PA	528	.11**	.01	.02, .20	515	.02	.73	-.07, .11
NA	528	-.09	.05	-.17, .01	515	-.01	.89	-.09, .07
Awe	528	.18***	< .001	.09, .27	515	.04	.35	-.04, .13

NNA	528	.16***	< .001	.08, .24	515	.16***	< .001	.08, .25
NR Item 8	528	.17***	< .001	.09, .24	515	.17***	< .001	.09, .24
NR item 21	528	.14**	.002	.05, .22	515	.14**	.002	.05, .22
NR Experience	528	.13**	.003	.04, .21	515	.16***	< .001	.08, .24
NR Self	528	.17***	< .001	.08, .25	515	.16***	< .001	.07, .26
NR Perspectives	528	.08	.06	-.01, .17	515	.16***	< .001	.07, .26
(iv) Only Images Positively Correlated with NR	*n*	*r*	*p*	95% CI	*n*	*r*	*p*	95% CI
NR	528	.21***	< .001	.13, .29	515	.22***	< .001	.13, .32
PA	528	.12**	.004	.04, .21	515	.02	.64	-.06, .11
NA	528	-.13**	.003	-.21, -.04	515	-.01	.90	-.09, .07
Awe	528	.18***	< .001	.09, .27	515	.04	.36	-.04, .13
NNA	528	.20***	< .001	.12, .28	515	.18***	< .001	.09, .27
NR Item 8	528	.20***	< .001	.12, .26	515	.19***	< .001	.11, .26
NR item 21	528	.17***	< .001	.08, .25	515	.16***	< .001	.07, .24
NR Experience	528	.16***	< .001	.08, .24	515	.18***	< .001	.10, .26
NR Self	528	.21***	< .001	.12, .28	515	.18***	< .001	.10, .27
NR Perspectives	528	.15***	< .001	.06, .23	515	.17***	< .001	.08, .27

Note: NR – Nature relatedness. PA – Positive Affect. NA – Negative Affect. NNA – Noticing Nature Activity. NR item 8: "Even in the middle of the city, I notice nature around me". NR Item 21: "I take notice of wildlife wherever I am". * $p < .05$, ** $p < .01$, *** $p < .001$.

Discussion

Study 3 aimed to replicate the findings from Study 2, which found that nature related individuals were more likely to *notice* nature in urban environments through the process of attention. A replication study was conducted to evaluate the generalizability and robustness of the results in a larger sample outside of the convenient recruitment of university students. Over 1100 Amazon Mturk workers were recruited through Cloud Research to participate in the study. The participants were randomly assigned to the first attention or voluntary attention task before completing the same questionnaire measures as Study 2. Psychometric analysis results provided that the images had enough variability to explore the association of the individual difference characteristic of nature relatedness with each of the attention items. Overall, the results of the current study did replicate results from Study 2. Thus, nature related individuals were more likely to first see nature and identify that nature was the thing in the image that they wanted to focus their attention on in images of urban environments; the results support hypotheses 2 and 3 .

Not all outcomes of Study 3 replicated Study 2 results. One of the biggest differences between the two studies was in the correlations with affect. For the first attention task, not only were there differences in the significance, but there was also a difference in the direction of the correlations. In Study 3, those who reported noticing the nature first in the image also reported more positive affect and less negative affect in the last four weeks. In Study 2, however, the associations were in opposite directions (negative for positive affect and positive for negative affect) and also non-significant. Similar discrepancies were found with the direction for the voluntary attention item, but the correlations were consistently non-significant in both studies. It is unclear why we are seeing these dispersions of the correlations. One explanation may be due to sampling error. To explore this, scatterplots of the correlations and Mahalanobis distance

values were computed. For the significant correlations between affect and first attention in Study 3, outliers were identified. Without the outliers included, the association was reduced to non-significant, but stayed in the same direction as the original result. Thus, this suggests that the small correlations were sensitive to outliers[8]. The difference in the pattern may be due to Study 2's student sample having, in general, lower levels of positive affect ($M = 3.45$, $SD = 0.67$) and higher negative affect ($M = 2.97$, $SD = 0.65$) than Study 3's Mturk sample (PA: $M = 3.60$, $SD = 0.87$; NA: $M = 2.22$, $SD = 0.86$). These mean differences are statistically different (PA: $t(1345) = 2.67$ $p = .01$ SE = 0.06; NA: $t(1345) = 13.79$ $p < .001$ SE = 0.05), but also relatively small. The correlations between the behavioural tasks and affect are fragile and may need more investigation for explanation with more comprehensive findings than can be provided in the current studies.

Differently from positive and negative affect, the direction of the association between awe and the first attention task was consistently positive, but differed in significance across the two. Only in Study 3 were participants who reported noticing nature first in the image also reporting feeling more awe in the last four weeks. Additionally, the correlation between first attention and awe in Study 2 was less consistent across the different sets, suggesting that the small correlation was fragile. In contrast, awe was not associated with voluntary attention in either study. These relationships were not dependent on outliers.

There were also mixed results in the replication of the face-valid items and measures correlating with the attention items. Consistently across both attention items in Study 2 and 3, those who paid more attention to nature also reported doing more noticing nature activities.

[8] As additionally exploratory analyses, the same outlier identification here was used to explore the robustness of the correlation of both attention tasks to nature relatedness for each study individually and combined. It was found that the correlation for both first attention and voluntary attention with nature relatedness was not sensitive to outliers, increasing the confidence in the association.

However, the two NRS items "Even in the city, I notice the nature around me" and "I take notice of wildlife wherever I am ") were not significantly with the first attention item in Study 2, but they were for Study 3. Across both studies, the NRS items were significantly associated with voluntary attention item.

Overall, across two studies, the attention behavioural tasks for noticing nature were associated with nature relatedness. With a consistent correlation in two samples, I am more confident in this association.

Combining Studies 2 and 3

As an exploratory step for the secondary goal of measurement building, I explored psychometric analyses previously explored in Studies 2 and 3 with a combined dataset of the two studies. This allows me to make decisions about which images (and scoring) are the best recommendations for other researchers to use. I have also explored additional analyses to make this decision. The sample size of this combined dataset was 1346[9].

The following paragraphs report on all psychometric results (similar to previous studies) in detail, but the results can be briefly summarized: the overall findings of this section is that there were three sets of images that may be better to use, but the results of these three sets and the overall set of 50 images were similar. Thus, for both attention tasks, future researchers could use the full sample of images (i.e. all 50 images), or if researchers need to save time, a reduced set of images also shows good internal consistency and predictive validity to a range of variables. The supplement on OSF provides a quick guide on how to use the tasks and a Qualtrics template. Efficient readers may wish to skip to the general discussion on p. 132.

[9] Data exclusions from each individual study were kept when combining the studies.

First Attention

First, preliminary psychometric analyses were run, including descriptive statistics, frequencies of click areas, reliability, inter-image correlations, and standardized distribution. All quantitative results can be seen in Table 20. Across all 50 images, the nature area was clicked as the first thing participant noticed approximately 28% of the time (*SD* = 20%). The average percentage of nature clicks ranged from 2% (Image 70) to 88% (Image 85). The aggregation of all 50 images was relatively stable, Cronbach's $\propto = .77$, Omega = .76, Split-half reliability $r(819) = .62$ $p < .001$ 95% CI [.56, .67]. The standardized distribution was identified as not normal, visually (see Figure 15), and quantitatively, Shapiro-Wilkes = 0.84 $p < .001$. As can be seen in Table 20 (Z column) and Figure 15, there is an image that was an outlier (3 *SD* above the mean; Image 85) or other images that were approximate outliers (above 2.70 *SD* above the mean; Image 23 & 83). When looking at the overall correlation with nature relatedness, the first attention behavioural measure correlated significantly with nature relatedness, $r(818) = .13$ $p < .001$ 95% CI [.05, .20]. This correlation is consistent with the results from Studies 2 and 3.

Table 20

Descriptive statistics of the first attention task for each image in the combined Studies 2 & 3 dataset (n = 822).

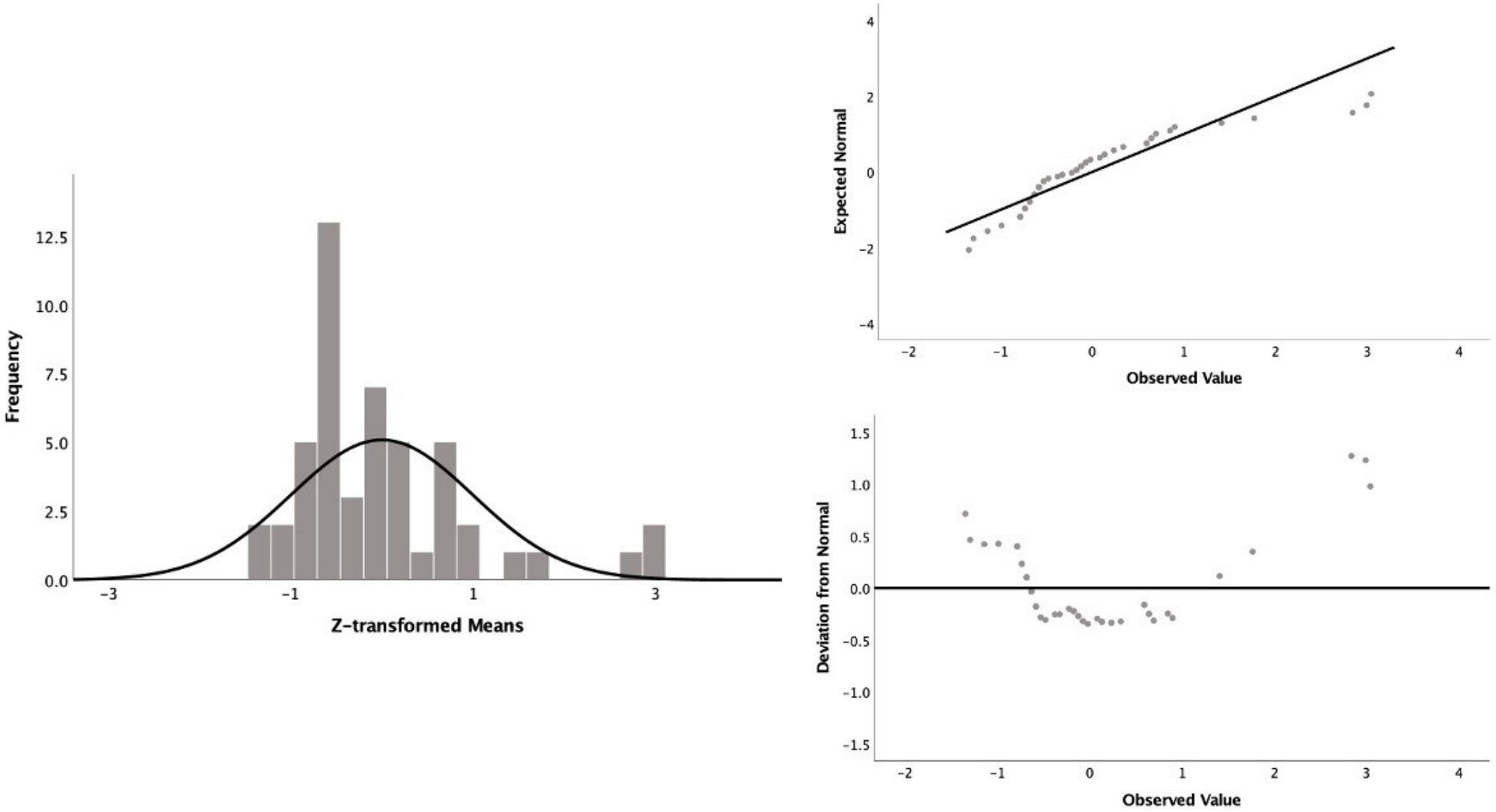
Image	N	*M*	*SD*	Frequency of Other Clicks	Frequency of Nature Clicks	% of Nature Clicks	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlations with NNA
2	794	0.40	0.49	447	317	39.92	0.59	.767	-.11	.16	.83*
4	800	0.19	0.39	648	152	19.00	-0.48	.766	-.06	.14	.03
7	804	0.28	0.45	580	224	27.86	-0.02	.765	-.07	.26	.01
8	808	0.63	0.48	303	505	62.50	1.77	.771	-.19	.23	.05
11	805	0.45	0.50	439	366	45.47	0.85	.766	-.11	.18	.04
12	797	0.40	0.49	481	316	39.65	0.59	.766	-.12	.23	.05
22	797	0.15	0.35	681	116	14.55	-0.68	.766	-.08	.19	.03
23	797	0.84	0.37	126	671	84.19	2.84	.769	-.20	.18	.05
25	796	0.17	0.37	662	134	16.83	-0.58	.763	-.07	.23	-.02
26	800	0.18	0.38	660	140	17.50	-0.53	.760	-.02	.25	.09*
28	801	0.56	0.50	349	452	56.43	1.41	.766	-.13	.18	.02
30	795	0.14	0.35	685	110	13.84	-0.73	.764	-.08	.20	.10**
35	802	0.41	0.49	470	332	41.40	0.64	.765	-.10	.23	.03
41	806	0.17	0.37	671	135	16.75	-0.58	.767	-.14	.20	.023
42	801	0.30	0.46	558	243	30.34	0.08	.758	-.07	.25	.01
43	797	0.25	0.43	597	200	25.09	-0.17	.767	-.07	.17	.04
46	799	0.26	0.44	592	207	25.91	-0.12	.766	-.04	.16	.02
47	803	0.27	0.44	590	213	26.53	-0.07	.761	-.03	.23	.11**
52	789	0.31	0.46	544	245	31.05	0.13	.767	-.08	.20	.04
53	801	0.03	0.17	776	25	3.12	-1.30	.764	-.02	.34	.05
54	809	0.15	0.35	691	118	14.59	-0.68	.762	-.18	.28	.01
55	797	0.13	0.33	697	100	12.55	-0.79	.764	-.06	.20	.06
57	803	0.24	0.42	614	189	23.54	-0.22	.762	-.04	.23	.03
58	807	0.18	0.38	662	145	17.97	-0.53	.765	-.19	.31	.06

59	808	0.06	0.24	759	49	6.06	-1.14	.766	-.20	.31	.02
60	803	0.14	0.34	693	110	13.70	-0.73	.765	-.10	.28	.03
62	802	0.46	0.50	437	365	45.51	0.90	.763	-.08	.22	.12**
63	802	0.22	0.41	629	173	21.57	-0.33	.761	-.06	.26	.01
64	805	0.42	0.49	464	341	42.36	0.69	.770	-.16	.22	.09*
67	803	0.31	0.46	556	247	30.76	0.13	.763	-.09	.24	.05
70	802	0.02	0.15	784	18	2.24	-1.35	.764	-.01	.34	.03
71	804	0.35	0.48	525	279	34.70	0.34	.760	-.01	.25	.13**
73	803	0.16	0.37	673	130	16.19	-0.63	.761	-.05	.24	.03
74	809	0.09	0.28	739	70	8.65	-0.99	.768	-.13	.28	.02
75	792	0.27	0.44	581	211	26.64	-0.07	.767	-.10	.22	.08*
77	809	0.16	0.36	682	127	15.70	-0.63	.760	-.02	.27	.09*
79	805	0.13	0.34	701	104	12.92	-0.79	.766	-.13	.25	-.03
81	798	0.17	0.38	662	136	17.04	-0.58	.761	-.04	.25	.02
82	793	0.15	0.36	672	121	15.26	-0.68	.766	-.05	.19	.10**
83	798	0.87	0.34	106	692	86.72	2.99	.766	-.14	.21	.04
84	805	0.21	0.41	635	170	21.12	-0.38	.762	-.09	.34	.04
85	806	0.88	0.33	99	707	87.72	3.04	.765	-.08	.21	.12**
86	792	0.41	0.49	465	327	41.29	0.64	.763	-.08	.24	.03
87	800	0.25	0.44	597	203	25.38	-0.17	.759	-.05	.27	.06
88	807	0.16	0.37	675	132	16.36	-0.63	.766	-.11	.27	.02
89	801	0.33	0.47	533	268	33.46	0.23	.761	-.04	.24	.01
90	800	0.33	0.47	535	265	33.13	0.23	.767	-.09	.15	.02
91	796	0.17	0.38	659	137	17.21	-0.58	.760	-.05	.34	.12**
92	797	0.13	0.33	696	131	16.44	-0.79	.762	-.03	.24	.15**
93	798	0.26	0.44	594	204	25.56	-0.12	.764	-.05	.20	.08*

Note: All images had a minimum of 0 and a maximum of 1; therefore, this information was redacted from the table. NNA = Noticing Nature Activity. * $p < .05$, ** $p < .01$, *** $p < .001$.

Figure 15

Visual representation of the standardized distribution of z-transformed means of nature clicks for the first attention item in the combined Studies 2 & 3.



To explore different sets of images, a few psychometric criteria were set. The first criterion to create a set of images was to include only images identified as good to aggregate through reliability analyses. To do this, leave-out item analyses were run as a robustness test to see how the Cronbach's ∝ value changes with the removal of each individual image. From this test, only three images were identified as poor for aggregation by showing that the Cronbach's ∝ would increase if the image was removed (Image 8, 23, & 64). However, the change in Cronbach's ∝ was minimal, with the biggest increase being by .003. A single image showed no change in Cronbach's ∝ (Image 74). All others showed a minimal decrease with its removal.

The second set was created with images identified as suitable through the standardized distribution robustness test. By suitable, images had to help create a *more* normal distribution. The robustness test was run to see how the distribution would change by removing low nature clicked images and outliers images (Z score above 2.70). To see robustness test results, see Table 21. The standardized distribution did not become normal visually or statistically when removing up to 20% or less nature clicked images, but the distribution became normal after removing the outliers from the sample of all 50 images (see Figure 16 for a visual of this distribution). This recommends keeping 46 images.

A third set of images was created with images that correlated positively (but not necessarily significantly) with the noticing nature activity scale. This was chosen because the attention tasks were meant to be behavioural tasks of noticing nature. Thus, the recommended images should positively correlate with a self-report noticing nature measure. The overall correlation between the first attention item and the noticing nature activity measure was significant, $r(818) = .16$ $p < .001$ 95% CI [.10, .23]. In relation to individual images, only two

correlations were in the negative direction (Images 25 & 79); these correlations were not significant. The rest of the images were in the positive direction, with 14 being significant.

Overall, across the three psychometric analyses criteria, the results suggest the removal of seven images: three images for their poor aggregation value (Images 8, 23 & 64), four for being outliers in a standardized distribution (Images 8, 23, 83, & 85), and two for negatively correlating with self-report noticing nature (Images 25 & 79).

Table 21

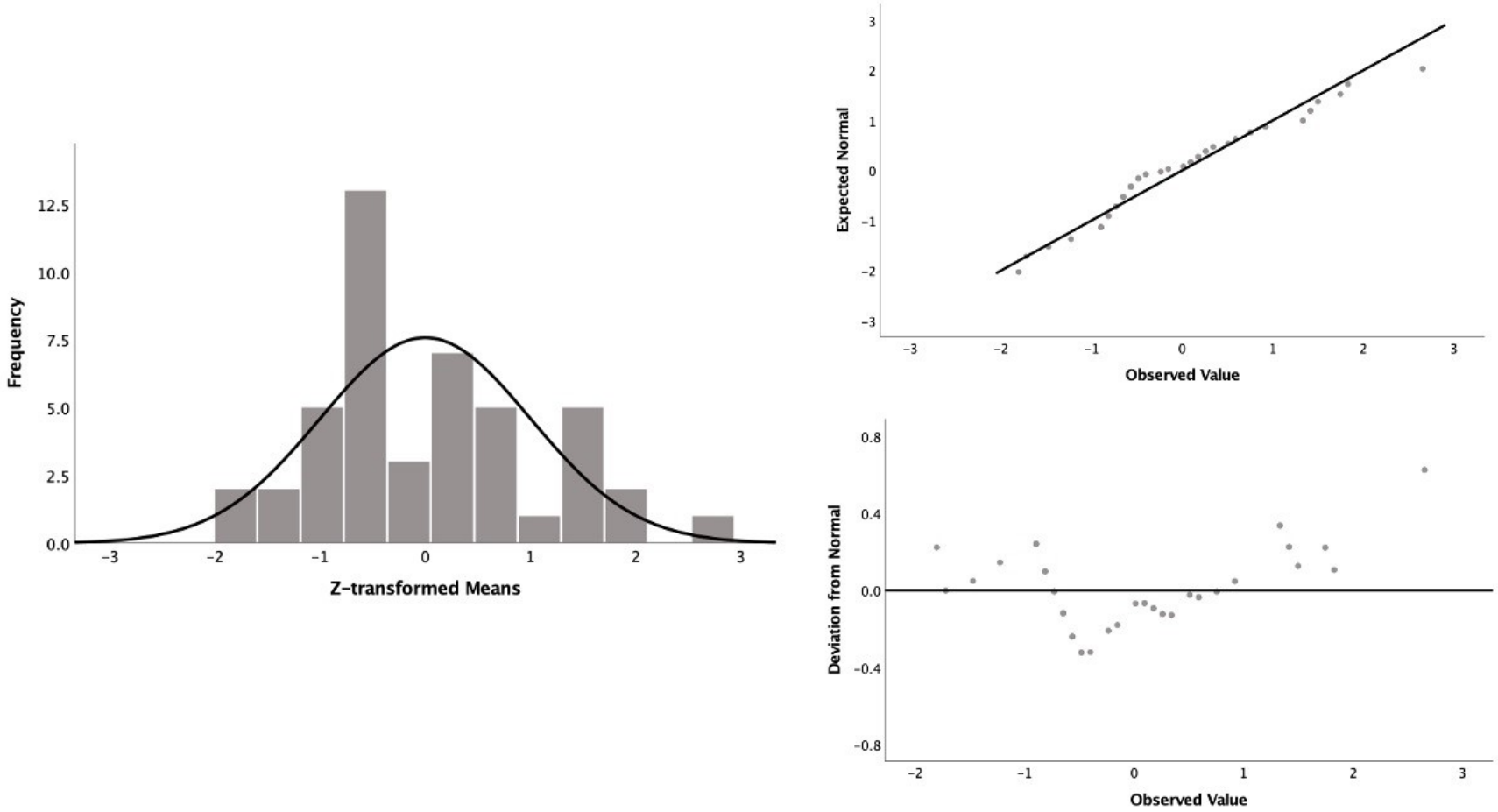
Robustness test for normality of the first attention task standardized distribution for the combined Studies 2 & 3 dataset.

	Low Percentage Image Removed				Outlier (~3 SD) Images Removed			
Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	Number of images	Images removed	Shapiro-Wilks Estimate	*p*
None	50	N/A	0.84	< *.001*	46	8[a], 23, 83, 85[a]	0.96	.11
Images with 5% or less clicks on the nature area	48	53, 70	0.81	< .001				
Images with 10% or less clicks on the nature area	46	53, 59, 70, 74	0.79	< .001				
Images with 15% or less clicks on the nature area	39	22, 30, 53, 54, 55, 59, 60, 70, 74, 79, 92	0.80	< .001				
Images with 20% or less clicks on the nature area	28	4, 22, 25, 26, 30, 41, 53, 54, 55, 58, 59, 60, 70, 73, 74, 77, 79, 81, 82, 88, 91, 92	0.79	< .001				

Note: Images were first removed for low percentages and then removed as outliers. [a]These images were not initially identified as outliers, but after the removal of the other images became outliers.

Figure 16

Visual representation of the first attention task standardized distribution of the 47 images identified as best suited for the distribution in the combined Studies 2 and 3.

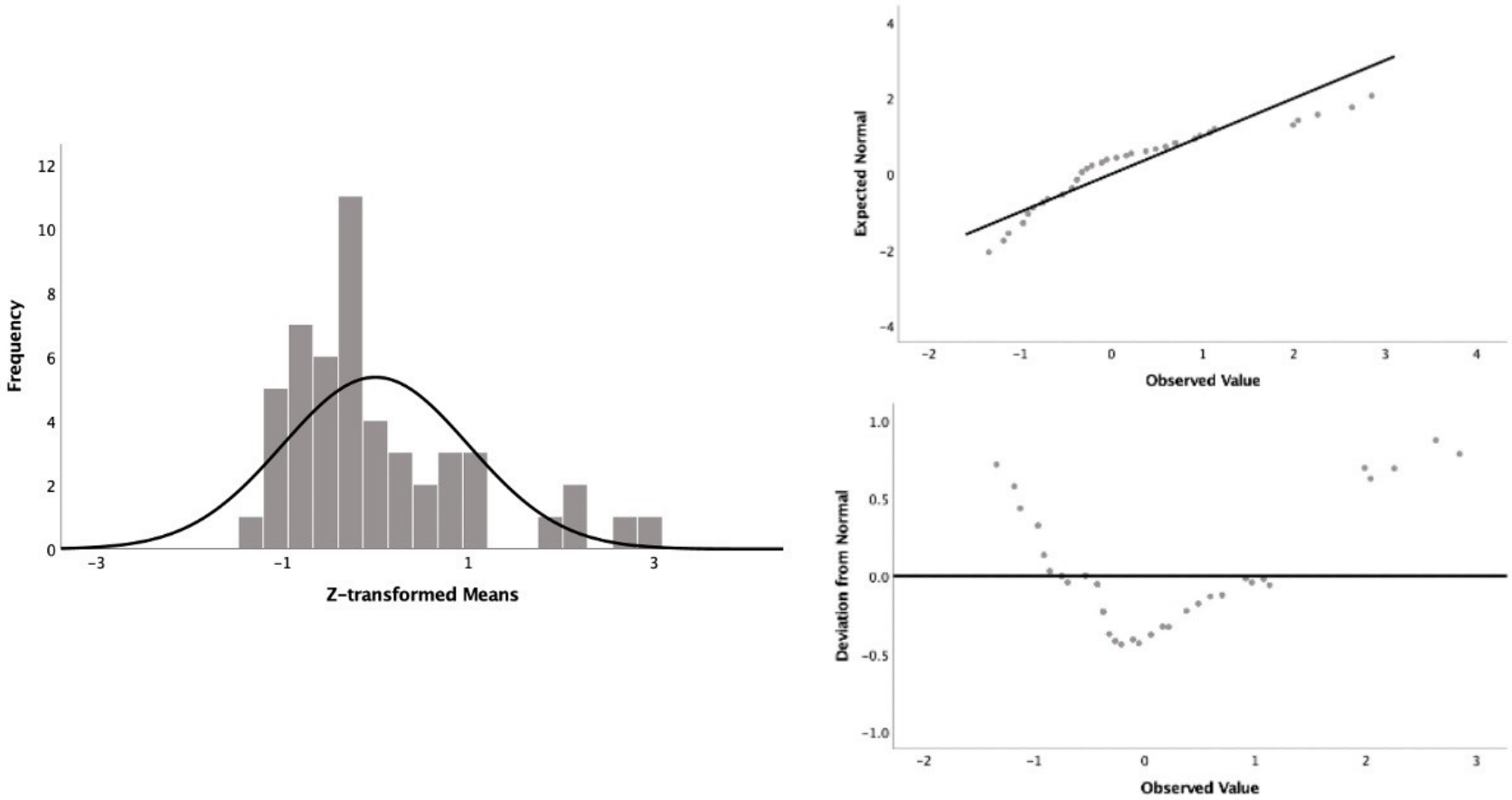


Voluntary Attention

The quantitative results of the psychometric analyses for the voluntary attention behavioural task can be seen in Table 22. Across all 50 images, the nature area of the images was clicked approximately 33% of the time (*SD* = 19%). The average percentage of nature clicks ranged from 8% (Image 70) to 86% (Image 83). The 50 images show good internal consistency, Cronbach's $\propto = .88$, Omega = .88, and split-half reliability = $r(806) = .77$ $p < .001$ 95% CI [.74, .80]. The standardized distribution of z-transformed means for all 50 images was not normal, visually (see Figure 15), or quantitatively (see Table 23). As seen in Table 22 (Z column) and Figure 15, an image was identified as an approximate outlier (above 2.80 *SD* above the mean; Image 83). There were images with a low percentage of clicks: one below 10%, another four below 15%, and another eight below 20%. When looking at the overall correlation with nature relatedness, the voluntary attention behavioural measure correlated significantly with nature relatedness, $r(805) = .15$ $p < .001$ 95% CI [.08, .22].

Figure 17

Visual representation of the standardized distribution of the z-transformed means of all 50 images on the voluntary attention item in the combined Studies 2 & 3.



To select the best images to use when exploring noticing nature through voluntary attention, a few different psychometric analysis criteria were examined. The first criterion to create an image set was to examine which images were best for aggregating based on the robustness test for Cronbach's alpha. The majority of images (i.e. 46 images) showed a reduction in the alpha value when removed, indicating that most images were suitable for aggregation. Five images showed no change in Cronbach's alpha when removed (i.e. also suitable for aggregation), whereas only Image 8 showed an increase. This increase in Cronbach's alpha was small (.002).

The second criterion explored was looking at images that made up the best standardized distribution. To explore this, the robustness test of removing images with low percentages in nature clicks and/or outliers (~3 SD above or below the mean). As seen in Table 23, none of the distributions were statistically normal. Thus, the closest distribution to normal with the fewest images removed was the highlighted as the recommended distribution. This distribution was with the outliers removed from the set of 50 images (i.e. 45 images; see distribution in Figure 18).

The last criterion was to explore how the images would positively associate (but not necessarily significantly correlate) with the noticing nature activities measurement. The majority of images had a positive correlation with the NNA. Of the 46 images with positive correlation, 24 were significant. Only four images had negative correlations with noticing nature activities (Images 4, 23, 73, & 90). Not one of these four correlations was significant.

Overall, across the different psychometric analysis criteria, the results suggest the removal of nine images: one for reliability (Image 8), four others for distribution (one image [8] was recommended for removal by both the reliability and distribution criteria; Images 23, 83, 85, & 86), and three others for individual correlations with self-report noticing nature (one image

[23] was recommended for removal by both the distribution and correlation criteria; Images 4, 73, & 90).

Table 22

Descriptive statistics of the voluntary attention task for each image in the combined Studies 2 & 3 dataset (n = 818).

Image	N	*M*	*SD*	Frequency of Other Clicks	Frequency of Nature Clicks	% of Nature Clicks	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations	Correlations with NNA
2	796	0.42	0.49	458	338	42.46	0.48	.874	-.04	.23	.04
4	803	0.19	0.39	653	150	18.68	-0.75	.873	-.05	.20	-.01
7	802	0.16	0.37	672	130	16.21	-0.91	.873	-.02	.26	.10**
8	800	0.75	0.43	197	603	75.38	2.26	.877	-.12	.14	.05
11	802	0.31	0.46	557	245	30.55	-0.11	.873	.02	.27	.07*
12	804	0.50	0.50	406	398	49.50	0.91	.874	-.01	.23	.05
22	802	0.32	0.47	545	257	32.04	-0.05	.874	-.01	.24	.07
23	797	0.71	0.45	232	565	70.89	2.04	.875	-.05	.16	-.02
25	798	0.23	0.42	616	182	22.81	-0.54	.874	-.03	.23	.08*
26	800	0.28	0.45	579	221	27.63	-0.27	.871	.003	.26	.12**
28	798	0.54	0.50	368	430	53.88	1.13	.875	-.03	.17	.02
30	798	0.15	0.36	678	120	15.04	-0.97	.874	.00	.24	.04
35	800	0.19	0.39	650	150	18.75	-0.75	.874	-.09	.25	.03
41	798	0.27	0.45	581	217	27.19	-0.32	.875	-.02	.22	.04
42	800	0.44	0.50	448	352	44.00	0.59	.871	.02	.29	.13**
43	800	0.28	0.45	577	223	27.88	-0.27	.873	-.01	.20	.12**
46	803	0.17	0.38	665	138	17.19	-0.86	.873	.03	.19	.06
47	803	0.36	0.48	513	290	36.11	0.16	.871	.03	.31	.13**
52	804	0.25	0.43	603	201	25.00	-0.43	.872	-.08	.27	.05
53	804	0.12	0.33	706	98	12.19	-1.13	.872	.03	.24	.08*
54	798	0.20	0.40	641	157	19.67	-0.70	.872	-.12	.37	.08*
55	802	0.26	0.44	594	208	25.94	-0.38	.873	-.03	.23	.06
57	802	0.23	0.42	619	183	22.82	-0.54	.873	.04	.29	.10**
58	803	0.46	0.50	436	367	45.70	0.70	.872	-.09	.52	.13**

59	801	0.37	0.48	501	300	37.45	0.21	.871	-.07	.52	.15**
60	799	0.31	0.46	551	248	31.04	-0.11	.875	-.04	.26	.08*
62	799	0.53	0.50	375	424	53.07	1.07	.873	.03	.24	.10**
63	799	0.26	0.44	591	208	26.03	-0.38	.872	-.03	.27	.10**
64	803	0.15	0.36	684	119	14.82	-0.97	.874	-.02	.20	.03
67	801	0.26	0.44	595	206	25.72	-0.38	.871	.05	.29	.07
70	802	0.08	0.27	741	61	7.61	-1.34	.873	-.03	.24	.07
71	800	0.51	0.50	394	406	50.75	0.97	.873	-.02	.25	.09*
73	800	0.16	0.37	670	130	16.25	-0.91	.873	-.07	.25	-.03
74	807	0.15	0.36	682	125	15.49	-0.97	.872	-.11	.35	.09**
75	798	0.23	0.42	613	185	23.18	-0.54	.872	-.02	.29	.04
77	804	0.26	0.44	529	212	26.37	-0.38	.872	-.03	.29	.09*
79	800	0.29	0.46	565	235	29.38	-0.21	.872	-.05	.37	.11**
81	804	0.11	0.32	712	92	11.44	-1.18	.873	-.02	.25	.03
82	804	0.34	0.48	527	277	34.45	0.05	.873	-.01	.23	.09*
83	800	0.86	0.34	110	690	86.25	2.85	.875	-.03	.27	.02
84	805	0.46	0.50	433	372	46.21	0.70	.871	.02	.31	.09*
85	798	0.82	0.38	141	657	82.33	2.63	.874	-.03	.27	.03
86	805	0.70	0.46	242	563	69.94	1.99	.872	-.01	.29	.05
87	801	0.40	0.49	480	321	40.07	0.38	.871	-.06	.25	.08*
88	800	0.17	0.37	666	134	16.75	-0.86	.872	-.09	.35	.07
89	801	0.27	0.44	588	213	26.59	-0.32	.873	.00	.30	.06
90	803	0.25	0.43	602	201	25.03	-0.43	.874	.01	.20	-.01
91	803	0.26	0.44	598	205	25.53	-0.38	.872	-.09	.30	.08*
92	800	0.25	0.43	600	200	25.00	-0.43	.871	.01	.27	.04
93	802	0.26	0.44	595	207	25.81	-0.38	.871	.04	.31	.08*

Note: All mages had a minimum of 0 and a maximum of 1; therefore, this information was redacted from the table. NNA = Noticing Nature Activity. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 23

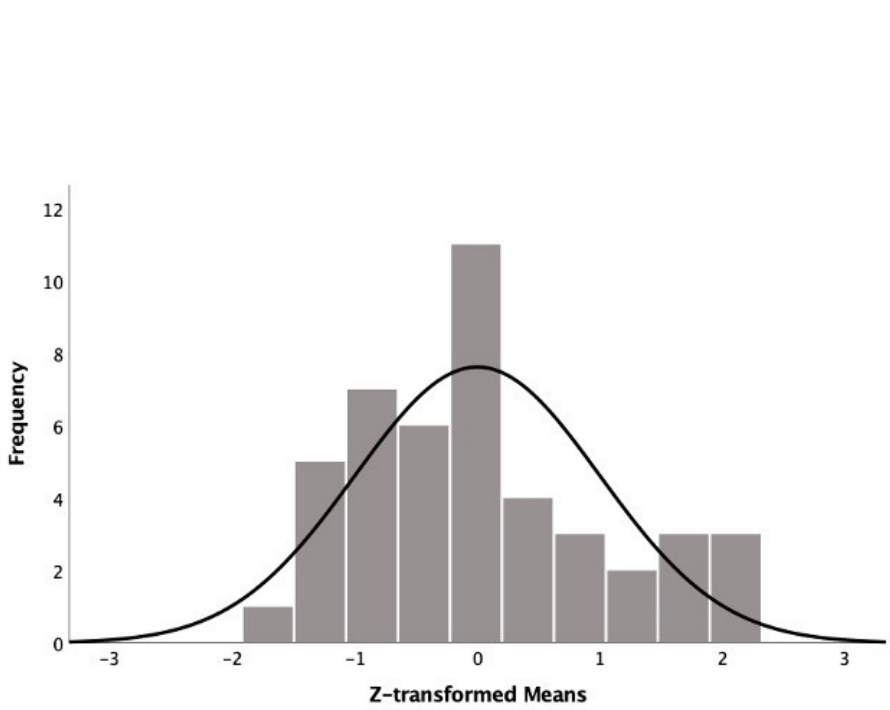
Robustness test for normality of the voluntary attention task standardized distribution for the combined Studies 2 & 3 dataset.

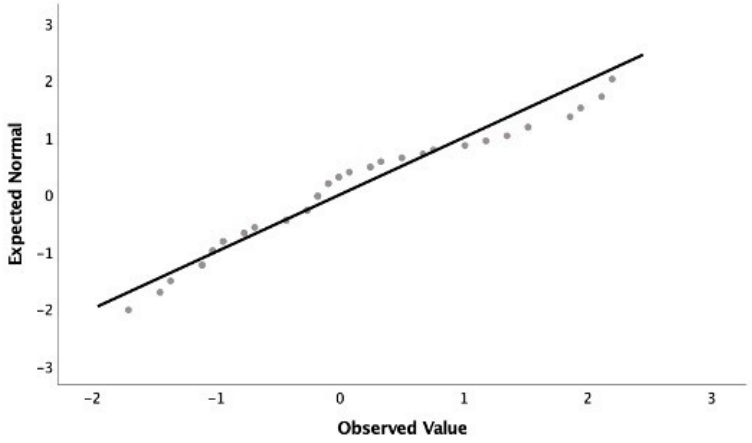
	Low Percentage Image Removed				Outlier (~3 SD) Images Removed			
Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	Number of images	Images removed	Shapiro-Wilks Estimate	*p*
None	50	N/A	0.87	< .001	45	8[a], 23[a], 83, 85[a], 86[a]	0.94	.03
Images with 10% or less clicks on the nature area	49	70	0.86	< .001	44	8[a], 23[a], 83, 85[a], 86[a]	0.93	.01
Images with 15% or less clicks on the nature area	46	53, 64, 70, 81	0.85	< .001	40	8 [a], 23 [a], 83, 85[a], 86[a]	0.94	.03
Images with 20% or less clicks on the nature area	37	4, 7, 30, 35, 46, 53, 54, 64, 70, 73, 74, 81, 88	0.81	< .001	-	-	-	-

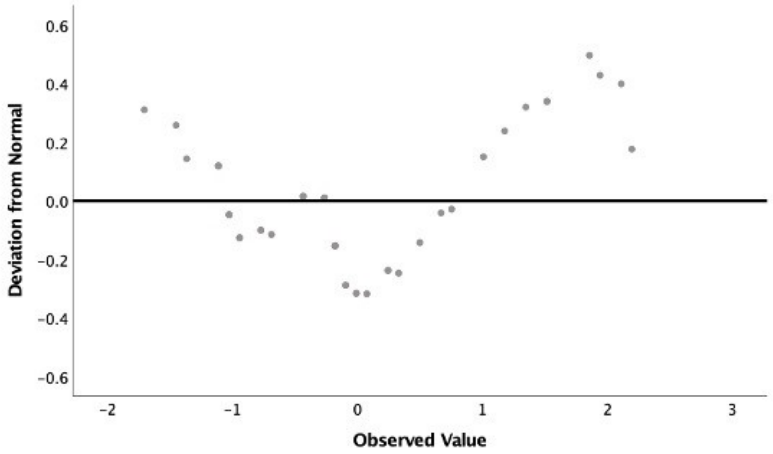
Note: Images were first removed for low percentages and then removed as outliers. [a]These images were not initially identified as outliers, but after the removal of the other images became outliers.

Figure 18

Visual representation of the voluntary attention task standardized distribution of the 45 images identified as best suited for the distribution in the combined Studies 2 and 3.







Criterion Validity

Lastly, to explore the best images to score or aggregate, correlations between the various variables included in the study with different scoring of the attention tasks. The first scoring sets were the aggregation of each attention task with all 50 images (Set A). Sets B of images was created with the images identified as suitable for each task individually from their psychometric analysis results (there are a different number of images used for each task; FA: 43 images; VA: 41 images). Future researchers may want to use both attention items in tandem, so it may be best to recommend a set of images that work well for both attention tasks. Thus, both Set C and D were created with this in mind. For Set C, only images that were identified as suitable across both psychometric analyses were used (i.e. if an images was recommended to be removed for either task, it was removed for this set; 11 images removed). Set D was created by only removing images that were identified as poor for *both* attentions tasks (i.e. if an image was identified as poor for one task but suitable for another it was kept for this set; 4 images removed). The results are in Table 24.

Across the three sets of image sets and the overall, there were minimal differences in the correlations. Additionally, there was minimal changes in the means and the reliability values. Due to this, no clear recommendation can be made. Rather, I recommend that future researchers use the set of images that work best for their design and available resources. The full set of 50 images allows for a wider breadth of images with varying degrees of nature, where some images have a smaller portion of nature and others with a larger portion. However, if time and cost are an issue, the smaller set of images (either images identified as best for the specific task or across both) have similar internal consistency and predictive validity to a range of variables. Thus, the decision should be made based on the researcher's situation. To help guide researchers on this

decision and the use of the tasks, a quick guide and Qualtrics template can be found in the online supplement (OSF).

Table 24

Correlations between the attention tasks and various variables across three set of images: A) overall; B) the best images to aggregate for each individual attention task; C) the best images to aggregate when looking across the psychometric analyses for both tasks; D) only the images that were identified as poor in both were removed.

First Attention	A(∝ = .77; *M* = .13.81, *SD* = 5.69)		B(∝ = .76; *M* = 9.97, *SD* = 5.14)		C(∝ = .74; ; *M* = 8.91, *SD* = 4.69)		D(∝ = .77; *M* = 10.68, *SD* = 5.40)	
Variables	*r*	CI	*r*	95% CI	*r*	95% CI	*r*	95% CI
NR	.13***	.05, .20	.12***	.04, .19	.12***	.05, .20	.11**	.04, .19
PA	.03	-.05, .10	.06	-.01, .14	.05	-.02, .12	.06	-.01, .13
NA	.04	-.03, .12	.07*	.003, .14	.05	-.02, .12	.08*	.002, .14
Awe	.13***	.05, .21			.16***	.08, .24	.18***	.11, .25
NNA	.16***	.10, .22	.17***	.11, .23	.16***	.10, .23	.17***	.10, .23
NR Item 8	.15***	.08, .21	.14***	.08, .20	.15***	.08, .22	.14***	.07, .20
NR item 21	.10**	.03, .17	.10**	.02, .16	.10**	.02, .16	.09**	.01, .16
NR Experience	.06	-.02, .13	.06	-.02, .13	.07	-.01, .14	.05	-.02, .12
NR Self	.14***	.06, .21	.15***	.07, .21	.14***	.07, .21	.14***	.07, .21
NR Perspectives	.10**	.03, .17	.07	-.01, .14	.08*	.07, .21	.06	-.02, .13
Voluntary Attention	A(∝ = .88; *M* = 16.35, *SD* = 8.05)		B(∝ = .87; *M* = 11.94, *SD* = 7.18)		C(∝ = .86; *M* = 11.29, *SD* = 6.77)		D(∝ = .88; *M* = 13.24, *SD* = 7.74)	
Variables	*r*	95% CI	*r*	95% CI	*r*	95% CI	*r*	95% CI
NR	.15***	.08, .22	.15***	.08, .22	.16***	.09, .22	.14***	.07, .21
PA	-.03	-.10, .04	-.02	-.09, .05	-.03	-.09, .04	-.02	-.09, .05
NA	.13***	.06, .19	.15***	.08, .22	.13***	.07, .21	.15***	.08, .22
Awe	.05	-.02, .12	.08*	.02, .15	.07	.00, .14	.08*	.01, .15
NNA	.18***	.11, .24	.20***	.13, .27	.19***	.13, .26	.18***	.12, .25
NR Item 8	.15***	.08, .21	.16***	.10, .23	.16***	.10, .22	.15***	.09, .22
NR item 21	.10**	.03, .16	.10**	.03, .17	.10**	.02, .16	.10**	.03, .16
NR Experience	.09*	.02, .15	.10**	.03, .17	.10**	.03, .16	.09**	.03, .16
NR Self	.12***	.06, .19	.13***	.07, .20	.13***	.07, .20	.13***	.06, .19
NR Perspectives	.14***	.06, .21	.12**	.05, .10	.13***	.07, .20	.11**	.04, .18

Note: NR – Nature relatedness. PA – Positive Affect. NA – Negative Affect. NNA – Noticing Nature Activity. NR item 8: "Even in the middle of the city, I notice nature around me". NR Item 21: "I take notice of wildlife wherever I am". * $p < .05$, ** $p < .01$, *** $p < .001$. Mean scores are computed from the average of the sum scores for the task aggregation for each set.

General Discussion

The overall goal of this dissertation was to explore if nature related people were more likely to *notice* nature in everyday moments in an urban environment. To control the amount of nature participants were exposed to, everyday nature was operationalized by showing participants images of urban environments with some degree of nature on a computer screen. Noticing nature was examined through two processes: perception (Phase 1) and attention (Phase 2). Perception was defined as a person's interpretation of nature (i.e., if a person does not recognize the nature, they will not *notice* it). Attention was defined in two ways: 1) if immediate attention was drawn towards nature and 2) if their preferred attention was oriented towards it. I hypothesized that nature related individuals would perceive more and pay more attention to nature in images of urban environments on a computer. Across both phases, mixed results were observed. In Phase 1, differing from the hypothesis, there was no relationship between participants' ratings of nature perception (either through the 7-point scale item or the percentage item) and their levels of nature relatedness. In contrast, hypotheses 2 and 3 were supported and replicated in Phase 2; nature related people were more likely to have their attention drawn toward the nature in images of urban environments and were also more likely to choose to focus their attention on it. Together, the results suggest a promising step in understanding nature related individuals' affinity for nature in their everyday life.

Noticing Nature Tasks: What did I measure, and what does it mean?

I created four tasks related to the two processes to assess the hypothesized correlation between noticing nature and nature relatedness. The perception process was measured through two rating scales that asked how much nature was in an image: 1) the 7-point rating scale; 2) a percentage rating item. The attention process was measured through two image-based click

assessments: 1) clicking on the first thing they noticed in the image (first attention); 2) clicking on the thing they wanted to focus their attention on in the image (voluntary attention). Though the results of the studies suggested two of the four noticing nature tasks (i.e., the attention noticing nature tasks) were associated with nature relatedness, it was essential to also calibrate how much noticing nature I could capture with my tasks.

There was a lack of support for the idea that nature related individuals noticed more nature through the two perception tasks. It is unclear whether the absence of an association with nature relatedness is due to no *true* relationship or rather that the measurement tool inaccurately assessed the concept of perception (or noticing nature in general) I intended to measure. The psychometric results corroborate that there is enough variability in the task to examine individual differences (identified through psychometric analyses and multilevel models). Still, the lack of association with face-valid self-report noticing nature measure (NNA; Richardson et al., 2021) and items (two individual NRS items; Nisbet et al., 2009) suggests that the tasks measured something outside of noticing nature. There was also a lack of predictive validity for the other variables (i.e., nature relatedness, affect[10], and frequency of noticing nature activities). Thus, the confidence in the perception tasks as behavioural measures of noticing nature is low.

In contrast, there was more support suggesting that the attention tasks were better measures of noticing nature than the perception tasks. Both attention items showed predictive validity in their association with nature relatedness. The two nature attention tasks were also related to the face-valid measure (NNA) and items (two items from the NRS) of noticing nature, supporting that the tasks were behavioural measures of noticing nature. More specifically,

[10] There was a single correlation between the 7-point scale rating item of perception and positive affect. This correlation is relatively small in size (.17) with a large confidence interval (.04 to .28). This suggests that the correlation may be due to sampling variation (i.e., chance). Thus, the confidence in the correlation is not high.

participants who indicated that they noticed nature first and voluntarily wanted to focus on it in the images also self-reported doing more noticing nature activities. Together, this gives me more confidence in the attention tasks as a better tool for measuring noticing nature and, thus, provide evidence that nature related people notice more nature in urban environments through the process of attention. Henceforth, the discussion will center mostly around the attention-based noticing nature behavioural tasks.

Attention and Perception

The definitions of the processes of perception and attention explored in the dissertation were more colloquial in nature. In cognitive psychology, perception is defined as the conscious experience that comes from sensory information and attention is defined as focusing on an object, place, thought, or activity (Goldstein, 2011). In the current studies, I defined perception as the interpretation of nature; attention as awareness being drawn and focused towards nature. Even though the colloquial definitions were an outside of cognitive psychology, the exploration of the "processes" are still related to the field, in general.

For example, the two "processes" of attention measured in the current studies are both considered to be explicit processing of attention (i.e., involves conscious processing) because of the way I measured them. I measured the two attentions by having participants click on the image. Thus, even if the first attention task may seem like it is measuring implicit/automatic/exogenous attention, because the participants were still consciously able to identify what they did notice first and actively clicked on the image, it is an explicit reporting of first attention. This is different than eye tracking techniques that measure fixation points (i.e., measuring where in the image the eye is pausing to focus on) of a scene. In Bingjing et al. (2022) study, the eye tracking measured fixation on the natural and urban elements of the 360° scene. In

this study (Bingjing et al., 2022), it was not clear that participants were aware that they were spending more time fixating on areas, such as nature related people fixated more on the natural elements of the scene.

The more explicit reporting of attention used in the dissertation may be biased by the implicit selective attention that is occurring subconsciously. As research in cognitive psychology would suggest, when perceiving or paying attention to anything there is a step of prioritization that occurs to bring information from the senses to the subjective mind; this prioritization can come from both bottom-up and top-down processing (van Ede & Nobre, 2023). Thus, my participants have gone through this prioritization subconsciously before being aware of the object there are noticing to respond to my questions. With my current results, I can not necessarily say that participants high in nature relatedness prioritize nature in a scene with multiple stimuli, but Bingjing and colleagues (2022) would suggest that they are fixating more on natural elements and my studies would support they are aware enough to explicit report the nature. Nevertheless, there may be another level of awareness that may not be occurring. For example, it is not clear if the people who click nature the most across the images would also self-report that they were noticing nature more often than not. In other words, they may not be aware that across all 50 clicks on the behavioural task their attention was being drawn towards the natural elements rather than the urban elements.

Implicit and Explicit Noticing Nature

Nature relatedness is an individual difference characteristic that would fit within an individual's self-concept (i.e., the description of who a person thinks they are). The self-concept can influence a person's attitudes and behaviours, where they have thoughts or do activities that fit their self-concept. Not only has previous research shown that some attitudes and behaviours

are associated with nature relatedness (Keaulana et al., 2021), but also the items used to assess the participant's characteristic level of their connectedness with nature are typical behaviours and attitudes linked with the individual difference characteristic. These items include, but are not limited to, attitudes towards sustainability (e.g. "I always think about how my actions affect the environment."), experiences with or exposure to nature ("I don't often go out in nature" – reverse-coded), and also noticing nature ("I take notice of wildlife wherever I am" and "Even in the middle of the city, I notice nature around me"). These items that assess nature relatedness highlight the attitudes and behaviours that are a part of the self-concept of highly nature related people. Thus, noticing nature is implied to be a fundamental aspect of their self-concepts, even though research has not behaviourally demonstrated this.

For the purpose of the dissertation, I created new behavioural tasks of noticing nature to be able to test if nature related individuals do, in fact, notice more nature (implicit behaviour) rather than relying on the already-created self-report measure (NNA; Richardson et al., 2021) or the responses to the two individual items of the NRS related to noticing nature. There are methodological limitations to using self-report measures. For example, the measure of nature relatedness (NRS; Nisbet et al., 2009) relies on participants to self-reflect, recall attitudes and behaviours, and then respond honestly to questions to identify their level of nature relatedness. A participant may report noticing nature because it aligns with their self-concept (i.e., "this is something I would do") rather than recalling actual behaviours (i.e., the discrepancy between the self and behaviours). Therefore, examining behavioural measures of key concepts that align with or help characterize nature relatedness is important.

The finding that the attention behavioural measure of noticing nature is associated with the total nature relatedness score supports the idea that noticing nature may be a fundamental

concept of nature relatedness, as implied. Additionally, the links of the two NRS items related to noticing nature with the explicit behavioural tasks reinforce that these two items of the NRS accurately assess psychological processes or self-reported behaviour of noticing nature. For instance, people who report they are likely to notice the nature around them, even in a city (i.e., item 8 of the NRS), were more likely to do so in an urban image task.

Even though there is an association between the implicit (the task) and the explicit noticing nature (the self-report measure and items), there are differences between the methods that are important to consider. The explicit measures are general assessments of attitudes and behaviours towards noticing nature, whereas the built tasks are momentary assessments. Thus, it is not clear how much the current tasks are momentary and unstable or if the act of noticing more nature measured with the tasks may be more trait-like, where the responses would be more stable. The replicated association between nature relatedness and the attention tasks would suggest that the task responses are more stable like an individual's self-concept. However, the small correlations indicate that other factors may associate with responses. Without clear test-retest assessments, the consistency in the tasks is not clear. The task would benefit from future research employing multiple administrations.

Nature Relatedness, Noticing Nature, and Affect

The notion that nature related individuals are more likely to notice nature in their daily lives stems from the proposition that noticing nature may lead to increased happiness. Thus, exploring the associations between nature relatedness, noticing nature, and affect in the dissertation studies extends the research on the connection between nature relatedness and happiness (Capaldi et al., 2014; Pritchard et al., 2020). There may be a few reasons why nature related individuals are generally happier. One of these reasons may be that they are spending

more time in nature, which has been previously shown (Colleony et al., 2017; Desrochers et al., 2022; Flowers et al., 2016; Lin et al., 2014; Nisbet et al., 2009; Shannahan et al., 2015b; Soga & Akasaka, 2019), but it may also be that they tend to notice nature in everyday moments. For example, are nature-related individuals better able to notice small parts of nature (micro-exposure to nature) in their everyday life to make them happy? No previous research has directly examined this, but there is suggestive support.

In the dissertation studies, I found mixed results for this notion. Overall, in the current studies, nature related people were happier (measured through positive affect; correlations ranged from .13 to .18 in the three studies, see Appendix F for more details). There was also an indication that nature related individuals do notice more nature when assessing it through the self-report measure of how often they engage in noticing nature activities (i.e., the NNA measure by Richardson et al., 2021; correlations ranged from .60 to .65, see Appendix F for more details) and the attention behavioural task of noticing nature[11]. However, it is not clear that noticing nature is associated with affect (i.e., there were mixed findings regarding the self-report and behavioural noticing nature measures being associated with positive affect). Thus, the links between noticing nature and positive affect are too inconsistent to see clear support.

Even though the current findings give mixed evidence for noticing nature being a reason nature related people are happier, it is still plausible that the association exists. The behavioural tasks were created to assess if participants notice more nature in their everyday life by simulating real-life views through images and were not necessarily meant to increase one's mood, as is often done in nature research. With this said, being able to assess participants' ability to notice

[11] Nature related individuals were not more likely to notice nature through the perception behavioural tasks. However, as previously discussed, it may be that the perception task was not an appropriate measure of noticing nature.

nature does not mean that they have noticed more nature recently, which could account for their mood scores over the last four weeks. As research shows, people spend a large majority of their time indoors (Matz et al., 2015), and if they spend time outdoors, their neighbourhoods, in general, tend to have minimal biodiversity (Turner et al., 2004); thus, even though nature related people may pay attention to nature, they may not have had the opportunity to do so. The fact that participants who reported doing more noticing nature activities reported that they had more positive moods over the last four weeks would help support the idea that noticing nature improved positive mood (correlations ranged from .14 to .32, see Appendix F for more details). The broader body of literature also provides support for the subject. The three studies by Passmore and colleagues (Passmore & Holder, 2016; Passmore et al., 2022ab) provide supportive evidence that performing a noticing nature activity over the course of two weeks improves well-being (i.e., there was a change in well-being outcomes post-intervention from the baseline values); further, noticing nature improved well-being regardless of time spent in nature.

Additionally, when exploring if noticing nature may be a reason that nature related individuals are happier, there is some conceptual support in this study and others, but it may be harder to see statistically in research. For instance, many things make up why a person may be happier, such as personality traits (e.g., extraversion, emotional stability; Diener et al., 1999), social relationships (Diener & Seligman, 2002), and even genes (Lykken & Tellegen, 1996; Nes et al., 2006). Plus, there may be other more specific predictors of happiness that a nature related person would do that make them happy, such as spending more time in nature, doing more pro-environmental behaviours, more physical activity, and reduced screen time, which have all have previously been associated with increased feelings of positive affect (Buecker et al., 2021; McKay & Schmitt, 2019; McMahan & Estes, 2015; Molina-Cando et al., 2021; Pedersen et al.,

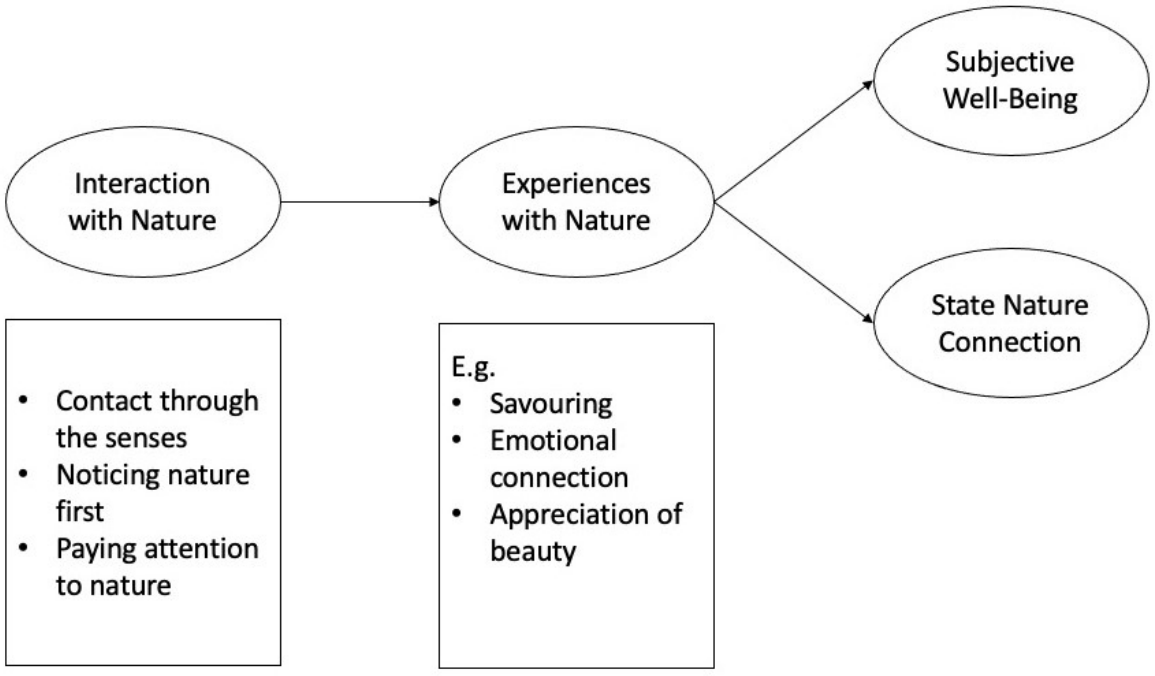
2022; Zawadzki et al., 2020). Thus, noticing nature may be trying to explain a small amount of variance in a substantial context and may get lost or overshadowed by other predictors. However, the little support for noticing nature is still important to understand as it may help, if even small, improve people's happiness.

So, even though I did not find a clear link between noticing nature and positive affect, it may be that noticing nature is only a small part of the path to happiness (see Figure 19). There may be other steps of engaging or augmenting experiences with nature that help see the benefit from it. As Gaston & Soga (2020) put it, interactions and experiences with nature are different; an *interaction* is a more superficial contact with nature through the senses, such as smell, sight, and sound (this could be considered more similar to my definition of noticing nature), whereas an *experience* is more of an engagement with nature. An example of an experience with nature may be savouring, characterized as actively doing something to enhance and prolong pleasurable feelings. Research has shown that savouring nature exposures can benefit well-being (Sato et al., 2017). Savouring activities, such as being present to absorb the moment, focusing on the moment to intensify it, taking a moment to feel grateful for the moment, and doing something to be able to share the moment with others (e.g., take an image of the natural element), could all be considered the next step of augmenting or enhancing nature experience after noticing it. In other words, a person first notices (or interacts with) nature and then may do something to augment the moments to turn it into an experience with nature. Thus, it may be that nature related individuals in an everyday moment notice the little bits of nature around them and then take a second to engage with it (e.g., potentially savour the moment with nature). To the best of my knowledge, no research has examined whether nature related individuals are better at savouring nature moments. One study found that high nature relatedness individuals identified that they valued

nature (had appreciation towards it) over the pandemic (Desrochers et al., 2022). Also, in Lomas's (2019) lexical enquiry of untranslatable words across the world that discuss one's eco-connection, appreciation and savouring were identified as relevant themes, where the act of paying attention and appreciating the nuances of nature was a way to connect with it. Together these findings, in combination with nature related people wanting to pay attention to nature (i.e., results for the current dissertation) and reporting doing more noticing nature activities (which may be more similar to augmenting noticing nature to an experience), suggest that nature related people may not only notice more nature, but also take the next step towards savouring the micro-exposure. This path (Figure 19) of interactions with nature to experiences with nature may improve state nature connection and subjective well-being. Having said that, it may also suggest that those low on nature relatedness may be missing out on the opportunity to savour nature because it is impossible to savour nature without noticing it (which they do less).

Figure 19

A visual representation of the conceptual framework model of how interactions with nature (i.e., noticing nature) is a part of the pathway to happiness.



Negative Affect

Negative affect was also measured in the three studies. Across all associations (i.e., nature relatedness, self-report and behavioural noticing nature) with negative affect, there was less consistency than for positive affect. For one, nature relatedness only showed significantly lower levels of negative affect in Study 3 (r[1053] = -.13 p < 001, see Appendix F for more details). Neither of the meta-analyses (Capaldi et al., 2014; Pritchard et al., 2020) previously examined the association between nature connectedness and negative affect; thus, it is less clear whether this result is unusual. However, it does seem to be in line with the previous finding in Nisbet et al. (2009), where nature relatedness was not significantly associated with negative affect in three studies. The lack of consistent negative affect associations is similar to McMahan and Estes's (2015) meta-analytic results showing that the benefits of nature exposure were stronger and more stable for positive affect than for negative affect. Exposure to nature is different from nature relatedness, but nature related individuals tend to spend more time in nature and (as suggested in this dissertation) notice more nature. This may imply that the association between nature relatedness and negative affect may also be less consistent.

Limitations and Future Directions

While the current studies of the dissertation have contributed valuable insights to the literature on how nature related individuals notice nature, it is essential to acknowledge that the current studies are not without limitations. For one, I used virtual nature, through images, in the current dissertation to control the amount of nature the participants were being exposed to. The use of virtual nature also made it easier to run the studies with both students and members of the larger community (i.e., MTurk workers from Canada and the United States). However, images do not capture the full essence of everyday moments. At all times, a person experiences a variety

of sensory inputs, including smells, sounds, tastes, textures, and many sights. The images used in the studies were limited to only displaying particular sights without any other sensory inputs to the participants. This may reduce the generalizability of the effect outside of a laboratory setting. It is unclear if nature related people would still be able to pay attention to nature amidst a multitude of competing information. Yet, the use of real nature exposure may prove difficult to control. I believe that noticing nature and the amount of nature exposure overlap, such that the more nature that one is exposed to will directly increase one's ability to notice it. Thus, it may be difficult to control the amount of nature each participant is being exposed to in real urban environments with some degree of nature. However, more creative ways of exploring noticing nature in real natural spaces should be explored next.

Further, the link of noticing nature with nature relatedness may rely on the specific images used for the tasks. The images used were of urban environments with varying degrees of nature. The natural elements in the images used were trees, plants (planted in pots or on the ground), bodies of water, green roof, mountains, and some wildlife (birds, ducks, wolves, dogs, and horses). These are only a few examples of what could be considered nature in an urban environment. As biologist Hanna Bjørgaas (2023) wants us to discover, many species live and thrive in urban spaces that we may not have noticed before, but are as fascinating as complex nature in pristine landscapes. Therefore, further considerations should be taken to broaden the scope of natural elements found in urban environments.

In a similar light, these natural elements found in the included images were natural elements that fit my definition of nature[12]. Due to the lack of a clear definition of nature across

[12] I acknowledge I had help finding the images for the studies by two research assistants that were aware of the purpose of the dissertation studies. Therefore, the assistants' understanding of the concept of nature was also reflected in the images they found.

people (Hartig & Kahn, 2016), I was faced with the decision of creating a priori boxes on Qualtrics of the natural elements in Phase 2. I chose to select the prototypical natural elements to include in the a priori boxes (i.e., the natural elements mentioned above). In these decisions, I did not consider the sky and clouds as natural elements for the images in the studies, but they could be defined as features of nature (Fehnker et al., 2021; Shaw, 2018). Additionally, when creating the a priori boxes, it was sometimes difficult to distinguish between the urban and natural areas in the images because natural elements often have irregular shapes. I did my best only to include the natural elements in the boxes (not the urban elements) and ensure I included all aspects of the natural element; however, it was not easily done. Some boxes may have a bit of overlap with other things in the image. This may create errors in the coded responses. All a priori boxes can be seen in the supplement on OSF.

There might also be cultural definitions of what counts as nature, which may impact the tool's generalizability or usefulness in cultures or geographical locations different than North America. In both studies of Phase 2, there was considerable consistency in the results, but these samples were of students and community members currently living in North America. The images used were meant to include typical natural elements found in North American cities. Previous research has shown that even the definition of nature is different around the world. Coscieme et al. (2019) found that in Western society, nature generally means pristine without humans, whereas in other languages (and cultures), nature may be defined by the integral connection between nature and humans (e.g., in many indigenous languages) or defined by its spiritual context. As language is important for how people see and understand the world, the different definitions of the word nature may impose different conceptualizations of nature. Additionally, as much as there are differences between different areas of North America, there

are also differences in the natural elements found around the world. For example, China is home to more flowing plant species than North and South America combined (Boudreau et al., 2023), and holds 10% of the world's flowering plant species (van Hoose, 2018). Thus, there may be native flower plant species that are expected in China (even in its city limits) that are not anticipated in other geographical areas. Together, this suggests that if the current tasks (or even the association of the tasks and nature relatedness) were explored in other cultural and geographical locations, further considerations should be taken regarding the natural elements of the images.

One of the goals of the dissertation was to explore the validity of the behavioural measure of noticing nature. I did this by exploring psychometric analyses; however, the behavioural measures of noticing nature I created may be further refined. For one, there was a lack of support for the first hypothesis stating that nature related individuals were more likely to perceive nature in an urban environment. As discussed, even though the measure of perception measured something reliably, it is not clear that this was the concept of noticing nature. Future research may attempt to explore a new tool for measuring noticing nature through perception. For instance, perception may be best examined through the limits of perception or the perceptual threshold by making the research question more implicit. This could be done by examining short-timed windows of exposure and asking participants to identify whether there was nature. For the behavioural task of attention, there was support that the measure reliably measured noticing nature. However, further refinement steps could be taken, such as test-retest stability, to explore if the measure is more like a stable trait. Attention could also be explored in more diverse images than the ones used in the current dissertation. For example, would those low in

nature relatedness be more likely to pay attention to the urban objects of a primarily natural setting?

Further, to extend the current support that nature related individuals were more likely to pay attention to nature, future research should also explore how the attention behavioural measure of noticing nature may be associated with other variables. One potentially related variable is pro-environmental behaviour. Previously, nature connection (relatedness) was associated with pro-environmental behaviours through both correlational and experimental explorations (McKay & Schmitt, 2019). Also, nature exposure is associated with pro-environmental behaviour (see review by DeVille et al., 2021). However, research has yet to examine how noticing nature may be associated with pro-environmental behaviour. In extra analyses not explored in the publication (Desrochers et al., 2022), my colleagues and I found associations between the single items, "I notice and see wildlife more," and "I value nature more (appreciation)," with doing more for the environment during the pandemic (Fall 2020) than before, $r(554) = .26$ $p < .001$ and $r(554) = .24$ $p < .001$. This may suggest that people who notice nature (and appreciate it) may be more likely to have pro-environmental attitudes and do something helpful for the environment. In the current studies, no pro-environmental attitudes or behaviour measures were included. However, the perspective subscale of the NRS is associated with topics of pro-environmental attitudes, making it the most similar measure in the dataset. The voluntary attention behavioural task was associated with the NRS perspective, suggesting that those who chose to pay attention to nature in urban environment images may also treat the environment better. Further, those who reported doing more noticing nature activities also had higher scores on the NRS perspective subscale (correlations ranged from .31 to .36, see Appendix F for more details). While these cross-sectional results do not establish a cause-and-

effect relationship, it serves as an initial step in investigating how noticing nature may help promote sustainable behaviours. However, further exploration should be taken with better measures of pro-environmental attitudes and behaviour.

Conclusion

In light of the ongoing conversation about the adverse effects of the extinction of experience, it is crucial to examine how individuals in urban areas can still engage with, connect with, and benefit from the natural world. As Shannahan et al. (2015a) suggest, two ways to combat extinction of experience is through: 1) increasing the amount of nature in urban environments and 2) assisting people in discovering how to connect with the nature readily available to them. Increasing the amount of nature in urban environments may be effective in promoting nature exposure, but it is also costly, and it is generally undervalued and under-implemented by cities' decisions makers. Showing people how to notice or connect with the nature they already have access to is a cost-effective, individual action, and an easy way to increase nature benefits in people. It shows people the "magic of the mundane" (Dobson et al., 2021). It may also, in turn, pressure policymakers to improve nearby nature to allow for better opportunities to notice (and maybe even savour) nature. Nisbet and Zelenski (2011) showed that people underestimated how a walk in nature would improve their mood, while overestimating their predicted mood for an urban walk. This may show that people are less likely to assume that exposing themselves to nature will benefit them, which will most likely reduce their likelihood of exposing themselves to nature. As other research shows, people tend to go to situations that provide them with improved positive affect (Gross, 2015; Mellers et al., 1999; Wilson & Gilbert, 2005).

The current dissertation showed preliminary support for the idea that nature related individuals may be better equipped to pay attention to small fragments of nature in urban environments that they already have available to them. They are also more likely to report doing noticing nature activities, like watching wildlife, listening to birdsongs, and smelling the wildflowers. Even though there were inconsistent findings on the behavioural noticing nature tasks associated with well-being in the current studies, the link between self-report noticing nature and well-being, in combination with the broader literature, proposes that noticing nature may still be the first step (process) on the path to happiness.

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Appendices
Appendix A
Study 1 Questionnaire

Task Block A*
You will be shown images of urban scenes (e.g. city street) and asked to report how much nature you see in the image.

(See Figure 3 for example images).

Rate how much nature you perceive in this urban scene?

1 – there is no nature for an urban scene	2	3	4	5	6	7 – there is a lot of nature for an urban scene
o	o	o	o	o	o	o

Task Block B*
In this block, you will also be shown nature images. Please report the percentage of nature you perceive in the image.

(See Figure 3 for example images)

What percentage (0=100%) of the image is nature? _____

Nature Relatedness:
For each of the following, please rate the extent to which you agree with each statement, using the scale from 1 to 5 as shown below. Please respond as you really feel rather than how you think "most people" feel.

1 – Disagree strongly	2 – Disagree a little	3 – Neither agree or disagree	4 – Agree a little	5 – Agree strongly
o	o	o	o	o

1. My connection to nature and the environment is a part of my spirituality
2. My relationship to nature is an important part of who I am
3. I feel very connected to all living things and the earth
4. I am not separate from nature, but a part of nature
5. I always think about how my actions affect the environment
6. I am very aware of environmental issues
7. I think a lot about the suffering of animals
8. Even in the middle of the city, I notice nature about me
9. My feelings about nature do not affect how I live my life
10. Humans have the right to use natural resources any way we want

11. Conservation is unnecessary because nature is strong enough to recover from any human impact
12. Animals, birds, and plants have fewer rights than humans
13. Some species are just meant to die out or become extinct
14. Nothing I do will change problems in other places on the planet
15. The state of nonhuman species is an indicator of the future for humans
16. The thought of being deep in the woods, away from civilization, is frightening
17. My ideal vacation spot would be a remote, wilderness area
18. I enjoy being outdoors, even in unpleasant weather
19. I don't often go out in nature
20. I enjoy digging in the earth and getting dirt on my hands
21. I take notice of wildlife wherever I am

Noticing Nature Activities:
For the following activities, please rate how frequent you do them:

1 – Never	2	3	4 – Often
o	o	o	o

1. Look at natural scenery from indoors or while on trips.
2. Watch wildlife.
3. Take photos or videos of wildlife, plants, or animals.
4. Listen to birdsong.
5. Smell wildflowers.
6. Take photos or draw/paint a picure of a natural view, plant, flower, or animals, etc.
7. Collect shells or pebbles on the beach.
8. Study nature with a microscope or binoculars.
9. Enjoy watching a wild animal playing.
10. Take time to notice butterflies and/or bees.
11. Stop to look at the moon and/or stars.
12. Watch the sunrise.
13. Watch clouds.

SPANE:
Below are 15 emotions with which you may have experience or not. Please think about what you have been dong and experiencing during the past four weeks. Then report how much you experienced each of the following feelings, using the 1-5 scale below.

1 – Very rarely or Never	2 – Rarely	3 – Sometimes	4 – Often	5 – Very often or Always
o	o	o	o	o

1. Positive
2. Negative

3. Good
4. Bad
5. Pleasant
6. Unpleasant
7. Happy
8. Sad
9. Afraid
10. Joyful
11. Angry
12. Contented
13. Awe
14. Wonder
15. Amazement

Demographic:

Please tell us a bit about your background.

1. Do you identify as: ☐Man ☐ Woman
☐Gender-fluid, non-binary, and/or two-spirited
☐Trans man ☐Trans woman
☐Prefer to self-describe: _____
☐Prefer not to say

2. Age _____ years

3. You may belong to one or more racial or cultural groups on the following list. Are you?

- White
- Indigenous (First Nations, Métis, Inuk/Inuit)
- South Asian (e.g. East Indian, Pakistani, Sri Lankan)
- Chinese
- Black
- Filipino
- Latin American
- Arab
- Southeast Asian (e.g. Vietnamese, Cambodian, Malaysian, Loatioan)
- West Asian (e.g. Iranian, Afghan)
- Korean
- Japanese
- Another racial or cultural group not included above: _____ (Please specify)
- Prefer not to answer

4. Where did you spend the most time while growing up? (Please choose only one response from the options below)

oCity (downtown)
oCity (suburbs)
oExurban area (development beyond suburbs)
oSmall town
oRural or farm
oOther (please specify) _____

5. Where do you live now?

oCampus residence
oCity (downtown)
oCity (suburbs)
oExurban area (development beyond suburbs)
oSmall town
oRural or farm
oOther (please specify) _____

6. Lastly, please indicate how you completed this survey. Select one of the following:

oPhone
oLaptop
oDesktop
oIpad/Tablet
oOther: _____

*The order of perceived nature items will be counterbalanced.

Appendix B
Study 1 Psychometric Analysis Results at 100 Participants

Due to using a new experimental tool, it is important to examine if the measures are working appropriately before moving on to data analysis. With this in mind, I used optional stopping technique to stop recruitment at 100 participants to allow me to test the variability in the ratings of the images. The variability of the ratings for the 7-point scale rating item and the percentage rating item were tested separately. The psychometric analyses were pre-registered on OSF.

The 7-point Scale Rating Item – "Rate how much nature you perceive in this urban scene?"

First, the variability of the 7-point scale rating item, "Rate how much nature you perceive in this urban scene?", was examined with a few different data analysis tools. I began by testing the distribution of the ratings of each image with descriptive statistics, such as mean, median, mode, standard deviation, minimum, and maximum. Distribution was also visually examined with histograms (Figure B-1). Next, reliability analyses were explored to see how well the images would aggregate together by computing an overall Cronbach's alpha, in addition to examining how each item would affect the Cronbach's alpha if the item were deleted. Lastly, inter-image correlations were examined. See Table B-1 for the results of these quantitative descriptions of each image.

Figure B-1

Histograms of the distribution of the 7-point scale rating item of nature perception for each image in the optional stopping step for Study 1 (N = 100).

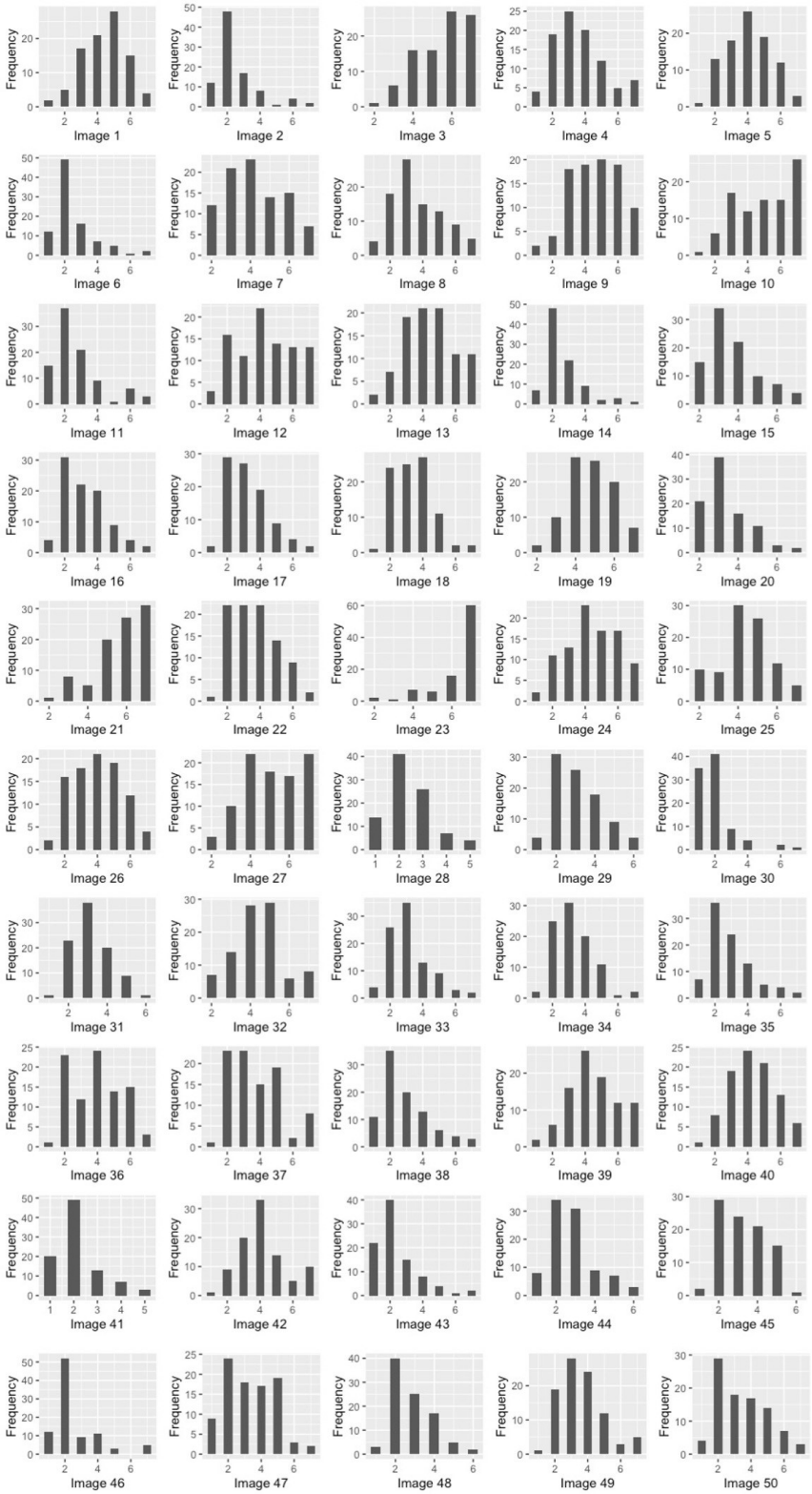


Table B-1

Descriptive statistics of the 5-point scale rating item of nature perception for each image in the optional stopping step of Study 1 (N =100).

Image	*N*	Mean	Median	Mode	SD	Min	Max	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
1	92	4.40	5	5	1.34	1	7	.951	-.09	.52
2	92	2.54	2	2	1.31	1	7	.951	-.26	.69
3	92	5.52	6	6	1.31	2	7	.952	-.11	.54
4	92	3.65	3	3	1.56	1	7	.951	.01	.56
5	92	4.05	4	4	1.38	1	7	.951	-.05	.52
6	92	2.51	2	2	1.25	1	7	.951	-.13	.65
7	92	4.22	4	4	1.49	2	7	.950	-.03	.59
8	92	3.67	3	3	1.56	1	7	.952	.04	.44
9	92	4.61	5	5	1.50	1	7	.952	-.13	.50
10	92	4.99	5	7	1.72	1	7	.953	-.12	.51
11	92	2.72	2	2	1.50	1	7	.952	-.03	.67
12	92	4.29	4	4	1.75	1	7	.950	-.01	.60
13	92	4.40	4	4[a]	1.53	1	7	.951	-.04	.58
14	92	2.61	2	2	1.16	1	7	.951	-.12	.69
15	92	3.70	3	3	1.32	2	7	.951	.12	.48
16	92	3.21	3	2	1.36	1	7	.951	-.01	.55
17	92	3.26	3	2	1.30	1	7	.950	.00	.64
18	92	3.40	3	4	1.21	1	7	.950	.07	.69
19	92	4.79	5	4	1.19	2	7	.951	.00	.56
20	92	3.37	3	3	1.18	2	7	.951	.12	.56
21	92	5.71	6	7	1.29	2	7	.952	-.17	.59
22	92	3.66	4	2[a]	1.39	1	7	.950	.04	.60
23	92	6.32	7	7	1.18	2	7	.952	-.33	.59
24	92	4.40	4	4	1.58	1	7	.951	-.06	.58
25	92	4.39	4	4	1.30	2	7	.951	-.05	.49
26	92	3.99	4	4	1.49	1	7	.951	.08	.50
27	92	5.11	5	4[a]	1.46	2	7	.951	-.11	.56
28	92	2.41	2	2	0.99	1	5	.951	.05	.60
29	92	3.10	3	2	1.22	1	6	.951	-.01	.62
30	92	1.95	2	2	1.13	1	7	.952	-.33	.58
31	92	3.17	3	3	0.99	1	6	.951	.08	.69
32	92	4.40	4	5	1.29	2	7	.951	.03	.59
33	92	3.15	3	3	1.28	1	7	.951	.15	.59
34	92	3.26	3	3	1.20	1	7	.951	.03	.64
35	92	2.92	3	2	1.35	1	7	.951	.04	.66
36	92	3.91	4	4	1.53	1	7	.951	-.08	.59
37	92	3.73	3	2[a]	1.55	1	7	.951	.08	.56
38	92	2.91	2.5	2	1.48	1	7	.951	-.03	.55
39	92	4.48	4	4	1.52	1	7	.951	.05	.51
40	92	4.29	4	4	1.40	1	7	.951	.09	.50

41	92	2.17	2	2	0.97	1	5	.952	-.17	.67
42	92	4.14	4	4	1.43	1	7	.951	.05	.49
43	92	2.38	2	2	1.32	1	7	.953	-.17	.58
44	92	2.80	3	2	1.18	1	6	.951	.08	.58
45	92	3.23	3	2	1.16	1	6	.952	-.03	.42
46	92	2.58	2	2	1.43	1	7	.952	-.25	.69
47	92	3.33	3	2	1.49	1	7	.951	.04	.54
48	92	2.86	3	2	1.08	1	6	.951	.04	.69
49	92	3.61	3	3	1.36	1	7	.951	.05	.56
50	92	3.45	3	2	1.52	1	7	.951	.05	.60

Note: [a] This identifies that there was more than one mode and that the reported value is the smallest.

The criteria pre-registered to examine the variability of the ratings were: mean relatively around the middle of the scale (value of 4), standard deviation above the value of 1, minimum/maximum approximately around the extreme anchors of the scale (value of 1 to 7), the removal of the item does not impact the reliability, and inter-image correlations are not problematic. Some of the criteria identified good variability. The overall mean value of how much nature there was in the urban scene across all the images was 3.67 (*SD* = 1.36), which is close to the middle value (4) of the scale. The mean did vary by images (minimum *M* = 1.95, maximum *M* = 6.32). Further, only three images had a standard deviation below the criteria threshold of 1 (Images 28, 31, & 41). The range of standard deviation values was 0.97 to 1.75. The three images with low standard deviations were abutting the threshold of 1, making this criterion less of a concern. The images reliably aggregate, Cronbach's $\propto$ = .95 and the criteria of how an item would impact the reliability demonstrate that the reliability is robust to removing each one of the images individually (as seen in the *Cronbach's* $\propto$ *if Item Deleted* column of Table B-1).

However, other methods, such as inter-image correlations demonstrated mixed results. It was also less clear about the interpretation of the inter-image correlations. For example, many of

the negative correlations were between images that contained different skewed ratings (i.e. images with lower nature ratings negatively correlated with images with higher nature ratings). Therefore, this criterion was observed, but not used as a strong case for the removal of the image.

From the different criteria, there were eight images that were flagged as potentially problematic (Images 2, 6, 14, 23, 28, 30, & 41). All but one image were left skewed with a low mean (also median and mode). Image 23 was right skewed with a mean of 6.32 (median and mode = 7). These eight images, however, did have some variability within their smaller range of values. It was less clear if the images should be removed, or rather be flagged as performing the least beneficially. These low performing images did not give any clear demonstration that they would be problematic for the hypothesis testing. At this point, the images were deemed okay to keep.

The percentage rating item - What percentage of the image is nature?

Similar to the 7-point scale rating item, distribution of the percentage rating item, "What percentage of the image is nature?", was explored through descriptive statistics (mean, standard deviation, minimum, maximum, percentiles) and histograms. Next, reliability analyses and inter-image correlations were explored. The quantitative values of these analyses can be seen in Table B-2 and the visual representation of the distribution in Figure B-2.

Table B-2

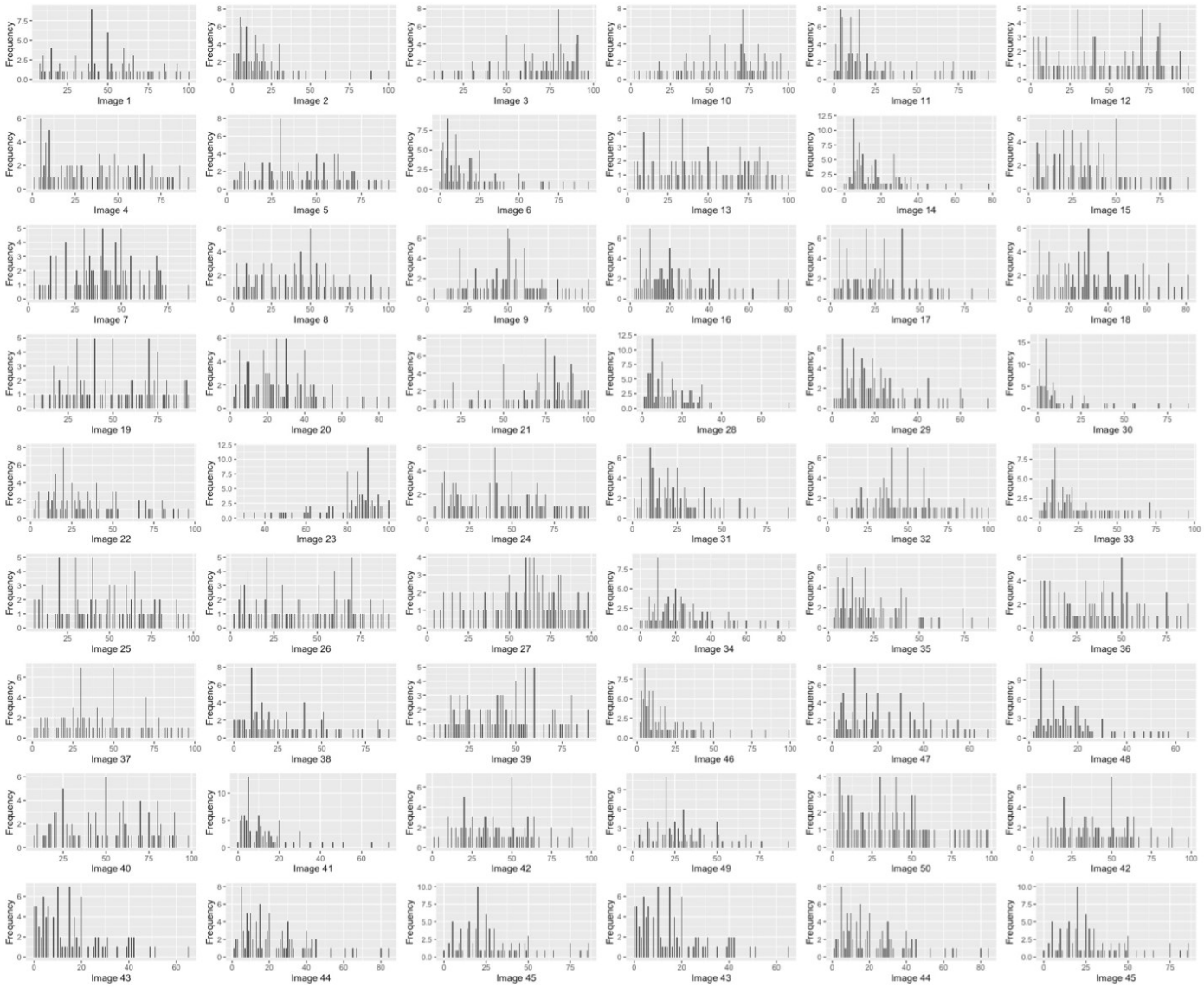
Descriptive statistics of the percentage rating item of nature perception for each image in the optional stopping step of Study 1 (N =100).

						Percentile					
Image	*N*	Mean	SD	Min	Max	25	50	75	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
1	88	47.52	23.96	8	100	29.25	50	64.75	.963	.18	.61
2	87	17.41	17.68	1	100	6	12	21	.964	-.06	.59
3	89	66.85	22.32	8	97	50.5	74	83	.964	.09	.64
4	89	37.04	25.41	1	91	11.5	35	58.5	.964	.20	.56
5	89	41.19	21.83	4	90	24.5	40	60	.964	.08	.63
6	87	19.29	20.32	0	94	5	12	25	.964	.00	.65
7	90	40.54	18.55	3	86	30	40	51.25	.964	.04	.64
8	89	41.30	25.57	1	100	17.5	44	59.5	.963	.16	.64
9	88	48.83	21.63	4	100	31.5	50	60.75	.964	.08	.61
10	90	58.76	24.42	2	100	39.5	67	76.25	.965	-.12	.59
11	86	22.98	23.81	0	93	5	14.5	30.25	.964	.06	.69
12	88	48.42	28.31	2	100	26.5	46.5	71	.963	.23	.59
13	89	46.78	26.75	4	100	24	44	72	.964	.10	.63
14	88	16.53	13.83	0	78	7	11.5	24.5	.964	.07	.63
15	89	32.61	20.28	3	91	16.5	30	43.5	.964	.02	.56
16	87	24.29	17.92	2	80	10	20	31	.964	.08	.56
17	87	27.80	18.44	2	88	13	25	40	.964	.09	.61
18	88	31.74	20.53	2	81	16.25	28.5	43.5	.963	.20	.65
19	89	49.83	23.22	6	92	30	50	70	.963	.22	.54
20	89	27.47	17.54	2	85	14.5	25	39	.964	.09	.64
21	89	69.45	23.27	9	100	55.5	75	88	.964	.09	.63
22	90	34.31	23.01	2	96	15.75	29	48.25	.963	.20	.60
23	88	80.57	15.67	30	100	78.5	85.5	90	.964	-.06	.64
24	88	43.24	24.67	4	95	20	41	64	.964	.10	.54
25	89	46.09	24.21	5	97	28	44	65	.964	.19	.56
26	89	43.15	25.90	2	91	20.5	42	67	.964	.12	.56
27	89	59.55	23.33	4	98	46.5	62	77	.963	.17	.64
28	87	13.52	11.16	1	74	5	10	20	.964	.07	.63
29	88	21.01	15.43	1	73	10	17.5	26.75	.963	.11	.61
30	86	12.23	17.26	0	87	3	5	11.25	.964	-.02	.63
31	88	23.58	16.23	1	87	11.25	20	30	.964	-.02	.65
32	89	45.46	21.98	4	100	32.5	41	60	.963	.25	.56
33	87	23.85	20.52	0	96	9	17	31	.964	.15	.59

34	88	23.66	17.63	0	84	10	20	30.75	.964	.03	.55
35	88	24.11	18.20	3	88	10	19.5	34.75	.964	.06	.68
36	89	37.20	22.05	1	87	18.5	36	51	.964	.11	.63
37	88	40.52	24.20	1	96	21.5	36	55.75	.964	.08	.54
38	89	26.29	21.95	0	88	10	19	40	.963	.25	.65
39	90	44.20	21.28	6	89	24	43	60	.964	.13	.58
40	88	50.30	24.52	8	98	27	50	70	.964	.09	.55
41	84	12.56	13.61	0	73	4	9	16	.964	.09	.69
42	90	40.01	20.64	1	98	23.75	38	53.5	.964	.09	.56
43	88	15.66	13.82	0	65	5	12.5	20	.964	-.06	.56
44	89	22.07	17.59	1	84	8.5	18	30.5	.964	.18	.60
45	89	28.31	20.08	0	86	15	24	37.5	.964	.19	.68
46	87	18.47	18.64	2	99	5	12	25	.964	-.06	.59
47	85	22.36	17.13	1	68	9	18	35	.964	.00	.51
48	87	17.21	13.43	2	65	8	14	21	.964	-.12	.58
49	90	29.17	17.65	1	92	18.75	26	39	.964	.10	.64
50	87	35.02	25.64	1	98	12	31	51	.964	.13	.65

Figure B-2

Histograms of the distribution of the percentage scale rating item of nature perception for each image in the optional stopping step for Study 1 (N = 100).



Distinctively from the 7-point scale rating item, the criteria pre-registered to identify images that were problematic and warranted removal was less clear. Pre-registered in the prospectus (and on OSF) was to examine the range of the scale used through both descriptives and visual representation. When examining the overall percentage ratings across all the images, the mean value of nature was 35.74% (*SD* = 20.57). The ratings were reliable across the 50 images, Cronbach's $\propto$ = .96. These values were taken into account when looking at the variability of each individual image.

The reliability of the percentage rating item was robust to the removal of each image (see *Cronbach's* $\propto$ *if Item Deleted* in Table B-2). Most of the items used the majority of the scale, in terms of minimum and maximum scores. Three of the images only used approximately 65% of the scale: Image 43 (*Min* = 0, *Max* = 65), Image 47 (*Min* = 1, *Max* = 68), and Image 48 (*Min* = 2, *Max* = 65). The images with the lowest averages were the same images previously identified with low means for the 7-point scale rating item (Images 2, 6, 14, 28, 30, 41, & 46).

Overall, across both items, there are 8 images identified as performing the poorest, in terms of variability. The majority of those images were left-skewed with a low mean (seven images), whereas a single item was right-skewed with a large mean. Despite these suspect means, the images showed appropriate standard deviations (the 7-point scale rating item: between 0.99 and 1.43; the percentage rating item: between 11.16 to 20.32) and a good range of values (the 7-point scale rating item: smallest range 1 to 5; the percentage rating item: smallest range 30 to 100). With this in mind, I chose to keep these images to explore their variability at the final sample size (*N* = 300) and potentially use them for examining the association with nature relatedness.

Exploratory Analysis

As an exploratory analysis (i.e. not pre-registered), the association between the ratings of items 1 and 2 for the same image was explored. The mean value of the 7-point scale rating item across all images is significantly correlated with the mean value of the percentage rating item across all images, $r(89) = .73$ $p < .001$. Image 48 did not have a significant correlation between the ratings on both items, $r(89) = .16$ $p = .14$. (This image was not one of the eight images identified as problematic). All other images had a significant association between the ratings between the two items (range: .34 to .79).

Another exploratory analysis was to explore if there were participants who responded in a string manner (i.e. all the same response across images). To do this, variance across all the images was computed for each item. Only two participants were identified to have 0 variance across the ratings for only the 7-point scale rating item. When exploring how robust the variability analyses were to excluding these participants, the results did not change. Therefore, this suggested the participants could be included in these psychometric analyses (i.e. at 100 participants); however, this will be examined again at the full sample and they may be excluded at that point.

Appendix C
Principle Component Analysis Results for Study 1

The 7-point Rating Scale Item

As an exploratory psychometric analysis for the full sample, I ran principle components analyses (PCA) to measure the underlying explanatory variables through different independent linear combinations (APA, 2015). Field (2013) outlines in the textbook *Discovering Statistics Using IBSM SPSS Statistics* the general procedure for PCA as followed: 1) check for appropriate sample size, 2) check correlations between variables (in our case images), 3) factor extraction, 4) factor rotation (however, because I only expect to have a single solution, rotation will be unnecessary) , and then 5) reliability. I will follow these procedure suggestions for both items.

For the 7-point scale rating item, the sample size was deemed appropriate for PCA through Kaiser-Meyer-Olkin measure of sampling adequacy (KMO), KMO = .93. Additionally, when examining the KMO for each individual image, all 50 images had a KMO of .80 or above, indicating an appropriate sample size for PCA. When examining inter-image correlations, Barlett's test was significant, $X^2(df = 1225) = 7400.13\, p < .001$, indicating that our inter-image correlation matrix is significantly different than an identity matrix. Field (2013) suggests that inter-item correlations should be significant, above the value of .3 to show that the images should are measuring the same thing, and below .8 to reduce multicollinearity. None of the images showed strong correlations (above .8). However, on the low end, every image had at least 11 correlations below .3. With this in mind, significance was examined instead. Twenty-two images had all significant inter-image correlations (this set was labelled best images). Twenty-two images of the 28 images that had non-significant inter-image correlations had less than 6 inter-image correlations that were non-significant (this set was labelled satisfactory), whereas the set labelled problematic were images with the range of 5 to 14 non-significant correlations (6

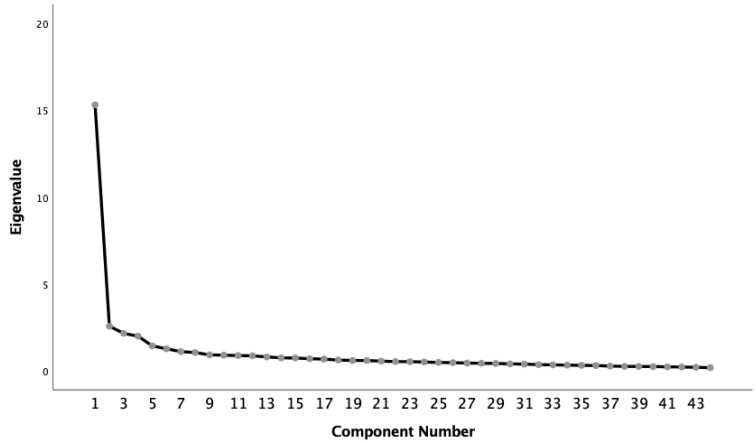
images fit this criteria; Images 3, 10, 21, 23, 30, & 46). Across all inter-image correlations, there were 19% (236) of the residuals (i.e. the difference between the inter-image correlations based on the model versus the real data) with values above .05. All, but three images had residuals above the .05 threshold (ranged between 0 to 7). Across all information, there were 6 images deemed unsuitable for PCA (i.e. images with more than four non-significant inter-image correlations; Images 3, 10, 21, 23, 30, & 46) and therefore, factor extractions were done on only the 44 images that were labelled best and satisfactory.

The scree plot (Figure C-1) suggests that there is one main factor (Eigenvalue = 15.32), with 8 other potential factors with Eigenvalues above 1 (next highest 2.56). For the single factor, all images had factor loadings of .40 or above. When examining factor loadings for the other factors, three images were identified as having higher loadings on other factors than on factor 1. These images (8, 9, & 43) were removed.

The reliability for the 7-point scale rating item of the 41 images, identified as appropriate, was identified as good: Cronbach's $\propto = .95$, Omega = .95 95% CI [.94, .96], split-half reliability $r(280) = .90$ $p < .001$ 95% CI [.88, .93]. All three of these values are reduced from the reliability of all 50 images.

Figure C-1

Scree plot for the 7-point scale rating item with 45 images of Study 1 (N = 284).



The Percentage Rating Item

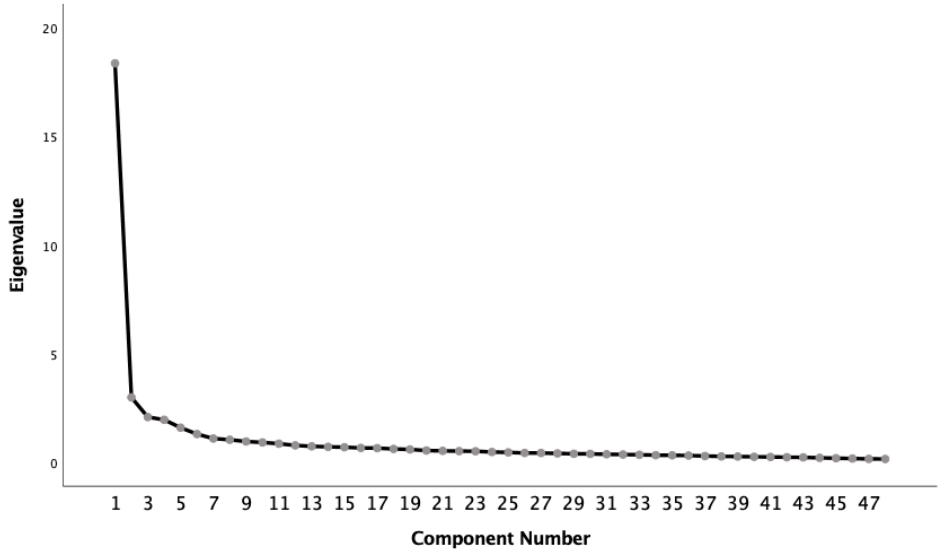
Next, I ran a PCA for the percentage rating item. The overall KMO suggests an appropriate sample size for PCA, KMO = .94, and the KMO value was above the threshold of .80 for all individual images. When examining the inter-image correlations, the Barlett's test was significant, $X^2(df = 1225) = 6793.21$ $p < .001$, indicating this matrix is significantly different from an identity matrix. When looking at non-significant inter-image correlations, there were only two images that had above five non-significant correlations. Image 10 had eight non-significant correlations, whereas Image 23 had five. These two images were removed from the set. None of the correlations were too high (above .8). Every image had at minimum 3 inter-image correlations that were below the threshold of .3. The highest number of inter-image correlations below .3 was 45 for Image 10. With all of this in mind, Images 10 and 23 were removed to run the PCA.

For the 48 images, 8 factors were identified as having Eigenvalues above 1 (see Figure C-2). However, there is a strong suggestion for a single factor, Eigenvalue = 18.38, with the next factor having a value of 2.98. The factor loadings for the single factor ranged from 0.46 to 0.76. All items loaded the strongest on the single factor. Therefore, no other images were identified to be removed.

The reliability for the percentage rating item, with 48 images identified as appropriate, was good: Cronbach's $\propto = .96$, Omega = .96 95% CI [.95, .97], split-half reliability $r(279) = .92$ $p < .001$ 95% CI [.89, .95]. The values of reliability through Cronbach's alpha and Omega are the same as when using the sample of 50 images; however, the split-half reliability was slightly better for this set.

Figure C-2

Scree plot for the percentage rating item with 48 images of Study 2 (N = 284).



By running these PCAs, I was able to identify a set of images to aggregate for exploring the association with nature relatedness. A different number of images were identified as the best suited for aggregations for each of the items. For the 7-point rating scale item, 41 images were identified as the best, whereas for the percentage rating item, 48 images were identified.

Appendix D
Studies 2 and 3 Questionnaire

Block A*

For this first task, you will be shown 50 images of urban environments. For each image you will be shown, you will be asked to click the first thing you notice.

But first, let's do a practice image. In the following image, please click on the first things you notice in the image.

Great. Now let's start the task at hand. The following images will be of urban environments that you might be familiar with. Please click the first thing in this image that you notice first. Do not think about it, just click on the image where your eyes first went. You will only have 7 seconds to click on the image. After 7 seconds, the survey will automatically move forward.

Click on the first thing you noticed and then click Next.

(See Figure 6 for example images)

Block B*

For the next task, you will be shown 50 images of urban environments. Please choose something in the image that you would want to focus your attention on. The something you chose can be anything in the image. Imagine you were in this environment, what would you want to focus your attention on? You can take as long as you need to look at each image. Please click on the thing you want to focus on and then click Next.

(See example images in Figure 6)

Nature Relatedness:

For each of the following, please rate the extent to which you agree with each statement, using the scale from 1 to 5 as shown below. Please respond as you really feel rather than how you think "most people" feel.

1 – Disagree strongly	2 – Disagree a little	3 – Neither agree or disagree	4 – Agree a little	5 – Agree strongly
o	o	o	o	o

1. My connection to nature and the environment is a part of my spirituality
2. My relationship to nature is an important part of who I am
3. I feel very connected to all living things and the earth
4. I am not separate from nature, but a part of nature
5. I always think about how my actions affect the environment

6. I am very aware of environmental issues
7. I think a lot about the suffering of animals
8. Even in the middle of the city, I notice nature about me
9. My feelings about nature do not affect how I live my life
10. Humans have the right to use natural resources any way we want
11. Conservation is unnecessary because nature is strong enough to recover from any human impact
12. Animals, birds, and plants have fewer rights than humans
13. Some species are just meant to die out or become extinct
14. Nothing I do will change problems in other places on the planet
15. The state of nonhuman species is an indicator of the future for humans
16. The thought of being deep in the woods, away from civilization, is frightening
17. My ideal vacation spot would be a remote, wilderness area
18. I enjoy being outdoors, even in unpleasant weather
19. I don't often go out in nature
20. I enjoy digging in the earth and getting dirt on my hands
21. I take notice of wildlife wherever I am

Noticing Nature Activities:
For the following activities, please rate how frequent you do them:

1 – Never	2	3	4 – Often
o	o	o	o

1. Look at natural scenery from indoors or while on trips.
2. Watch wildlife.
3. Take photos or videos of wildlife, plants, or animals.
4. Listen to birdsong.
5. Smell wildflowers.
6. Take photos or draw/paint a picure of a natural view, plant, flower, or animals, etc.
7. Collect shells or pebbles on the beach.
8. Study nature with a microscope or binoculars.
9. Enjoy watching a wild animal playing.
10. Take time to notice butterflies and/or bees.
11. Stop to look at the moon and/or stars.
12. Watch the sunrise.
13. Watch clouds.

SPANE:
Below are 15 emotions with which you may have experience or not. Please think about what you have been dong and experiencing during the past four weeks. Then report how much you experienced each of the following feelings, using the 1-5 scale below.

1 – Very rarely or Never	2 – Rarely	3 – Sometimes	4 – Often	5 – Very often or Always

o o o o o

1. Positive
2. Negative
3. Good
4. Bad
5. Pleasant
6. Unpleasant
7. Happy
8. Sad
9. Afraid
10. Joyful
11. Angry
12. Contented
13. Awe
14. Wonder
15. Amazement

Demographic:

Please tell us a bit about your background.

1. Do you identify as: ☐Man ☐ Woman
☐Gender-fluid, non-binary, and/or two-spirited
☐Trans man ☐Trans woman
☐Prefer to self-describe: _____
☐Prefer not to say

2. Age _____ years
3. You may belong to one or more racial or cultural groups on the following list. Are you?

- White
- Indigenous (First Nations, Métis, Inuk/Inuit)
- South Asian (e.g. East Indian, Pakistani, Sri Lankan)
- Chinese
- Black
- Filipino
- Latin American
- Arab
- Southeast Asian (e.g. Vietnamese, Cambodian, Malaysian, Loatioan)
- West Asian (e.g. Iranian, Afghan)
- Korean
- Japanese
- Another racial or cultural group not included above: _____ (Please specify)

- Prefer not to answer

4. Where did you spend the most time while growing up? (Please choose only one response from the options below)

oCity (downtown)
oCity (suburbs)
oExurban area (development beyond suburbs) oSmall town
oRural or farm
oOther (please specify) _____

5. Where do you live now?

oCampus residence
oCity (downtown)
oCity (suburbs)
oExurban area (development beyond suburbs)
oSmall town
oRural or farm
oOther (please specify) _____

6. Lastly, please indicate how you completed this survey. Select one of the following:

oPhone
oLaptop
oDesktop
oIpad/Tablet
oOther: _____

*For Study 2, the order of Block A & B were presented in the same order. For Study 3, participants were randomly assigned to see one of two blocks.

Appendix E
Psychometric Analysis Results for Optional Stopping Steps of Study 2

Similar to Phase 1, optional stopping was used to examine the variability of the attention items for each item. This was done at a sample of 100 participants before recruiting the full sample target of 300 participants. After recruiting 100 participants, the variability of each attention item was explored separately. As pre-registered (see OSF), if more than seven images were identified as having poor variability, they were removed, replaced, and recruitment began again at zero. For Study 2, three unique optional stopping psychometric analyses were run before identifying appropriate images to continue to recruiting the full sample target of 300. The results of the three rounds are described below.

Round 1

The variability of the two attention items were explored separately. To test variability of the images for both items, descriptive statistics (minimum, maximum, mean, standard deviation, and frequency of the are clicked), standardized distribution (Z-scores and visual representation), reliability analysis and inter-image correlations were explored.

First Attention

The first attention mean across all images was 0.18 (*SD* = 0.19; Min = 0.00, Max = 0.81). This suggesting, across all clicks, nature was clicked for the first attention item 18% of the time. The amount of nature area clicks varied per image, ranging from 0% to 81%. All quantitative results can be seen in Table E-1.

When exploring the variability in the first attention clicks on either the nature area or the other areas, three images were identified to have no variance in the responses (i.e. all participants chose the other area instead of the nature area of the image). This warrants the removal of

Images 13, 15, and 49. When running the reliability analysis, three other images were identified to lack variance to be able to run the analysis and therefore were removed for this reason (Images 18, 34, & 44).

The reliability of the remaining 44 images was quite low for aggregation, Cronbach's $\propto$ = .61. When examining how robust the reliability is for the removal of each image individually, it identified that 28 images were good (i.e. the Cronbach's $\propto$ decreases when the image is removed), 13 images were poor (i.e. the Cronbach's $\propto$ increases when the image is removed), and three images showed no change to the Cronbach's $\propto$ when removed. Although, the 13 images that were identified as poor by showing an increase in the Cronbach's $\propto$ when removed, the values of the Cronbach's $\propto$ only increased by a range of .001 and .023. Therefore, this metric was not used to influence the removal of any more images.

Standardized distribution of the mean scores for each image was created by creating Z-scores. When exploring the overall standardized distribution of the Z-transformed mean scores for each image, the visual representation shows a right-skewed distribution (See Figure E-1). This visual demonstrating that more other areas were clicked as being first noticed versus the nature areas, making more mean values for each image closer to 0, skewing the distribution. The distribution was tested for normality by running the examine analysis in SPSS. The distribution was identified as not normal, quantitatively (Shapiro-Wilkes estimate = .84 $p < .001$) and visually (see QQ plots in Figure E-1).

Table E-1

Descriptive statistics of the first attention item for Study 2 optional stopping Round 1 (N = 100).

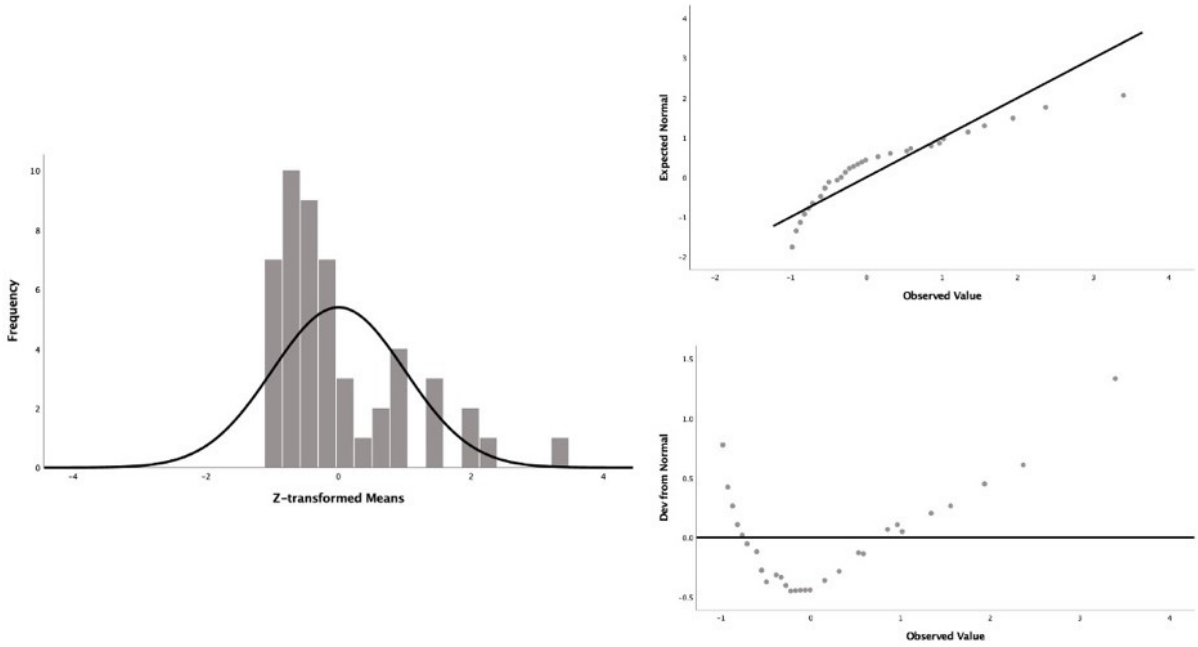
Image	*N*	Min	Max	*M*	*SD*	Other	Nature	~ % of nature	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
1	88	0	1	0.03	0.18	85	3	3	-0.82	.606	-.18	.70
2	93	0	1	0.54	0.50	43	50	54	1.94	.588	-.32	.33
3	92	0	1	0.07	0.25	86	6	7	-0.60	.597	-.18	.56
4	92	0	1	0.24	0.43	70	22	24	0.31	.628	-.35	.28
5	92	0	1	0.08	0.27	85	7	8	-0.55	.609	-.32	.48
6	91	0	1	0.02	0.15	89	2	2	-0.87	.601	-.30	.70
7	90	0	1	0.37	0.48	57	33	37	1.02	.586	-.21	.36
8	92	0	1	0.62	0.49	35	57	62	2.37	.586	-.34	.31
9	92	0	1	0.08	0.27	85	7	8	-0.55	.606	-.30	.39
10	93	0	1	0.03	0.18	90	3	3	-0.82	.612	-.27	.31
11	93	0	1	0.43	0.50	53	40	43	1.34	.605	-.27	.40
12	86	0	1	0.36	0.48	55	31	36	0.96	.605	-.23	.40
13	90	0	0	0.00	0.00	90	0	0	-0.98	N/A	N/A	. N/A
14	92	0	1	0.13	0.34	80	12	13	-0.28	.597	-.24	.49
15	92	0	0	0.00	0.00	92	0	0	-0.98	N/A	.00	.00
16	92	0	1	0.05	0.23	87	5	5	-0.71	.602	-.16	.70
17	92	0	1	0.12	0.33	81	11	12	-0.33	.585	-.26	.38
18	93	0	1	0.01	0.10	92	1	1	-0.93	N/A	N/A	N/A
19	91	0	1	0.16	0.37	76	15	16	-0.12	.613	-.23	.45
20	88	0	1	0.08	0.27	81	7	8	-0.55	.594	-.25	.70
21	90	0	1	0.07	0.25	84	6	7	-0.60	.597	-.19	.47
22	92	0	1	0.17	0.38	76	16	17	-0.06	.609	-.24	.48
23	89	0	1	0.81	0.40	17	72	81	3.40	.595	-.30	.30
24	93	0	1	0.13	0.34	81	12	13	-0.28	.604	-.25	.31

25	92	0	1	0.29	0.46	65	27	29	0.58	.587	-.24	.40
26	91	0	1	0.21	0.41	72	19	21	0.15	.606	-.27	.34
27	89	0	1	0.15	0.36	76	13	15	-0.17	.568	-.20	.49
28	89	0	1	0.54	0.50	41	48	54	1.94	.621	-.26	.40
29	91	0	1	0.12	0.33	80	11	12	-0.33	.589	-.19	.56
30	91	0	1	0.18	0.38	75	16	18	-0.01	.622	-.34	.33
31	92	0	1	0.08	0.27	85	7	8	-0.55	.591	-.27	.61
32	91	0	1	0.08	0.27	84	7	8	-0.55	.600	-.23	.69
33	89	0	1	0.07	0.25	83	6	7	-0.60	.602	-.20	.42
34	91	0	1	0.01	0.10	90	1	1	-0.93	N/A	N/A	N/A
35	90	0	1	0.37	0.48	57	33	37	1.02	.596	-.27	.32
36	91	0	1	0.11	0.31	81	10	11	-0.39	.588	-.15	.42
37	91	0	1	0.03	0.18	88	3	3	-0.82	.599	-.13	.70
38	90	0	1	0.09	0.29	82	8	9	-0.50	.602	-.19	.33
39	93	0	1	0.14	0.35	80	13	14	-0.23	.583	-.35	.36
40	91	0	1	0.02	0.15	89	2	2	-0.87	.596	-.13	.70
41	92	0	1	0.34	0.48	61	31	34	0.86	.605	-.27	.26
42	92	0	1	0.43	0.50	52	40	43	1.34	.576	-.13	.44
43	90	0	1	0.21	0.41	71	19	21	0.15	.614	-.27	.36
44	91	0	1	0.05	0.23	86	5	5	-0.71	N/A	N/A	N/A
45	92	0	1	0.04	0.21	88	4	4	-0.77	.606	-.16	.56
46	89	0	1	0.28	0.45	64	25	28	0.53	.618	-.26	.23
47	92	0	1	0.47	0.50	49	43	47	1.56	.578	-.25	.35
48	91	0	1	0.05	0.23	86	5	5	-0.71	.603	-.18	.56
49	90	0	0	0.00	0.00	90	0	0	-0.98	N/A	N/A	N/A
50	92	0	1	0.13	0.34	80	12	13	-0.28	.597	-.26	.38

Note: N/A = Not applicable. This is specified for images that showed no to little variability in the ratings and due to this, were not included in the reliability analysis (which provided the Cronbach's ∝ if item deleted and inter-image correlations).

Figure E-1

Normality plots for the standardized distribution of the Z-transformed first attention mean scores for all 50 images in Study 2 optional stopping Round 1.



A pre-registered criteria for removal of image was to examine Z-scores of ±3, indicating that the image means were three standard deviations away from the overall mean and therefore an outlier. The only image that had a Z-score of ±3 was the image 23 with a Z-score of 3.40. In this image, 81% of participants clicked on the nature area. However, due to the skewness of the Z distribution, other images with similar frequency in the opposite direction (~80% chose the other area) was not identified as an outlier. Due to this, I have decided to keep image 23 going forward.

The skewedness of the standardized distribution suggests that many of the images are working improperly for the new experimental tool because the majority of individuals are only noticing the other areas first and less the nature. This will make it difficult to examine individual differences in the first attention item (i.e. the correlation with nature relatedness). Due to this, I

explored how the standardized distribution with z-transformed means looked when removing images with varying degrees of percentage of nature clicks. I did this by examine four different set of images: 1) images with no variance removed; 2) images with 5% or less of the clicks for first attention as on the nature area removed; 3) images with the nature area been chosen 10% or less were removed; 4) images with 15% or less was removed. See Table E-2 for all normality robustness test results. The distribution became normal (quantitatively and visually) after the removal of images with 15% or less nature clicks (see Table E-2 and Figure E-2). This would suggest that I should remove the images with 15% or less of nature clicks and replace with new images.

Figure E-2

Normality plots for the standardized distribution of the Z-transformed first attention mean scores for images that have at least 16% nature chosen (19 images) in Study 1 optional stopping Round 1.

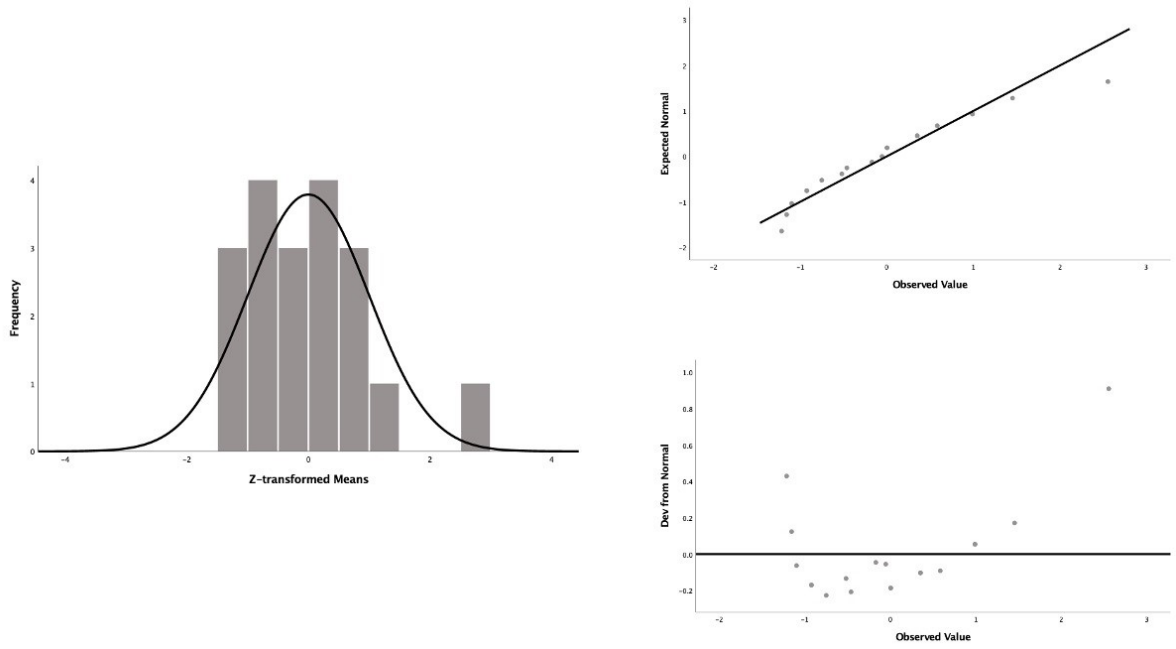


Table E-2

Study 2 optional stopping Round 1 robustness test for normality of the standardized distributions for the first attention item.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	*M*	*SD*	Min	Max
Images without variance	44	13, 15, 18, 34, 44, 49	0.84	< *.001*	0.21	0.19	0.02	0.81
Images with 5% or less clicks on the nature area	36	1, 6, 10, 16, 37, 40, 45, 48	0.85	< .001	0.24	0.18	0.07	0.81
Images with 10% or less clicks on the nature area	27	3, 5, 9, 20, 21, 31, 32, 33, 38	0.88	.01	0.30	0.18	0.11	0.81
Images with 15% or less clicks on the nature area	19	14, 17, 24, 27, 29, 36, 39, 50	0.93	.16	0.37	0.17	0.16	0.81

Voluntary Attention

Overall, the mean of the voluntary attention item across all images was 0.25 (*SD* = 0.19); therefore, nature was clicked for the first attention item, on average, 25% of the time. This did differ per image, with a range of 0% to 75%. All quantitative information about the voluntary attention for each image van be seen in Table E-3. The standardized distribution of Z-transformed means for each image was right-skewed (see Figure E-3).

Compared to the first attention item, only two images had no variance. Images 34 and 40 did not have anyone choose the nature as the area on which they wanted to focus their attention. Therefore, these images were identified as poor images and were removed.

The reliability was good across the 48 items (Images 34 and 40 removed), Cronbach's $\propto$ = .82. There were 34 items that were identified as good by showing that the alpha decreases if the item is removed, whereas 14 items were identified as bad by showing that the alpha would decrease if removed. However, the decrease was very minimal (the maximum increase was by .007). Therefore, this criterion will not be used to highlight images to removal.

Table E-3

Descriptive statistics of the voluntary attention item for Study 2 optional stopping Round 1 (N = 100).

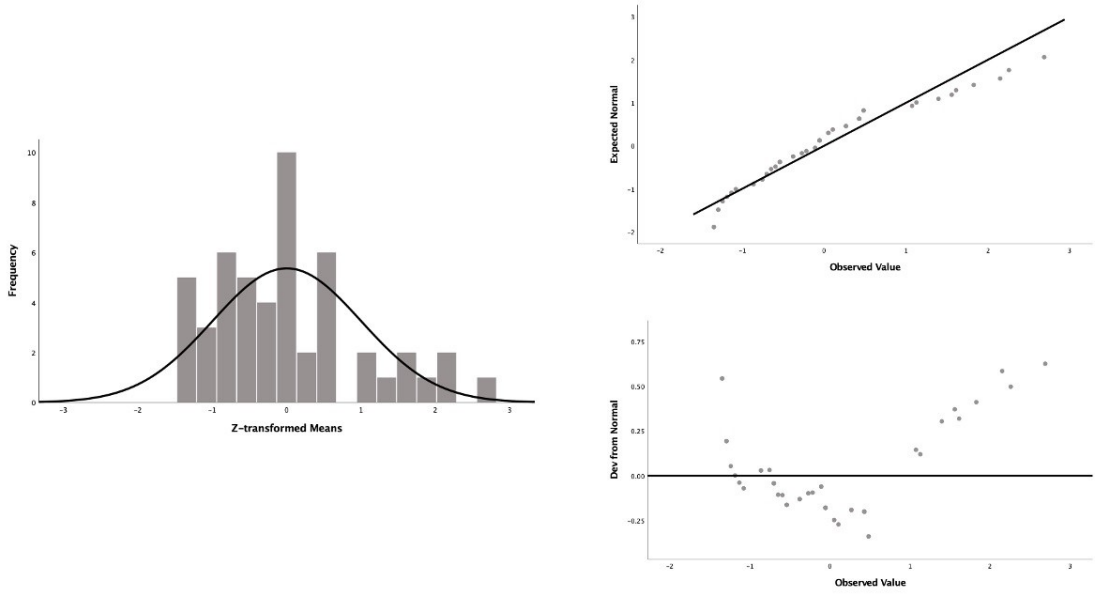
Image	*N*	Min	Max	*M*	*SD*	Other	Nature	~ % of nature	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
1	92	0	1	0.05	0.23	87	5	5	-1.08	.823	-.18	.29
2	91	0	1	0.54	0.50	42	49	54	1.56	.820	-.12	.39
3	92	0	1	0.18	0.39	75	17	18	-0.38	.826	-.17	.26
4	91	0	1	0.24	0.43	69	22	24	-0.06	.820	-.17	.36
5	92	0	1	0.13	0.34	80	12	13	-0.65	.822	-.14	.31
6	92	0	1	0.09	0.28	84	8	9	-0.86	.820	-.15	.42
7	92	0	1	0.23	0.42	71	27	29	-0.11	.821	-.24	.37
8	92	0	1	0.75	0.44	23	69	75	2.69	.826	-.20	.26
9	90	0	1	0.26	0.44	67	23	26	0.05	.820	-.26	.37
10	92	0	1	0.04	0.21	88	4	4	-1.13	.822	-.14	.49
11	91	0	1	0.27	0.45	66	25	27	0.11	.825	-.14	.26
12	92	0	1	0.55	0.50	41	51	55	1.61	.820	-.12	.33
13	92	0	1	0.03	0.18	89	3	3	-1.19	.822	-.14	.57
14	92	0	1	0.24	0.43	70	22	24	-0.06	.815	-.12	.41
15	91	0	1	0.01	0.10	90	1	1	-1.29	.822	-.12	.57
16	92	0	1	0.01	0.10	91	1	1	-1.29	.826	-.20	.19
17	92	0	1	0.21	0.41	73	19	21	-0.22	.819	-.10	.37
18	92	0	1	0.15	0.36	78	14	15	-0.54	.816	-.08	.43
19	91	0	1	0.12	0.33	80	11	12	-0.70	.820	-.17	.42
20	92	0	1	0.09	0.28	84	8	9	-0.86	.826	-.15	.25
21	92	0	1	0.11	0.31	82	10	11	-0.76	.821	-.18	.40
22	92	0	1	0.33	0.47	92	30	33	0.43	.821	-.07	.33
23	92	0	1	0.67	0.47	30	62	67	2.26	.825	-.24	.32
24	92	0	1	0.30	0.46	64	28	30	0.27	.815	-.11	.44

25	92	0	1	0.33	0.47	62	30	33	0.43	.819	-.17	.39
26	92	0	1	0.46	0.50	50	42	46	1.13	.816	-.13	.32
27	90	0	1	0.12	0.33	79	11	12	-0.70	.818	-.12	.40
28	92	0	1	0.59	0.50	38	54	59	1.83	.825	-.26	.36
29	92	0	1	0.30	0.46	64	28	30	0.27	.816	-.18	.41
30	92	0	1	0.24	0.43	70	22	24	-0.06	.820	-.09	.42
31	92	0	1	0.24	0.43	70	22	24	-0.06	.815	-.13	.45
32	91	0	1	0.12	0.33	80	11	12	-0.70	.829	-.24	.20
33	92	0	1	0.15	0.36	78	14	15	-0.54	.821	-.19	.40
34	92	0	0	0.00	0.00	92	0	0	-1.35	N/A	N/A	N/A
35	92	0	1	0.15	0.36	78	14	15	-0.54	.825	-.16	.37
36	91	0	1	0.45	0.50	50	41	45	1.07	.819	-.19	.44
37	92	0	1	0.02	0.15	90	2	2	-1.24	.826	-.20	.20
38	92	0	1	0.23	0.42	71	21	23	-0.11	.818	-.13	.35
39	92	0	1	0.33	0.47	62	30	33	0.43	.821	-.23	.39
40	92	0	0	0.00	0.00	92	0	0	-1.35	N/A	N/A	N/A
41	92	0	1	0.34	0.48	61	31	34	0.48	.831	-.23	.20
42	92	0	1	0.65	0.48	32	60	65	2.15	.818	-.16	.41
43	92	0	1	0.34	0.48	61	31	34	0.48	.825	-.19	.37
44	92	0	1	0.14	0.35	79	13	14	-0.59	.825	-.17	.26
45	92	0	1	0.24	0.43	70	22	24	-0.06	.819	-.13	.38
46	92	0	1	0.33	0.47	62	30	33	0.43	.819	-.16	.37
47	92	0	1	0.51	0.50	45	47	51	1.40	.819	-.15	.38
48	92	0	1	0.26	0.44	68	24	26	0.05	.815	-.24	.45
49	92	0	1	0.20	0.40	74	18	20	-0.27	.820	-.19	.39
50	92	0	1	0.18	0.39	75	17	18	-0.38	.826	-.16	.23

Note: N/A = Not applicable. This is specified for images that showed no to little variability in the ratings and due to this, were not included in the reliability analysis (which provided the Cronbach's ∝ if item deleted and inter-image correlations).

Figure E-3

Normality plots for the standardized distribution of the Z-transformed voluntary attention mean scores for all 50 images in Study 2 optional stopping Round 1.



The standardized distribution of z-transformed means for the 48 images (with the two images with no variance removed) can be seen in Figure E-4. Visually, it can be seen that there are more values at the mean value or below the mean (i.e. more other areas clicked than nature); however, visually it does not seem too far off a normal distribution. The Shapiro-Wilke's estimate = .924 p = .004, indicating that the distribution is significantly different from a normal distribution. Similar to the first attention item, robustness tests for normality were examined by exploring different sets of images based on low percentage of nature areas being clicked. Five different sets were explored: 1) the removal of images with no variance; 2) images with 5% or less nature clicks removed; 3) images with 10% or less nature clicks removed; 4) images with 15% or less nature clicks removed; and 5) images with 20% or less removed. When exploring each robustness test, the distribution becomes more skewed and thus, less normal. This can be seen visually (Figure E-5), but it also is demonstrated by the Shapiro-Wilks test for normality (Table E-4).

Figure E-4

Normality plots for the standardized distribution of the Z-transformed voluntary attention means for all images that have variance in Study 2 optional stopping Round 1 (48 images).

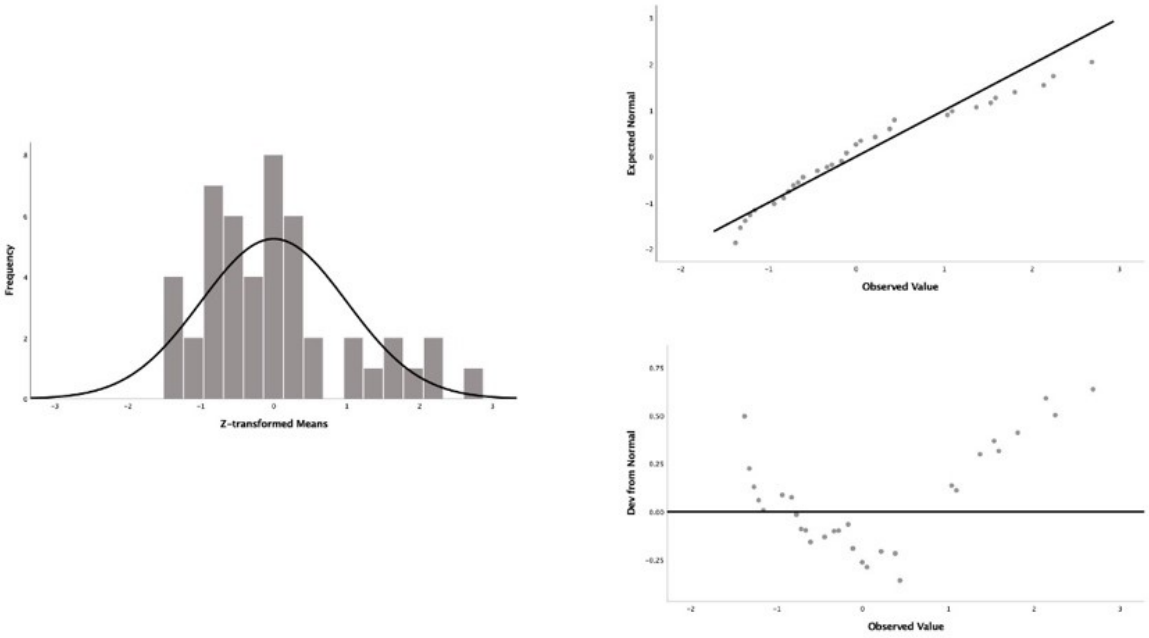


Figure E-5

Visual representation of the voluntary attention standardized distribution for the four robustness tests of Study 1 optional stopping Round 1: a) removing 5 or less % nature images (42 images), b) removing 10 or less % nature images (40 images), c) removing 15 or less % nature images (31 images), and d) removing 20 or less % nature images (28 images).

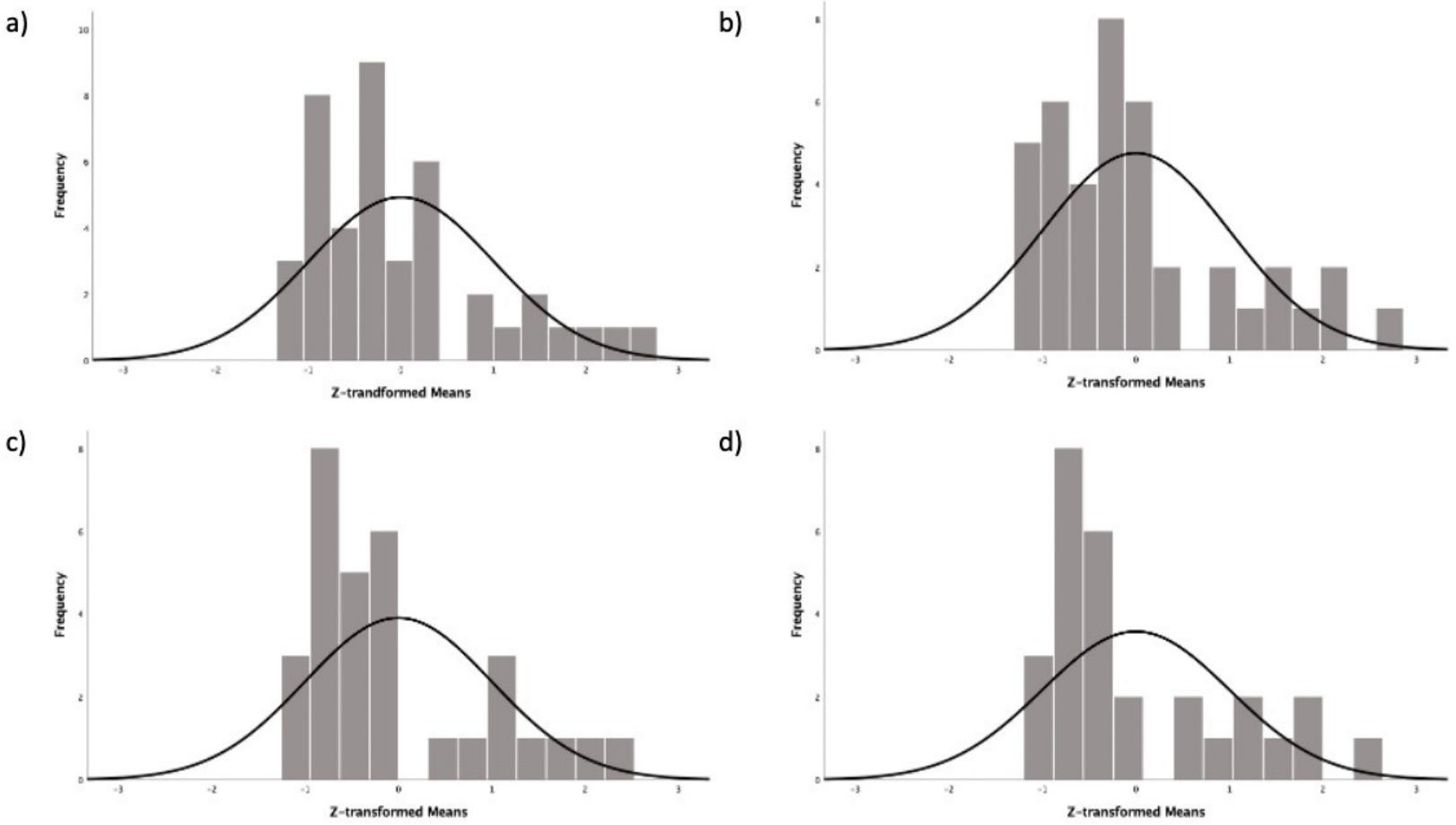


Table E-4

Study 2 optional stopping Round 1 robustness test for normality of the standardized distributions for the voluntary attention item.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	*M*	*SD*	Min	Max
Images without variance	48	34, 40	0.92	.004	0.26	0.18	0.01	0.75
Images with 5% or less clicks on the nature area	42	1, 10, 13, 15, 16, 37	0.89	< .001	0.29	0.17	0.09	0.75
Images with 10% or less clicks on the nature area	40	6, 20	0.88	< .001	0.3	0.17	0.11	0.75
Images with 15% or less clicks on the nature area	31	5, 18, 19, 21, 27, 32, 33, 35, 44	0.86	< .001	0.35	0.16	0.18	0.75
Images with 20% or less clicks on the nature area	28	3, 49, 50	0.85	< .001	0.37	0.16	0.21	0.75

Looking Across Both Items

Now that images were highlighted as poor for either for the first attention item or the voluntary attention item, to make a decision about what images will be replaced before restarting the recruitment, I have also examined how images performed across both items. In the psychometric analysis for the first attention item, it was shown that the distribution becomes more normal when using only images with more than 15% nature clicks. For voluntary attention, when removing images with up to less than 20% nature clicks, the distribution does not significantly change. With this in mind, I have chosen to select images to keep depending on at least 15% nature clicks across both (i.e. the image must have performed with at least 15% for each item). With this criteria, 18 images had a good percentage of nature clicks and therefore, would be kept. These images were: 2, 4, 7, 8, 11, 12, 22, 23, 25, 26, 28, 30, 35, 41, 42, 43, 46, and 47. The 32 images identified a poor were replaced.

Round 2

After the replacement of the 32 images, a new set of 100 participants were recruited to explore the variability of the new set of images (18 old images and 32 new images).

First Attention

On average, across all the clicks, the nature areas were clicked 26% of the time (M = 0.26, SD = 0.18; Min = 0.00, Max = 0.81). This is a larger mean than Round 1 (M = 0.18, SD. = .19), even though the minimum and maximum are the same. All 49 images with variance showed both positive and negative correlations with each other. The inter-image correlations ranged from -0.41 to 0.61. All quantitative results of the psychometric analyses for all 50 images can be seen in Table E-5.

When exploring the variability of all 50 items, one image (Image 61) was identified as having no variance and therefore was removed from the analysis. The overall Cronbach's alpha of these 49 images was again quite low at .67, but this was larger than Round 1 (Cronbach's α = .61). The reliability was quite robust to the removal of a single image, where the alpha was reduced by a maximum of .02 or increased by a maximum of .03. In this robustness test, 10 images were highlighted as poor images (i.e. where the Cronbach's alpha was increased if the image was removed), 38 images were good images (i.e. where the Cronbach's alpha was decreased if removed), and a single image showed no change in Cronbach's alpha if removed. Due to the small amount of change in Cronbach's alpha, this criteria was dropped for identifying images to be replaced.

Table E-5

Descriptive statistics of the first attention item for Study 2 optional stopping Round 2 (N = 100).

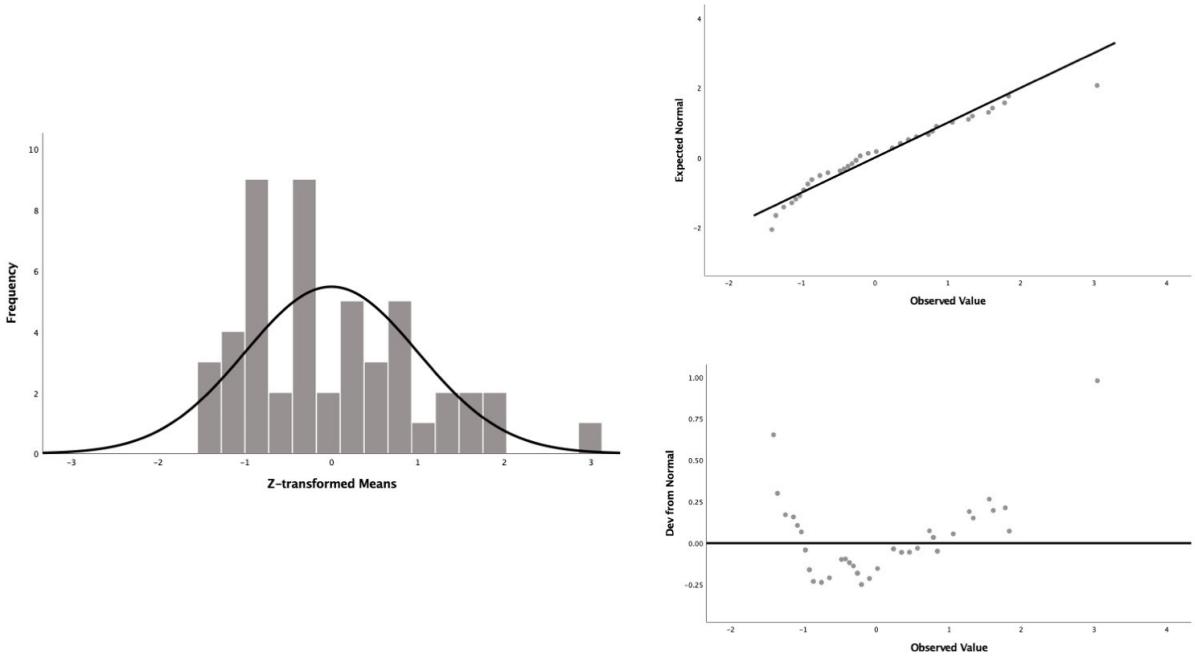
Image	*N*	Min	Max	*M*	*SD*	Other	Nature	~ % of Nature	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
2	99	0	1	0.55	0.50	45	54	55	1.61	.678	-.30	.33
4	99	0	1	0.24	0.43	75	24	24	-0.09	.659	-.27	.34
7	96	0	1	0.39	0.49	59	37	39	0.73	.658	-.25	.30
8	99	0	1	0.58	0.50	42	57	58	1.78	.673	-.37	.26
11	98	0	1	0.41	0.49	58	40	41	0.84	.657	-.27	.51
12	94	0	1	0.49	0.50	48	46	49	1.28	.662	-.27	.33
22	98	0	1	0.19	0.40	79	19	19	-0.37	.648	-.18	.61
23	98	0	1	0.81	0.40	19	79	81	3.04	.676	-.41	.31
25	98	0	1	0.26	0.44	73	25	26	0.02	.662	-.29	.31
26	98	0	1	0.21	0.41	77	21	21	-0.26	.653	-.28	.34
28	98	0	1	0.59	0.49	40	58	59	1.83	.652	-.33	.51
30	98	1	1	0.18	0.39	80	18	18	-0.42	.656	-.32	.37
35	96	0	1	0.54	0.50	44	52	54	1.56	.677	-.28	.36
41	99	0	1	0.32	0.47	67	32	32	0.35	.662	-.31	.38
42	97	0	1	0.40	0.49	58	39	40	0.79	.657	-.39	.37
43	98	0	1	0.30	0.46	69	29	30	0.24	.657	-.25	.37
46	98	0	1	0.41	0.49	58	40	41	0.84	.662	-.18	.31
47	96	0	1	0.34	0.48	63	33	34	0.46	.651	-.18	.36
51	97	0	1	0.12	0.33	85	12	12	-0.75	.644	-.13	.59
52	98	0	1	0.40	0.49	59	39	40	0.79	.695	-.39	.33
53	95	0	1	0.08	0.28	87	8	8	-0.97	.663	-.18	.49
54	97	0	1	0.22	0.41	76	21	22	-0.20	.651	-.24	.36
55	96	0	1	0.14	0.34	83	13	14	-0.64	.666	.25	.34
56	98	0	1	0.09	0.29	89	9	9	-0.92	.661	-.19	.48

57	97	0	1	0.32	0.47	66	31	32	0.35	.654	-.31	.37
58	99	0	1	0.17	0.38	82	17	17	-0.48	.673	-.41	.26
59	98	0	1	0.05	0.22	93	5	5	-1.14	.667	-.26	.26
60	98	0	1	0.20	0.41	78	20	20	-0.31	.663	-.24	.38
61	99	0	0	0.00	0.00	99	0	0	-1.41	.663	-.24	.38
62	98	0	1	0.50	0.50	49	49	50	1.34	.660	-.30	.35
63	99	0	1	0.30	0.46	69	30	30	0.24	.649	-.17	.35
64	96	0	1	0.45	0.50	53	43	45	1.06	.656	-.23	.33
65	99	0	1	0.10	0.30	89	10	10	-0.86	.655	-.19	.36
66	99	0	1	0.10	0.30	89	10	10	-0.86	.648	-.17	.61
67	97	0	1	0.36	0.48	62	35	36	0.57	.649	-.26	.37
68	97	0	1	0.08	0.28	89	8	8	-0.97	.674	-.30	.25
69	98	0	1	0.01	0.10	97	1	1	-1.36	.662	-.14	.57
70	98	0	1	0.03	0.17	95	3	3	-1.25	.658	-.23	.57
71	95	0	1	0.34	0.48	63	32	34	0.46	.665	-.24	.39
72	95	0	1	0.06	0.24	89	6	6	-1.08	.664	-.27	.26
73	98	0	1	0.22	0.42	76	22	22	-0.20	.656	-.17	.38
74	98	0	1	0.08	0.28	90	8	8	-0.97	.670	-.24	.36
75	96	0	1	0.30	0.46	67	29	30	0.24	.656	-.24	.36
76	94	0	1	0.12	0.32	83	19	19	-0.75	.664	-.22	.36
77	99	0	1	0.19	0.40	80	19	19	-0.37	.659	-.19	.32
78	98	0	1	0.01	0.10	97	1	1	-1.36	.664	-.17	.57
79	97	0	1	0.09	0.29	88	9	9	-0.92	.664	-.23	.43
80	98	0	1	0.07	0.26	91	7	7	-1.03	.659	-.25	.59
81	96	0	1	0.21	0.41	76	20	21	-0.26	.656	-.22	.34
82	97	0	1	0.21	0.41	77	20	21	-0.26	.673	-.26	.26

The standardized distribution was right-skewed (see Figure E-6), indicating that more clicks were on the other areas than the nature areas. Statistically, the distribution was not normal, Shapiro-Wilks = .94 p = .02; this was also visually seen in the normality plots (see Figure E-5). There is a single outlier (Image 23) of the distribution, where nature was chosen more frequently than for the other images.

Figure E-6

Normality plots for the standardized distribution of the Z-transformed first attention mean scores for all 50 images in Study 1 optional stopping Round 2.



Similar to Round 1, I examined the robustness of the distribution by looking at the normality of the distribution when removing different set of images (see Table E-6). The first set was removing images with no variance. The distribution was again not normal. Next, the distribution was examined when removing the images with the lowest amount of nature clicks (i.e. 5% or less, 10% or less, 15% or less, and 20% or less). At each step, the removal of images did not improve the normality of the distribution (Shapiro-Wilks test for normality were all p > .05). However, at each of these stages, an outlier was identified. Image 23 had a Z-score of above

3. These robustness tests were run again without Image 23. The distribution became normal when removing the images with less than 10% nature clicks and the outlier. This suggests the removal of 15 images. The standardized distribution of this set of images can be seen in Figure E-7.

Figure E-7

The visual of the statistically normal standardized distribution of the first attention item for Study 2 optional Round 2 (35 images).

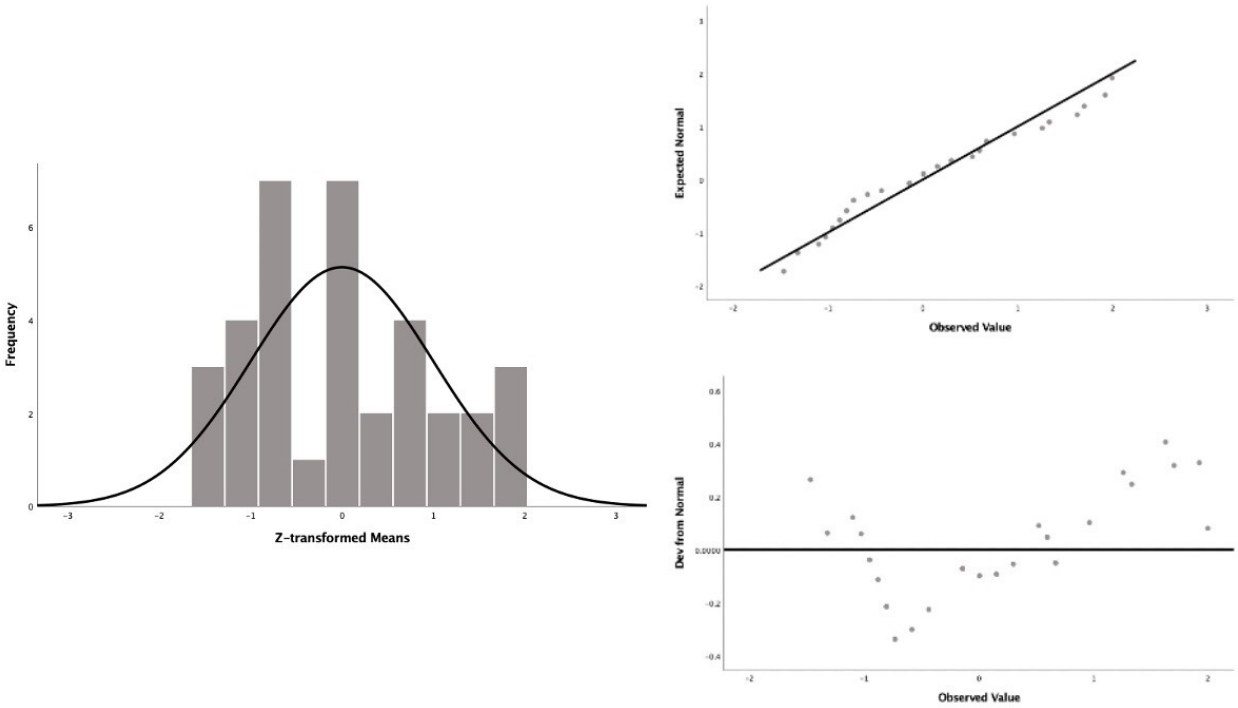


Table E-6

Study 2 optional stopping Round 2 robustness test for normality of the standardized distributions for the first attention item.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	*M*	*SD*	Min	Max
Images without variance	49	61	0.94	.02	0.26	0.18	0.01	0.81
Images with 5% or less clicks on the nature area	45	59, 69, 70, 78	0.93	.01	0.28	0.17	0.06	0.81
Images with 10% or less clicks on the nature area	36	53, 56, 65, 66, 68, 72, 74, 79, 80	0.93	.03	0.33	0.16	0.12	0.81
Images with 15% or less clicks on the nature area	34	51, 55	0.93	.03	0.34	0.15	0.12	0.81
Images with 20% or less clicks on the nature area	28	22, 30, 58, 60, 76, 77	0.92	.03	0.38	0.14	0.21	0.81
Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	*M*	*SD*	Min	Max
Images without variance + Outlier	48	61, 23	0.95	.03	0.25	0.16	0.01	0.59
Images with 5% or less clicks on the nature area + Outlier	44	59, 69, 70, 78, 23	0.94	.02	0.27	0.15	0.06	0.59
Images with 10% or less clicks on the nature area + Outlier	35	53, 56, 65, 66, 68, 72, 74, 79, 80, 23	0.95	.08	0.32	0.14	0.12	0.59

Voluntary Attention

The mean across all 50 images was 0.33 (*SD* = 0.20; Min = .05, Max = 0.80), indicating that, on average, the nature area was clicked 33% of the time. This is higher than the 25% from Round 1. This varied per image with a minimum percentage of 5% and a maximum of 80%. All quantitative results can be seen in Table E-7.

The reliability of all images (i.e. no images were identified as having no variance) was $\propto = .87$, which was .05 points above the reliability of Round 1. When exploring the robustness of the reliability, the majority of images were identified as good images with 41 images showing the Cronbach's alpha was reduced when the image was removed (i.e. identified as good images for aggregation) and 6 images where the removal had no effect on the alpha value. For the three poor images, the removal of the image showed a maximum improvement of the Cronbach's $\propto$ by .003, which is very minimal. Thus, this was not used a criteria for identifying images to be replaced.

The standardized distribution for the 50 images was not normal visually (see Figure E-8) nor quantitatively, Shapiro-Wilkes estimate = .93 $p = .004$. Similar to previous psychometric analyses, robustness tests were run to examine if the standardized distribution would become normal when removing images on the lower end of the distribution (i.e. images with the smallest percentage of nature chosen as the area participants wanted to focus their attention on). The quantitative results of these robustness tests can be seen in Table E-8. When running the robustness removing images with 5%, 10%, 15%, and 20% or less nature, the distribution did not become normal visually or statistically (see Table E-8; Shapiro-Wilkes $p > .05$). None of means were above or below 3 *SD*, indicating that there were no outliers identified. Thus, no images were highlighted to be removed to improve normality of the distribution.

Table E-7

Descriptive statistics of the voluntary attention item for Study 2 optional stopping Round 2 (N = 100).

Image	*N*	Min	Max	*M*	*SD*	Other	Nature	~ % of Nature	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
2	98	0	1	0.58	0.50	41	57	58	1.29	.888	-.36	.26
4	97	0	1	0.26	0.44	72	25	26	-0.33	.886	-.17	.31
7	98	0	1	0.19	0.40	79	12	13	-0.68	.883	-.11	.42
8	97	0	1	0.80	0.40	19	78	80	2.41	.888	-.16	.17
11	98	0	1	0.39	0.49	60	38	39	0.33	.881	-.03	.36
12	98	0	1	0.56	0.50	43	55	56	1.19	.885	-.14	.32
22	98	0	1	0.28	0.45	71	27	28	-0.23	.881	-.03	.40
23	98	0	1	0.71	0.45	28	70	71	1.95	.885	-.15	.31
25	97	0	1	0.22	0.41	76	21	22	-0.53	.882	-.08	.44
26	98	0	1	0.31	0.46	68	30	31	-0.08	.882	-.16	.40
28	97	0	1	0.60	0.49	39	58	60	1.39	.886	-.12	.29
30	98	0	1	0.18	0.39	80	18	18	-0.74	.882	-.10	.50
35	98	0	1	0.23	0.43	75	23	23	-0.48	.885	-.22	.44
41	98	0	1	0.33	0.47	66	32	33	0.03	.889	-.16	.21
42	98	0	1	0.57	0.50	42	56	57	1.24	.883	-.06	.37
43	98	0	1	0.33	0.47	66	32	33	0.03	.886	-.12	.34
46	98	0	1	0.19	0.40	79	19	19	-0.68	.884	-.16	.38
47	98	0	1	0.42	0.50	57	41	42	0.48	.881	.01	.39
51	98	0	1	0.09	0.29	89	9	9	-1.19	.884	-.15	.45
52	98	0	1	0.38	0.49	61	37	38	0.28	.884	-.03	.37
53	97	0	1	0.23	0.42	75	22	23	-0.48	.882	-.10	.38
54	97	0	1	0.31	0.46	67	30	31	-0.08	.883	-.08	.37
55	98	0	1	0.39	0.49	60	38	39	0.33	.886	-.17	.44
56	98	0	1	0.11	0.32	87	11	11	-1.09	.884	-.10	.38

57	97	0	1	0.30	0.46	68	29	30	-0.13	.883	-.08	.34
58	98	0	1	0.59	0.49	40	58	59	1.34	.886	-.22	.55
59	98	0	1	0.65	0.48	34	64	65	1.65	.886	-.11	.55
60	98	0	1	0.35	0.48	64	34	35	0.13	.885	-.36	.35
61	98	0	1	0.07	0.26	91	7	7	-1.29	.883	.01	.38
62	98	0	1	0.68	0.47	31	67	68	1.80	.885	-.12	.35
63	98	0	1	0.29	0.45	70	28	29	-0.18	.882	-.11	.46
64	98	0	1	0.17	0.38	81	17	17	-0.79	.883	-.11	.44
65	97	0	1	0.15	0.36	82	15	15	-0.89	.882	-.02	.42
66	98	0	1	0.08	0.28	90	8	8	-1.24	.884	-.14	.42
67	98	0	1	0.35	0.48	64	34	35	0.13	.883	-.02	.46
68	98	0	1	0.16	0.37	82	16	16	-0.84	.884	-.12	.48
69	98	0	1	0.18	0.39	80	18	18	-0.74	.885	-.11	.39
70	98	0	1	0.24	0.43	74	24	24	-0.43	.883	-.10	.40
71	98	0	1	0.72	0.45	27	71	72	2.00	.885	-.09	.37
72	98	0	1	0.09	0.29	89	9	9	-1.19	.884	-.08	.40
73	98	0	1	0.17	0.38	81	17	17	-0.79	.881	-.06	.50
74	98	0	1	0.28	0.45	71	27	28	-0.23	.884	-.15	.29
75	98	0	1	0.26	0.44	73	25	26	-0.33	.885	-.10	.40
76	98	0	1	0.10	0.30	88	10	10	-1.14	.884	-.11	.38
77	98	0	1	0.35	0.48	64	34	35	0.13	.881	-.04	.48
78	97	0	1	0.05	0.22	92	5	5	-1.39	.885	-.13	.41
79	98	0	1	0.51	0.50	48	50	51	0.94	.883	-.32	.37
80	98	0	1	0.10	0.30	88	10	10	-1.14	.883	-.08	.46
81	98	0	1	0.17	0.38	81	17	17	-0.79	.885	-.11	.35
82	98	0	1	0.53	0.50	46	52	53	1.04	.885	-.12	.33

Figure E-8

Normality plots for the standardized distribution of the Z-transformed voluntary attention mean scores for all 50 images in Study 2 optional stopping Round 2.

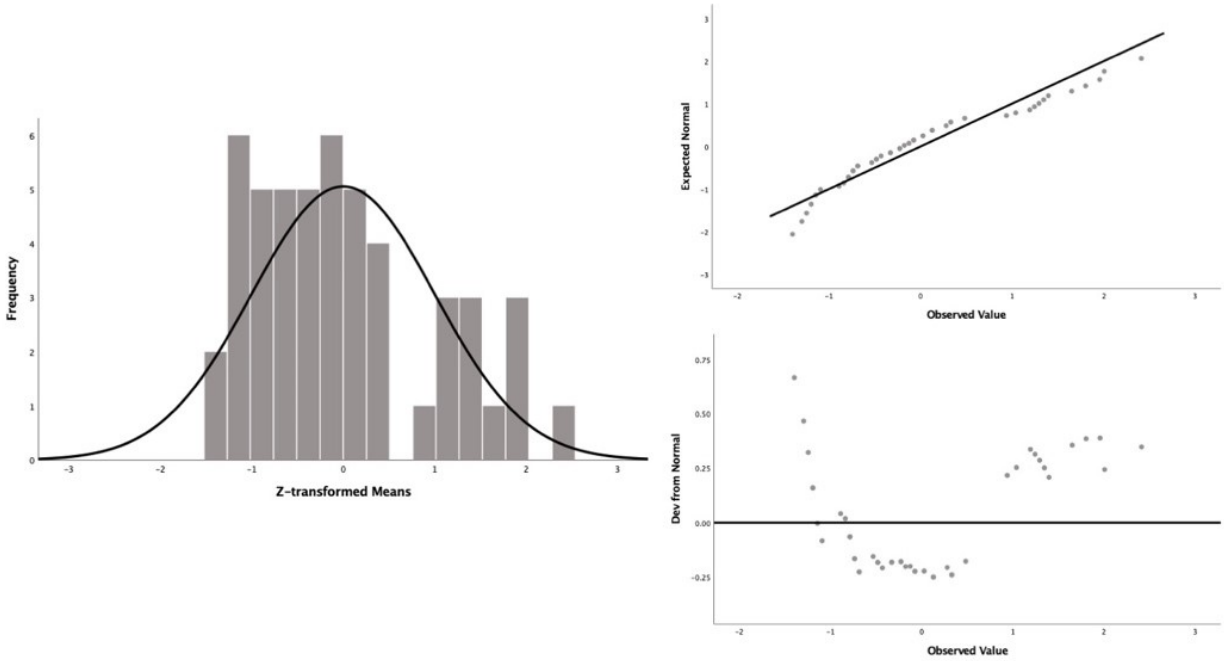


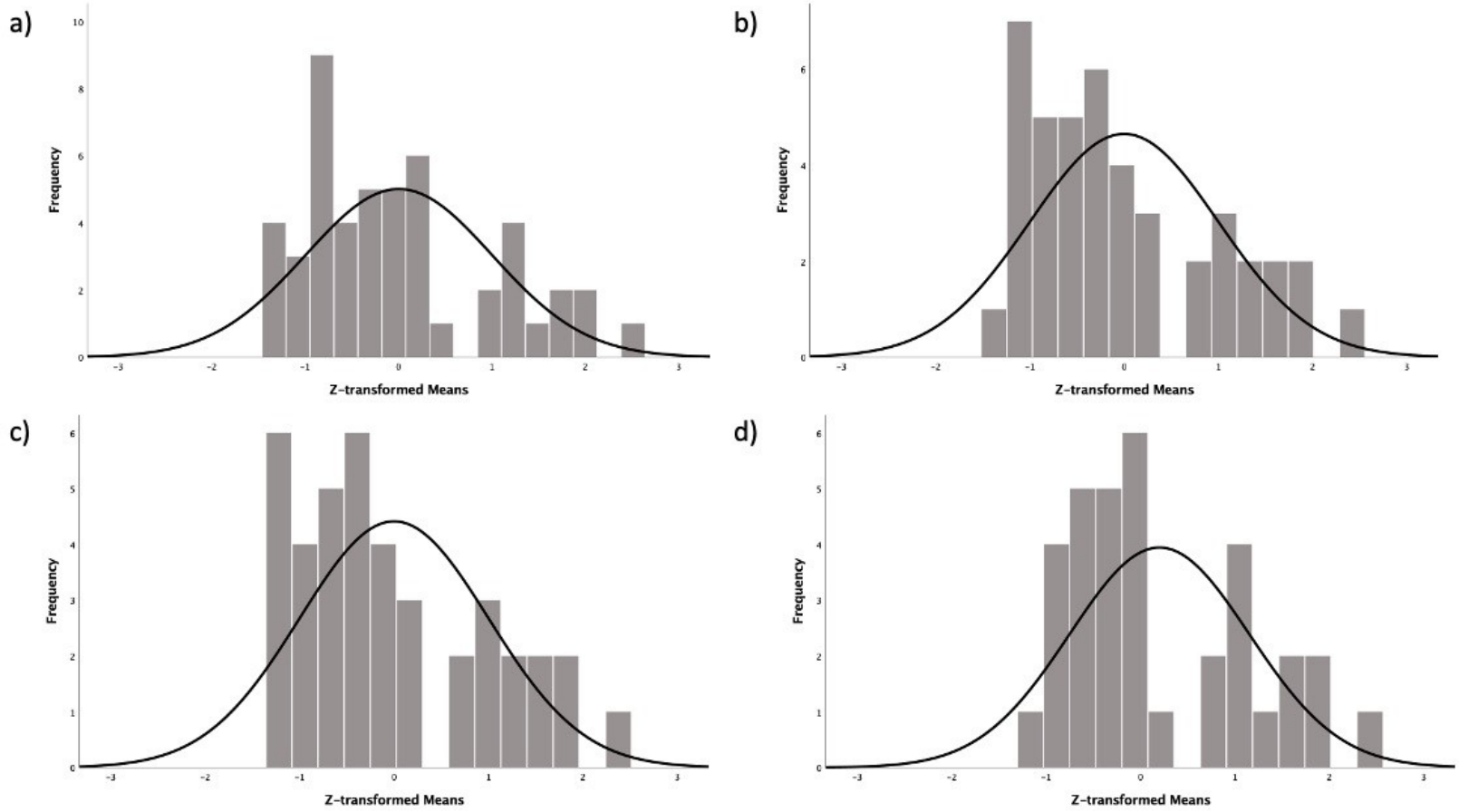
Table E-8

Study 2 optional stopping Round 2 robustness test for normality of the standardized distributions for the voluntary attention item.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*	*M*	*SD*	Min	Max
Images with 5% or less clicks on the nature area	49	78	0.92	.004	0.33	0.2	0.07	0.8
Images with 10% or less clicks on the nature area	43	51, 61, 66, 72, 76, 80	0.91	.003	0.36	0.18	0.11	0.8
Images with 15% or less clicks on the nature area	40	7, 56, 55	0.91	.003	0.34	0.18	0.16	0.8
Images with 20% or less clicks on the nature area	33	30, 46, 64, 68, 69, 73, 81	0.91	.01	0.42	0.17	0.17	0.8

Figure E-9

Visual representation of the voluntary attention standardized distribution for the four robustness tests of Study 1 optional stopping Round 1: a) removing 5 or less % nature images (49 images), b) removing 10 or less % nature images (43 images), c) removing 15 or less % nature images (40 images), and d) removing 20 or less % nature images (33 images).



Looking at both

Similar to round 1, I have decided to make decisions about the images by examining how they performed across both items. To make the decision on the criteria, I will look towards my robustness tests. For the first attention item, the standardized distribution becomes more normal when removing images with 10% or less nature clicks. For voluntary attention, when removing images with 20% or less, there was no changes to the normality of the distribution. Therefore, I have chosen to select keeping images which have above 10% nature clicks across both images. If I use this criterion, I would be removing 16 images (51, 53, 56, 59, 61, 65, 66, 68, 69, 70, 72, 74, 76, 78, 79, 80). However, when exploring how these 16 images performed for each item, there are a few items that underperformed for first attention item, but did well on the voluntary attention item. There are five images that fit this description. These images were retained. This reduces my number of images to replace to 11. These images were: 51, 56, 61, 65, 66, 68, 69, 72, 76, 78, and 80. These 11 images were replaced.

Round 3

After replacing the 11 images, a new sample of 100 participants were recruited to explore the variability of the new set of images (i.e. 11 new ones and 39 old ones).

First Attention

The mean value for the first attention across all images was 0.31 (*SD* = 0.22; Min.= 0.00, Max = 0.95), indicating that, across all the clicks made by participants, participants clicked the nature area approximately 31% of the time. This mean is the largest between the three rounds. This percentage did vary across the images (Min = 0%, Max = 95%). All quantitative results can be seen in Table E-9.

When examining the variability, a single image was identified as having no variance (Image 70) and was removed from analyses. The reliability for the 49 images was again low at .67 and unchanged from round 2. The reliability was robust to the removal of each individual image. Twelve images increased the Cronbach's alpha when removed, but as much as by .01. The majority of images (34 images) were identified as good by the robustness test of the reliability, with a decrease of at least .002 and as much as .03. Three images showed no changes to Cronbach's alpha when removed. The minor changes in Cronbach's alpha suggest that this criteria does not highly suggest the removal of any images.

When exploring the inter-image correlations, all images had both positive and negative correlations. Five images had more negative correlations than positive (Image 41, 52, 59, 64, & 74) and four had equal amounts (Images 2, 35, 46, & 71). Out of the 49 images, 24 had at least one inter-image correlation equal of 0. Image 43 had up to 10 correlations that were equivalent to 0.

Table E-9

Descriptive statistics of the first attention item for Study 2 optional stopping Round 3 (N = 100).

Image	*N*	Min	Max	*M*	*SD*	Other	Nature	~ % of Nature	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
2	97	0	1	0.51	0.50	48	49	51	0.91	.678	-.33	.28
4	97	0	1	0.23	0.42	75	22	23	-0.36	.671	-.28	.46
7	98	0	1	0.30	0.46	69	29	30	-0.04	.668	-.26	.45
8	97	0	1	0.63	0.49	36	61	63	1.48	.682	-.25	.35
11	97	0	1	0.46	0.50	52	45	46	0.73	.666	-.19	.36
12	94	0	1	0.51	0.50	46	48	51	0.94	.674	-.19	.26
22	93	0	1	0.22	0.41	73	20	22	-0.41	.665	-.35	.33
23	96	0	1	0.88	0.33	12	84	88	2.61	.664	-.26	.27
25	97	0	1	0.21	0.41	77	20	21	-0.45	.665	-.16	.50
26	95	0	1	0.16	0.37	80	15	16	-0.67	.669	-.23	.37
28	98	0	1	0.60	0.49	39	59	60	1.36	.672	-.28	.30
30	96	0	1	0.09	0.29	87	9	9	-0.97	.671	-.56	.35
35	100	0	1	0.47	0.50	53	47	47	0.75	.675	-.23	.52
41	98	0	1	0.16	0.37	82	16	16	-0.65	.674	-.27	.35
42	100	0	1	0.34	0.48	66	34	34	0.16	.655	-.27	.31
43	98	0	1	0.20	0.41	78	20	20	-0.46	.684	-.22	.30
46	98	0	1	0.33	0.47	66	32	33	0.10	.676	-.23	.38
47	97	0	1	0.32	0.47	66	31	32	0.07	.654	-.18	.45
52	93	0	1	0.33	0.47	62	31	33	0.13	.688	-.35	.30
53	96	0	1	0.04	0.20	92	4	4	-1.20	.671	-.26	.32
54	99	0	1	0.14	0.35	85	14	14	-0.75	.666	-.24	.56
55	97	0	1	0.09	0.29	88	9	9	-0.97	.668	-.25	.33
57	97	0	1	0.32	0.47	66	31	32	0.07	.664	-.23	.40
58	98	0	1	0.13	0.34	85	13	13	-0.79	.670	-.20	.47

59	99	0	1	0.04	0.20	95	4	4	-1.21	.675	-.23	.47
60	98	0	1	0.13	0.34	85	13	13	-0.79	.667	-.16	.39
62	98	0	1	0.58	0.50	41	57	58	1.26	.660	-.19	.46
63	99	0	1	0.22	0.42	77	22	22	-0.38	.659	-.27	.46
64	99	0	1	0.56	0.50	44	55	56	1.15	.688	-.27	.38
67	97	0	1	0.35	0.48	63	34	35	0.21	.676	-.29	.34
70	98	0	0	0.00	0.00	98	0	0	-1.39	.676	N/A	N/A
71	99	0	1	0.36	0.48	63	36	36	0.27	.677	-.31	.52
73	97	0	1	0.22	0.41	76	21	22	-0.40	.665	-.23	.35
74	99	0	1	0.06	0.24	93	6	6	-1.12	.675	-.25	.48
75	93	0	1	0.20	0.41	74	19	20	-0.46	.670	-.27	.50
77	98	0	1	0.09	0.29	89	9	9	-0.98	.669	-.17	.42
79	97	0	1	0.07	0.26	90	7	7	-1.06	.669	-.28	.39
81	98	0	1	0.23	0.43	75	23	23	-0.32	.649	-.22	.56
82	96	0	1	0.14	0.34	83	13	14	-0.78	.672	-.22	.24
83	97	0	1	0.89	0.32	11	86	89	2.66	.665	-.27	.27
84	97	0	1	0.20	0.40	78	19	20	-0.50	.662	-.33	.37
85	96	0	1	0.95	0.22	5	91	95	2.94	.666	-.56	.27
86	92	0	1	0.42	0.50	53	39	42	0.54	.661	-.34	.37
87	98	0	1	0.22	0.42	76	22	22	-0.37	.654	-.26	.42
88	100	0	1	0.26	0.44	74	26	26	-0.21	.679	-.31	.30
89	97	0	1	0.40	0.49	58	39	40	0.44	.668	-.14	.35
90	98	0	1	0.36	0.48	63	35	36	0.24	.664	-.27	.34
91	98	0	1	0.15	0.36	83	15	15	-0.69	.660	-.26	.37
92	99	0	1	0.16	0.37	83	16	16	-0.66	.668	-.22	.41
93	96	0	1	0.31	0.47	66	30	31	0.03	.674	-.18	.32

Note: N/A = Not applicable. This is specified for images that showed no to little variability in the ratings and due to this, were not included in the reliability analysis (which provided the Cronbach's ∝ if item deleted and inter-image correlations).

Lastly, I examined the standardized distribution of the Z-transformed first attention means for each image. The distribution of all 50 images was right-skewed (see Figure E-9) and was identified as non-normal, visually and statistically, Shapiro-Wilks = .90 $p < .001$. This indicates that, overall, participants first noticed the non-nature areas of the image more often than the nature area. The robustness of this distribution was examined by removing the images with the lowest percentage of nature clicks. Five robustness tests were done: i) removing the image with no variance; ii) removing images with 5% or less nature clicks; iii) removing images with 10% or less nature clicks; iv) removing images with 15% or less nature clicks; v) removing images with 20% or less nature clicks. The removal of images with 20% or less clicks on the nature areas the distribution did not become normal (see Table E-10). None of means were above or below 3 *SD*, indicating that there were no outliers identified. Thus, no images were identified to be removed to improve normality of the distribution.

Figure E-10

Normality plots for the standardized distribution of the Z-transformed first attention mean scores for all 50 images in Study 2 optional stopping Round 3.

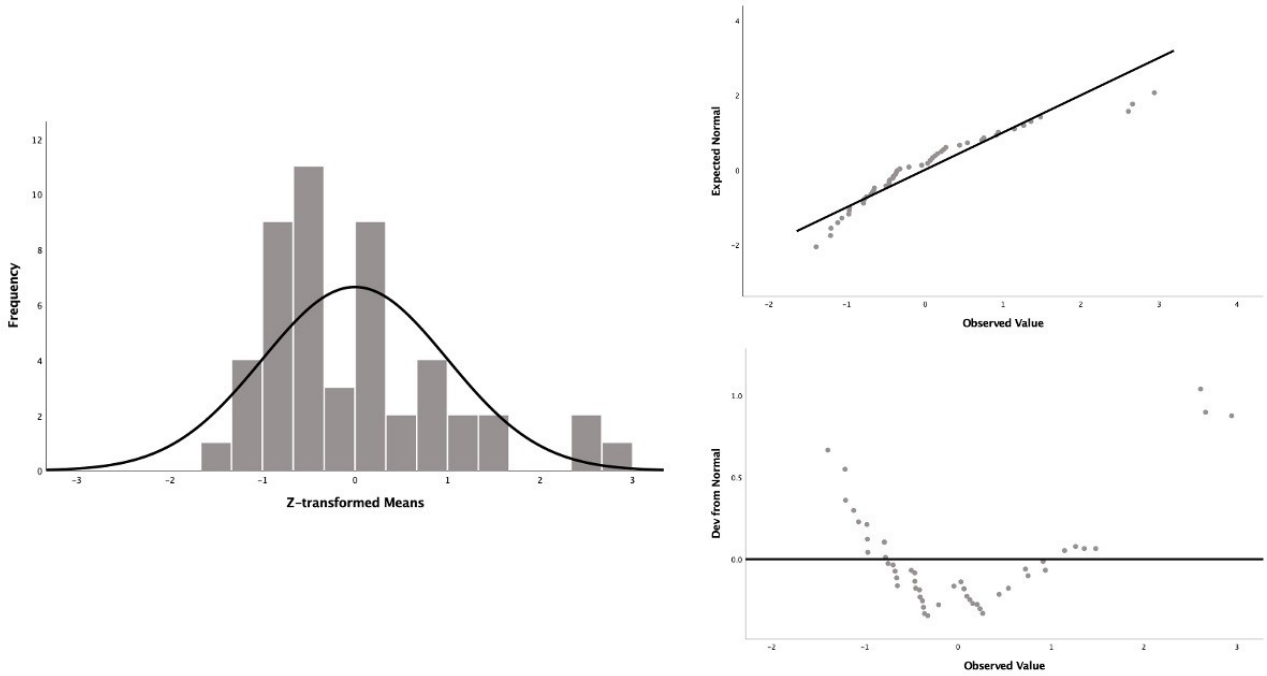


Table E-10
Study 2 optional stopping Round 3 robustness test for normality of the standardized distribution for first attention.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	*p*
Images with no variance	49	70	0.89	< .001
Images with 5% or less clicks on the nature area	47	53, 59	0.88	< .001
Images with 10% or less clicks on the nature area	42	30, 55, 74, 77, 79	0.86	< .001
Images with 15% or less clicks on the nature area	37	54, 58, 60, 82, 91	0.85	< .001
Images with 20% or less clicks on the nature area	33	26, 41, 84, 92	0.85	< .001

Voluntary Attention

The mean voluntary attention across all 50 images was 0.41 (*SD* = 0.21; Min = 0.12, Max = 0.94), indicating that, on average, across all clicks, participants clicked the nature area approximately 41% of the time. This percentage did vary across images, with a minimum of 12% and a maximum of 94%. All quantitative results can be seen in Table E-11. The reliability across all images was .84, and this was robust to the removal of each individual image (maximum change of ± 0.009). The majority of images were identified as good items (38 images; i.e. decreasing Cronbach's alpha if removed), whereas nine were identified as poor (i.e. increasing Cronbach's alpha if removed), and three did not change the alpha. When exploring inter-image correlations, every image had both positive and negative correlations. Only two images had more negative correlations than positive ones (Images 35 & 84).

Table E-11

Descriptive statistics of the voluntary attention items for Study 2 optional stopping Round 3 (N = 100).

Image	*N*	Min	Max	Mean	*SD*	Other	Nature	%	*Z*	Cronbach's ∝ if Item Deleted	Smallest Inter-Image Correlations	Largest Inter-Image Correlations
2	99	0	1	0.52	0.50	48	51	52	0.51	.841	-.23	.31
4	100	0	1	0.17	0.38	83	17	17	-1.12	.840	-.25	.27
7	100	0	1	0.17	0.38	83	17	17	-1.12	.840	-.16	.29
8	100	0	1	0.75	0.44	25	75	75	1.61	.844	-.25	.18
11	99	0	1	0.46	0.50	53	46	46	0.27	.837	-.06	.33
12	99	0	1	0.55	0.50	45	54	55	0.65	.838	-.19	.29
22	100	0	1	0.35	0.48	65	35	35	-0.27	.842	-.13	.30
23	100	0	1	0.76	0.43	24	76	76	1.66	.844	-.19	.23
25	100	0	1	0.23	0.42	77	23	23	-0.84	.840	-.18	.27
26	100	0	1	0.41	0.49	59	41	41	0.01	.836	-.16	.39
28	100	0	1	0.54	0.50	46	54	54	0.62	.840	-.18	.40
30	100	0	1	0.14	0.35	86	14	14	-1.26	.844	-.25	.23
35	100	0	1	0.20	0.40	80	20	20	-0.98	.844	-.31	.36
41	100	0	1	0.44	0.50	56	44	44	0.15	.840	-.20	.39
42	100	0	1	0.55	0.50	45	55	55	0.67	.837	-.22	.39
43	100	0	1	0.29	0.46	71	29	29	-0.55	.840	-.13	.30
46	100	0	1	0.24	0.43	76	24	24	-0.79	.837	-.03	.32
47	100	0	1	0.52	0.50	48	52	52	0.53	.832	-.18	.42
52	100	0	1	0.35	0.48	65	35	35	-0.27	.840	-.13	.28
53	100	0	1	0.15	0.36	85	15	15	-1.21	.837	-.15	.40
54	100	0	1	0.30	0.46	70	30	30	-0.51	.833	-.11	.44
55	100	0	1	0.35	0.48	65	35	35	-0.27	.842	-.25	.28
57	100	0	1	0.34	0.48	66	34	34	-0.32	.843	-.20	.24
58	100	0	1	0.63	0.49	37	63	63	1.05	.834	-.29	.67

59	100	0	1	0.60	0.49	40	60	60	0.91	.832	-.31	.67
60	100	0	1	0.44	0.50	56	44	44	0.15	.842	-.18	.36
62	100	0	1	0.76	0.43	24	76	76	1.66	.838	-.13	.31
63	100	0	1	0.31	0.46	69	31	31	-0.46	.836	-.16	.31
64	100	0	1	0.13	0.34	87	13	13	-1.31	.839	-.12	.30
67	99	0	1	0.27	0.45	72	27	27	-0.64	.839	-.09	.36
70	100	0	1	0.12	0.33	88	12	12	-1.36	.838	-.09	.36
71	100	0	1	0.66	0.48	34	66	66	1.19	.839	-.14	.44
73	100	0	1	0.17	0.38	83	17	17	-1.12	.837	-.20	.30
74	100	0	1	0.25	0.44	75	25	25	-0.74	.835	-.25	.40
75	100	0	1	0.21	0.41	79	21	21	-0.93	.838	-.18	.41
77	100	0	1	0.35	0.48	65	35	35	-0.27	.835	-.22	.31
79	100	0	1	0.43	0.50	57	43	43	0.11	.834	-.15	.35
81	100	0	1	0.13	0.34	87	13	13	-1.31	.839	-.18	.49
82	100	0	1	0.44	0.50	56	44	44	0.15	.835	-.09	.40
83	100	0	1	0.94	0.24	6	94	94	2.51	.842	-.16	.44
84	100	0	1	0.65	0.48	35	65	65	1.14	.837	-.25	.18
85	100	0	1	0.93	0.26	7	93	93	2.46	.841	-.23	.47
86	100	0	1	0.79	0.41	21	79	79	1.80	.837	-.16	.29
87	100	0	1	0.47	0.50	53	47	47	0.29	.837	-.12	.36
88	100	0	1	0.26	0.44	74	26	26	-0.70	.836	-.09	.37
89	100	0	1	0.34	0.48	66	34	34	-0.32	.841	-.25	.46
90	100	0	1	0.32	0.47	68	32	32	-0.41	.840	-.21	.31
91	99	0	1	0.34	0.48	65	34	34	-0.30	.839	-.14	.31
92	100	0	1	0.29	0.46	71	29	29	-0.55	.839	-.13	.31
93	100	0	1	0.37	0.49	63	37	37	-0.18	.835	-.25	.35

The distribution was right-skewed (see Figure E-10) and non-normal, Shapiro-Wilkes estimate = 0.94 p = .01. The distribution did not become normally distributed during the robustness tests. In this round, only two robustness tests were run because of the range of nature clicks available (15% or less nature clicks & 20% or less nature clicks). The standardized distribution did not become normal after the removal of the lower nature clicked images (see Table E-12 for the Shapiro-Wilkes estimates and list of removed images).

Figure E-11

Normality plots for the standardized distribution of the Z-transformed voluntary attention mean scores for all 50 images in Study 2 optional stopping Round 3.

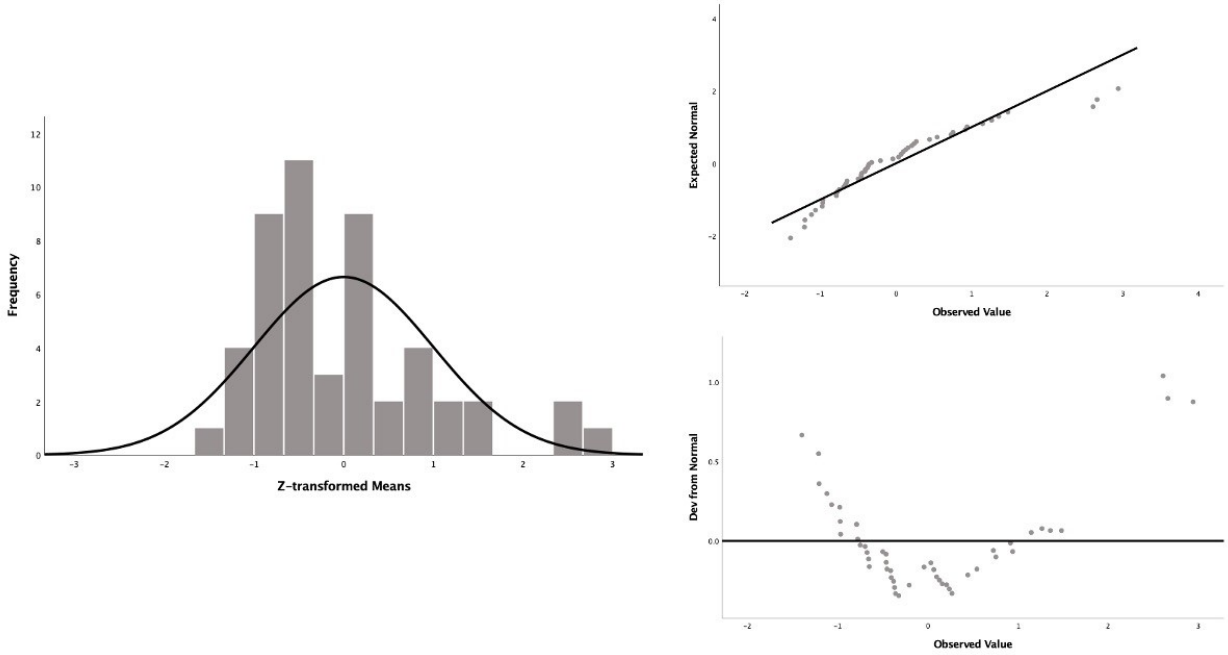


Table E-12

Study 2 optional stopping Round 3 robustness test for normality of the standardized distribution of the voluntary attention item.

Removal	Number of images	Images removed	Shapiro-Wilks Estimate	p
Images with 15% or less clicks on the nature area	46	30, 53, 64, 70, 81	0.93	.007
Images with 20% or less clicks on the nature area	42	4, 7, 35, 73	0.92	.005

Looking at both

Across both items, there was only a single image that had no variance. Image 70 showed no variance in the first attention item. In the previous round, this image had a small percentage of nature clicks (3% in Round 2). For voluntary attention item, this is the image with the least amount of nature clicks, with 12%.

In contrast with the other rounds, the robustness tests for normality were unchanged across all tests. This suggested that there was not an appropriate cut off point to identify images for removal. With this in mind, I have decided to move on with recruitment of the full sample (N = 300) with the current set of images (i.e. all images are retained).

Extra Analyses

To examine if the means across the three rounds of psychometric analyses are in fact different, a one-way ANOVA was run with each item's mean as the dependent variable and the psychometric rounds was the independent variable. For both attention items, there was a significant difference in means across Rounds (first attention: $F(2, 149) = 7.84$ $p = .008$ $\eta^2 = 0.6$; voluntary attention: $F(2, 149) = 7.84$ $p < .001$ $\eta^2 = 0.10$. The only significant Scheffe post-hoc comparison for both items was between Rounds 1 and 3, where the mean of Round 3 was significantly larger than the mean of Round 1 (item 1: $M_{difference(3-1)} = 0.12$ $p = .008$; item 2: $M_{difference(3-1)} = 0.16$ $p < .001$).

I also explored the correlation between the first attention and voluntary attention items across all images in the three rounds. There was a significant correlation $r(148) = .70$ $p < .001$. This indicates that there was a strong association between how much nature clicks there were in each of the attention items.

Lastly, I also examined unconditional multilevel models where the mean of each of the rounds was nested within the image. For the first attention item, the interclass correlation was 94.87%, indicating that the mean across the three rounds were consistent. This was similar for the voluntary attention. The interclass correlation for the voluntary attention item was 95%.

Appendix F
Correlation Matrices from Studies 1, 2, and 3

As exploratory analyses, the correlation between all included variables outside of the created behavioural noticing nature tasks were explored. The variables included are: the averaged nature relatedness, item 8 of NRS ("Even in the middle of the city, I notice nature around me"), item 21 of NRS "I take notice of wildlife wherever I am"), the Self subscale of NRS, the Perspective subscale of NRS, the Experience subscale of NRS, positive affect, negative affect, awe, and noticing nature activity. There is a separate table for Study 1 (Table F-1), 2 (Table F-2), and 3 (Table F-3).

Table F-1

Correlation matrix of all included variables from Study 1, except for the perception items (because they are included in the main text).

Variables		1	2	3	4	5	6	7	8	9
1. NR		-								
2. Item 8 NRS	*r*	.58***								
	p	<.001								
	N	281								
3. Item 21 NRS	*r*	.61***	.47***							
	p	<.001	<.001							
	N	281	281							
4. NRS Self	*r*	.88***	.63***	.58***						
	p	<.001	<.001	<.001						
	N	281	281	281						
5. NRS Perspective	*r*	.65***	.28***	.23***	.43***					
	p	<.001	<.001	<.001	<.001					
	N	281	281	281	281					
6. NRS Experience	*r*	.72***	.36***	.55***	.49***	.14*				
	p	<.001	<.001	<.001	<.001	.02				
	N	281	281	281	281	281				
7. PA	*r*	.13*	.17**	.17***	.23***	-.03	.06			
	p	.03	.01	.004	<.001	.61	.31			
	N	281	281	281	281	281	281			
8. NA	*r*	.05	-.07	.01	.01	.07	.04	-.47***		
	p	.44	.27	.87	.94	.23	.50	<.001		
	N	281	281	281	281	281	281	281		
9. Awe	*r*	.08	.12*	.10	.19**	-.14*	.08	.37***	-.06	
	p	.19	.05	.11	.001	.02	.20	<.001	.34	
	N	281	281	281	281	281	281	281	281	
10. NNA	*r*	.60***	.44***	.45***	.55***	.33***	.46***	.14*	.16**	.18**
	p	<.001	<.001	<.001	<.001	<.001	<.001	.02	.01	.002
	N	280	280	280	280	280	280	280	280	280

Note: NR – Nature relatedness. NRS – Nature Related scale. PA – Positive Affect. NA – Negative Affect. NNA – Noticing Nature Activity. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table F-2

Correlation matrix of all included variables from Study 2, except for the attention tasks (because they are included in the main text).

Variables		1	2	3	4	5	6	7	8	9
1. NR		-								
2. Item 8 NRS	*r*	.46***								
	p	<.001								
	N	294								
3. Item 21 NRS	*r*	.61***	.45***							
	p	<.001	<.001							
	N	293	293							
4. NRS Self	*r*	.81***	.46***	.44***						
	p	<.001	<.001	<.001						
	N	294	294	293						
5. NRS Perspective	*r*	.62***	.19***	.26***	.28***					
	p	<.001	<.001	<.001	<.001					
	N	294	294	293	294					
6. NRS Experience	*r*	.74***	.31***	.60**	.42***	.13*				
	p	<.001	<.001	<.001	<.001	.02				
	N	293	293	293	293	293				
7. PA	*r*	.14*	.09	.07	.16**	.05	.09			
	p	.02	.13	.23	0.01	.36	.12			
	N	293	293	293	293	293	293			
8. NA	*r*	.01	-.08	-.003	0.03	.07	-.08	-.38***		
	p	.89	.20	.97	0.57	.26	.20	<.001		
	N	293	293	293	293	293	293	293		
9. Awe	*r*	.24***	.04	.08	.27***	.04	.19**	.42***	-.12*	
	p	<.001	.46	.15	<.001	.46	.001	<.001	.05	
	N	293	293	293	293	293	293	293	293	
10. NNA	*r*	.65***	.39***	.51***	.52***	.36***	.51***	.18**	.04	.27***
	p	<.001	<.001	<.001	<.001	<.001	<.001	.002	.53	<.001
	N	293	293	293	293	293	293	293	293	293

Note: NR – Nature relatedness. NRS – Nature Related scale. PA – Positive Affect. NA – Negative Affect. NNA – Noticing Nature Activity. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table F-3

Correlation matrix of all included variables from Study 3, except for the attention tasks (because they are included in the main text).

Variables		1	2	3	4	5	6	7	8	9
1. NR		-								
2. Item 8 NRS	*r*	.60***								
	p	<.001								
	N	1052								
3. Item 21 NRS	*r*	.61***	.49***							
	p	<.001	<.001							
	N	1052	1052							
4. NRS Self	*r*	.90***	.68***	.57***						
	p	<.001	<.001	<.001						
	N	1052	1052	1052						
5. NRS Perspective	*r*	.78***	.31***	.31***	.56***					
	p	<.001	<.001	<.001	<.001					
	N	1052	1052	1052	1052					
6. NRS Experience	*r*	.74***	.46***	.62***	.58***	.30***				
	p	<.001	<.001	<.001	<.001	<.001				
	N	1052	1052	1052	1052	1052				
7. PA	*r*	.18***	.20***	.14***	.26***	-.02	.20***			
	p	<.001	<.001	<.001	<.001	.60	<.001			
	N	1052	1052	1052	1052	1052	1052			
8. NA	*r*	-.13***	-.05	-.04	-.11***	-.05	-.17***	-.60***		
	p	<.001	.08	.23	<.001	.08	<.001	<.001		
	N	1052	1052	1052	1052	1052	1052	1052		
9. Awe	*r*	.30***	.28***	.24***	.41***	.02	.29***	.60***	-.20***	
	p	<.001	<.001	<.001	<.001	.50	<.001	<.001	<.001	
	N	1052	1052	1052	1052	1052	1052	1052	1052	
10. NNA	*r*	.62***	.51***	.55***	.63***	.31***	.57***	.32***	-.07*	.44***
	p	<.001	<.001	<.001	<.001	<.001	<.001	<.001	.02	<.001
	N	1052	1052	1052	1052	1052	1052	1052	1052	1052

Note: NR – Nature relatedness. NRS – Nature Related scale. PA – Positive Affect. NA – Negative Affect. NNA – Noticing Nature Activity. * $p < .05$, ** $p < .01$, *** $p < .001$.